Will I have signal out there?

Documentation set, October 9, 2026

This PDF is built from the files in the Will I have signal out there folder. Links to other folders appear as plain text.

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Part 1: Overview

Will I have signal out there/README.md

Will I have signal out there?

A trip planner for hikers, campers and backpackers. Pick a trail and see where each phone carrier has coverage, where it drops out, when a Starlink Direct-to-Cell satellite text could get through, and how much to trust the coverage map at each point.

This folder is the documentation for Project 1. It is written so that someone who has never seen the project can build it from these documents alone.

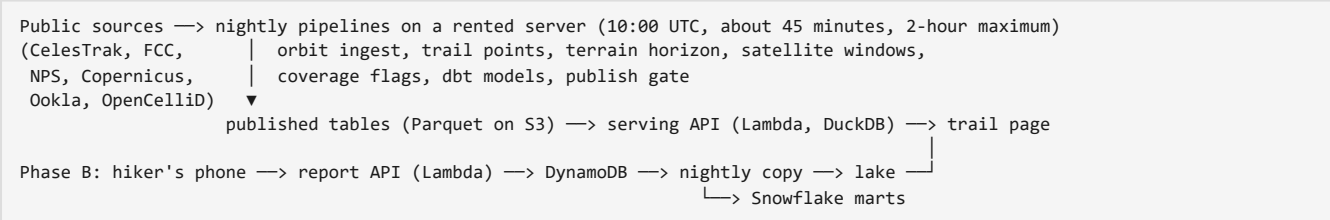
Field	Value
Status	Documents complete for review. Draft pending John's approval. **No code exists yet**
Owner	John
Last updated	Oct 9, 2026
Live link	None yet. Phase A (live) has not started

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1. The project in plain words

- **Who it is for:** hikers, campers and backpackers who want to know whether their phone will work on a trail, and whether a satellite text could reach someone.
- **What it shows:** for each point along a trail, whether the carrier's coverage map says there is signal, and whether a satellite overhead could reach the phone, with canyon walls and mountains taken into account.
- **How far to trust it:** the coverage map is a model's claim. Where measured speed tests exist nearby, the page gives a confidence level. Where they do not, the page says "Not enough measurements here; the coverage map's claim is unverified." It does not guess.
- **Late data is never hidden.** If the nightly update fails, the page keeps the old data and shows a visible late notice with the time of the last good run.
- **Hikers can report.** In Phase B, a hiker with no signal can tap "I had signal here" or "I had no signal here". The phone reduces its location to a cell about 350 m across, and exact coordinates never leave the phone.
- **Phase A (first live version):** Grand Canyon's Bright Angel, South Kaibab and North Kaibab trails, with T-Mobile, for the next 24 hours of satellite passes. Phase B adds every carrier, measured speeds, a warehouse, the confidence model, a trip assistant and hiker reports. Phase C (second round) adds every national park, route upload and an offline trip pack.

2. How it works, in one picture



The page never waits for the nightly run. It reads the last published tables, so it answers while the rented server is switched off.

3. Documents in this folder

Document	What it is for	When it was written
README.md	This page: what the project is, the document map and the reading order	Before coding
requirements.md	Who the project is for; features by phase; functional and quality requirements; acceptance criteria	Before coding

Document	What it is for	When it was written
design-doc.md	The design in 17 sections: context, architecture (C4 diagrams), every component with its functions, runtime flows, data model, API, deployment, cross-cutting concerns, testing, alternatives, risks, open questions and glossary	Before coding
adrs/	Architecture decision records: one short record per significant decision, with alternatives and consequences. The superseded record 0006a was removed on Oct 9, 2026 as a duplicate old version; 0006 is the amended record	Before coding
contracts/	Data contracts (ODCS v3, YAML) for every source and every published table: schema, rules, freshness, owner, version and storage location	Before coding
openapi.yaml	The API specification (OpenAPI 3): every endpoint, request, response, error and limit	Before coding
../Platform/platform-requirements.md	The shared platform parts Project 1 depends on: repository and CI, cloud base, the session box, the serving template, status and incidents, engineering standards, tools and keys	Before coding
operations.md	Schedules, freshness targets, monitoring, cost limits, runbooks for each known failure, incidents and backups	Kept current while building
security-and-privacy.md	Threat model, personal-data inventory with retention, access matrix, licence and attribution register, and open privacy questions	Kept current while building
model-card.md	For the two AI components: intended use, out-of-scope use, data, evaluation plan, known failure cases and updates	Kept current while building
changelog.md	What changed in the documents and, later, in each phase	Kept current while building
Reference from code	Function reference from docstrings, the dbt docs site and the API docs page	**Not written yet.** Generated from the code once the code exists

4. Reading order

1. **This page**, then requirements.md sections 1 to 3 for scope.
2. design-doc.md sections 2, 3 and 6: the summary, the constraints, and the diagrams.
3. design-doc.md section 7: every component, with its functions and the four hardest parts in full.
4. design-doc.md sections 8 and 9: the runtime flows and the data model. Then the contracts.
5. openapi.yaml and design-doc.md section 10: the API.
6. ../Platform/platform-requirements.md: the platform parts, before the first pipeline.
7. adrs/: read the decision behind any value that looks odd.
8. operations.md, security-and-privacy.md and model-card.md before anything goes live.
9. design-doc.md section 16: the open questions. Check each one that touches the part you are building.

5. Build order, by dependency

This is the order the parts depend on each other. It has no dates. The documents say what to build and in what order it must happen. They do not say when.

Before Project 1 (platform, from ../Platform/platform-requirements.md) 1. Repository, CI, dev container (section 2). 2. Cloud base in Terraform: buckets, Glue, CloudFront, budget, roles, Parameter Store, state (section 3). 3. Session box with k3s, Airflow and Spark, the nightly schedule, and the Airflow database backup (section 4). 4. Serving template: Parquet publish, Lambda with DuckDB behind API Gateway, the site template (section 5). 5. Status page, metrics table and alerts (section 6).

Phase A, in order 1. C8 contract diff, and C1 orbit ingest with C2 selector (the [DTC] tag and the cross-check). Needs the first-run check on a real CelesTrak file (OQ-6, OQ-20). 2. C3 trail loader and C4 terrain horizon. Needs the NPS and Copernicus first-run checks (OQ-8, OQ-36). 3. C5 satellite windows. Needs C2 and C4, and the measured size (OQ-1). 4. C6 FCC coverage. Needs the FCC file size (OQ-7). 5. C7 dbt models on DuckDB, then C9 publish gate, then C10 publisher. The gate stops a publish when a check fails. 6. C11 serving API and C12 trail page. Gate: the page answers for every main trail with the session box off (AC-1A.6).

Phase B, after Phase A is live 1. C13 all carriers, Ookla and OpenCelliD. Needs the Ookla first-run check (OQ-10). 2. C14 orbit update stream (Kafka, availableNow), which needs C1. 3. C15 Snowflake warehouse and C16 lineage (Marquez, dbt docs). Needs the Snowflake and ARM Marquez first-run checks (OQ-4, OQ-5). 4. C21 report button on the phone, C22 report API with limits, C23 nightly copy and set-aside. These need the limits and N decisions (OQ-12, OQ-26). 5. C17

confidence model. Needs the labels from C23, Databricks access (OQ-3) and the model family (OQ-18). 6. C18 AI foundations (call log, traces, MCP tools), then C19 assistant, then C20 evals in CI. The eval thresholds are set by John (OQ-16). 7. C24 operations hardening: clean-up rules, cold-start measurement, weekly smoke test, incident issues.

Second round: Phase C features (national scope, route upload, offline trip pack, model comparison). Their acceptance checks are written when Phase C is designed.

6. Accounts and keys needed

Account or key	Needed for	Phase	Status
AWS account, in us-east-1 for dev and live	Everything	Platform	Region chosen by John (OQ-42, ADR 0034)
GitHub, with a public repository	Code, CI and the public site	Platform	Needed from the first commit
Databricks Free Edition	Training the confidence model	B	Upload is untested (OQ-3)
Snowflake trial	Warehouse and metrics	B	Connection untested (OQ-4)
Model API account, with a $10 monthly spend limit set before the first call	Assistant and LLM judge	B	Anthropic API, cheapest Claude model (ADR 0034)
Public data sources (CelesTrak, NPS, FCC, Copernicus, Ookla, OpenCelliD, U.S. Census TIGER/Line)	Inputs	A and B	No key is recorded. Confirm in the first-run checks

Every key is stored in AWS Parameter Store and read at run time. None is written in code or in GitHub. See ../Platform/platform-requirements.md, section 9.

7. Conventions

- **Times** are stored in UTC and converted to local time only on the page.
- **Identifiers:** FR is a functional requirement, QR a quality requirement, AC an acceptance criterion, ADR an architecture decision record, OQ an open question, C a component of the design doc. All are defined in requirements.md and design-doc.md.
- **Values** are either recorded decisions (cited by ADR or by "Decided by John") or confirmed facts. Anything else is an open question in design-doc.md, section 16, with a matching GitHub issue. The issues are not yet created.
- **Names** of tables, functions, endpoints and buckets are proposals until John approves them (OQ-21).
- **Environments:** `dev` and `live` are fully separate. A dev role cannot write live data.
- **Licences:** the Ookla notice is shown exactly as written in ADR 0013. Every source is credited.

8. What this set does not contain

- **A schedule.** The documents say what to build, in what order and against what checks. The schedule is kept apart from them.
- **The platform's own document set.** The platform's documents are not written yet. ../Platform/platform-requirements.md holds the parts Project 1 needs.
- **The reference generated from code.** It does not exist until the code does.
- **Setup commands.** These are written with the code, in the repository's own README, once there is code to run.
- **GitHub issues for the open questions.** These are to be created. They are not created yet.

9. Approval

John approves each document before coding of that part begins. Coding of Project 1 does not start until the requirements, design doc, ADRs, contracts and API specification are approved.

Part 2: Requirements

`Will I have signal out there/requirements.md`

Will I have signal out there? Requirements

Field	Value
Status	Draft for review
Owner	John
Reviewers	John (approves before coding of this project begins)
Last updated	Oct 9, 2026
Sources	Build Plans for the Five Projects (Oct 8, 2026), Build Plan Decisions (Oct 9, 2026), Documentation Plan (Oct 9, 2026)
Related	design-doc.md, adrs/, contracts/, openapi.yaml, ../Platform/platform-requirements.md, operations.md, security-and-privacy.md, model-card.md, README.md, changelog.md

Conventions used in this document:

- A value is either a recorded decision (cited as an ADR, an architecture decision record in adrs/) or a confirmed fact from the three source files. Anything else is an open question, listed as OQ-n (open question n) in section 16 of the design doc.
- Terms are defined in the glossary in section 17 of design-doc.md. Phase A, B and C and the platform skeleton are defined in section 1 of design-doc.md. "Satellite window" is a period when a satellite is reachable; "prediction span" is the 26 hours the pipeline predicts; "log retention period" is how long logs may be kept with internet addresses. "Valid until" in prose is `valid_until` in code.
- The rules cited by number are listed with their content in section 1 of design-doc.md. In short: engineering standard 1 keep raw files unchanged; 2 event time and source key on every record; 3 idempotent and atomic tasks; 4 write, audit, publish; 5 contracts and schema diff; 6 dbt layers; 13 least privilege; 15 UTC timestamps; platform step 0.2 cloud base with separate dev and live; platform step 0.5 status page and incident records. "Engineering standard N" means rule N in the build plan's "Engineering standards" table. "The page" means the public trail page.
- Each functional requirement names who acts (a component from design-doc.md section 7, or a person) and is checked by the acceptance criterion of its feature (section 6) or by the quality scenario that names it (section 5).
- "Done when" checks from the build plan are reproduced as the acceptance criteria (AC) of each feature, without change.
- Requirement identifiers: F = feature, FR = functional requirement, QR = quality requirement, AC = acceptance criterion.
- Names of tables, endpoints and fields that appear below are proposals of the design documents unless a source file names them. They are approved with the documents (OQ-21).

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1. Introduction and goals (arc42 section 1)

1.1 What the project is

A trip planner for hikers, campers and backpackers. The visitor picks a trail or campsite and sees:

- where each carrier has coverage and where it drops out;
- when a Starlink Direct-to-Cell satellite text could get through, accounting for canyon walls and mountains that block the sky;
- how much to trust the coverage map at each point.

Direct-to-Cell texting also needs T-Mobile service (in the United States) and a supported phone, and the page says so. T-Mobile is the Phase A carrier.

Story, as decided by John: the project is about people going out in nature and whether they will have a connection (Build Plan Decisions, "Projects and scope").

1.2 Quality goals

The top quality goals, in priority order for this project. Each is expanded as a quality requirement in section 5.

Priority	Goal	Why it matters here	Basis
1	Never present late or unverified information as current or certain	A hiker may act on a satellite window or a coverage claim	ADR 0006, ADR 0008
2	Hikers' locations stay private	Exact coordinates never leave the phone; no internet addresses used	ADR 0009, ADR 0011
3	The public page answers while the session box (the rented AWS server that runs the pipelines) is off	The box runs only about 45 minutes a night	Build plan 1A.6, platform skeleton (the first build stage of the platform)
4	Cost never surprises	Budget under $50 a month, a hard limit on every paid item	Decisions, "Cost, limits and operations"

1.3 Stakeholders

Stakeholder	Expectation
Hikers, campers, backpackers (public visitors)	Accurate, honest signal and satellite information for the trail they pick; ability to report signal from the trail
John (owner and operator)	A live page that does not break while building; costs under limits; documents to build from
Data providers (CelesTrak, FCC, NPS, Copernicus, Ookla)	Credit and license terms respected (section 5, QR-12)

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2. Users and needs

User	Situation	Needs	Features that serve it
Hiker planning a trip at home	Choosing a trail and wanting to know where to check in with family	See coverage by carrier along the trail; see when a satellite text could get through; see how far to trust the map	F-1A.4, F-1A.6, F-1B.1, F-1B.4, F-1B.6
Hiker on the trail with no signal	Has just found that a place has or lacks signal	Report "I had signal here" or "I had no signal here" without signal at that moment, without giving away exact position	F-1B.8
Camper or backpacker packing for a trip	Will lose signal before reaching camp	Save the map, satellite windows and check-in plan to the phone before losing signal	F-1C.2 (second round)
Hiker using a trail app	Has a route as a GPX file	Upload a route and get the same information	F-1C.1 (second round)
Owner and operator (John)	Runs the nightly pipeline and the site	Know when a run fails; know cost; rerun safely; see data quality	F-1A.5, F-1B.3, F-1B.5, F-1B.9

Needs that apply to every visitor:

- The page shows the time until which the predictions are valid, in the hiker's local time, and a late notice when the predictions are late.
- The page states what Direct-to-Cell texting requires (T-Mobile service, supported phone).
- The page says so when the data cannot answer.

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3. Features by phase

Phases follow the build plan: A is live and small, B adds depth, C is the second-round scale phase. Feature identifiers equal the build plan step numbers.

3.1 Phase A: live

Scope (ADR 0001, ADR 0020): Grand Canyon National Park's Bright Angel, South Kaibab and North Kaibab trails, one carrier (T-Mobile), the next 24 hours of satellite passes (26 hours are predicted so there is no daily gap, ADR 0019). Decided by John, Oct 9, 2026.

Feature	Name	Summary
F-1A.1	Orbit ingest	Airflow DAG pulls CelesTrak GP data as JSON and selects Direct-to-Cell satellites by the [DTC] name tag, cross-checked against altitude and inclination
F-1A.2	Trails and terrain	NPS trail lines for Grand Canyon sampled every 50 meters; terrain horizon per point from Copernicus 30-meter elevation, stored in Iceberg
F-1A.3	Satellite windows	SGP4 propagation at 1-minute steps for 26 hours (the page says the next 24 hours); satellite windows per trail point when a satellite is above 10 degrees and above the terrain horizon; each run stamped with a valid until time
F-1A.4	Cell coverage	FCC mobile coverage (H3 resolution 9 files) for one carrier in Arizona; each trail point marked covered or not
F-1A.5	Models and checks	dbt on DuckDB (staging, intermediate, small star schema); contracts for CelesTrak, FCC, NPS trails and elevation; Great Expectations publish gate
F-1A.6	Trail page	Gold tables published; React page with a strip map (bars where the carrier has coverage, no bars where it does not) and the next satellite windows along the trail; "Predictions valid until [time]"; late data shown with a visible late notice and the time of the last good run

3.2 Phase B: depth

Feature	Name	Summary
F-1B.1	Every carrier and measured speeds	FCC coverage for every carrier for the park; Ookla open mobile speed-test data for Arizona; license notice
F-1B.2	Orbit update stream	New element sets published to Kafka; Structured Streaming job with availableNow trigger finds changed satellites and triggers re-propagation
F-1B.3	Warehouse, metrics and lineage	Snowflake over Glue, dbt marts with SCD Type 2 carrier coverage, MetricFlow metrics, Superset dashboard, resource monitor, OpenLineage with Marquez, dbt docs
F-1B.4	Coverage confidence model	Model of how likely the FCC map is right; confidence only where enough speed tests exist; trained in Databricks, tracked in MLflow, exported to the repo by hand
F-1B.5	AI foundations	Call log, tracing to AWS X-Ray, MCP server (tool interface for AI models) for metrics, read-only logged SQL fallback, versioned prompts, structured output, eval harness, daily cap, $10 spend limit
F-1B.6	Trip-planning assistant	Plain-language question returns a cited check-in plan
F-1B.7	Evals in CI	30 or more graded questions, calibrated LLM judge, red-team set of 20 or more prompts, votes in DynamoDB, A/B test of two prompt versions
F-1B.8	Hiker signal reports	"I had signal here" and "I had no signal here" buttons; H3 cell computed on the phone; offline queue; rate limits; DynamoDB; labels for the model
F-1B.9	Operations hardening	S3 clean-up rules, cold-start measurement, weekly source smoke test, de-duplicated GitHub issues for failures

3.3 Phase C: scale (second round)

Starts only after every project has reached Phase B.

Feature	Name	Summary
F-1C.1	National scope and route upload	Every national park; a route upload (a GPX file from any trail app)
F-1C.2	Offline trip pack	Save the map, satellite windows and check-in plan to the phone before losing signal
F-1C.3	Model comparison	Compare the cheapest model with a stronger one in the eval suite; switch only if the scores justify the cost

Phase C features have no "Done when" checks in the build plan. Their requirements and acceptance criteria will be written when the second round is planned; this is a recorded gap (Decided by John, Oct 9, 2026: write them when Phase C is designed), not guessed here.

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4. Functional requirements

Each requirement is traceable to a build plan step (the Feature column) and to a decision where one exists.

4.1 Phase A

Numbering note: IDs are never renumbered. Some IDs appear out of order because requirements were added or split later: FR-63 to FR-65 (Phase C) are listed after Phase B; FR-66 to FR-69 follow FR-62; FR-70 to FR-74 follow FR-54 and replace what FR-54 used to say about limits; FR-75 and FR-76 follow FR-6.

ID	Requirement	Feature	Decision
FR-1	The orbit ingest DAG (an Airflow job, component C1) pulls CelesTrak GP data (orbital elements) as JSON no more than once per 2-hour update, and stops on any response other than HTTP 200	F-1A.1	Build plan 1A.1
FR-2	Every ingest job keeps each raw file unchanged, with its source URL, fetch time and hash	F-1A.1 and all sources	Engineering standard 1
FR-3	The Direct-to-Cell selector (C2) selects Direct-to-Cell satellites by the [DTC] tag in their names	F-1A.1	ADR 0002
FR-4	The selector (C2) cross-checks each selected satellite against altitude (about 330 to 370 km) and inclination (43 or 53 degrees), and logs and lists every disagreement	F-1A.1	ADR 0002
FR-5	The trail loader (C3) loads the NPS trail lines for Grand Canyon and samples one point every 50 meters along them	F-1A.2	ADR 0004
FR-6	The terrain horizon engine (C4) computes, for each trail point, the terrain horizon (how high terrain rises in each compass direction, out to about 15 km) from the Copernicus 30-meter elevation model on AWS	F-1A.2	ADR 0004
FR-75	The terrain horizon engine (C4) runs the horizon computation for many trail points at the same time in Spark	F-1A.2	ADR 0004
FR-76	The terrain horizon engine (C4) stores the horizon of every trail point in Iceberg	F-1A.2	ADR 0004
FR-7	The satellite window engine (C5) propagates the selected satellites with SGP4 (the standard orbit model) at 1-minute steps for 26 hours (24 hours plus the 2-hour nightly maximum), so the page has no daily gap. The 24 hours is the promise to the visitor, and the extra 2 hours are a safety margin; the page still says "the next 24 hours"	F-1A.3	Build plan 1A.3; ADR 0019
FR-8	The satellite window engine (C5) counts a satellite as reachable from a trail point only when the satellite is above both 10 degrees and the terrain horizon in its direction	F-1A.3	ADR 0003
FR-9	The satellite window engine (C5) stores satellite windows as start and end times per trail point, and never as one row per minute	F-1A.3	ADR 0005
FR-10	The satellite window engine (C5) stamps each run with the time its predictions stop being valid, and a rerun of a day replaces only that day	F-1A.3	ADR 0006
FR-11	The satellite window engine (C5) records in this project's design document how the satellite windows change at 5 and 20 degrees	F-1A.3	ADR 0003
FR-12	The FCC coverage loader (C6) loads FCC mobile coverage (H3 resolution 9 files) for one carrier in Arizona and flags each trail point as covered or not	F-1A.4	ADR 0007
FR-13	The dbt project (C7) builds staging, intermediate and a small star schema (trail point, time, coverage and satellite windows) on DuckDB; `source()` is used only in staging; unique and not_null tests are on keys	F-1A.5	Engineering standard 6
FR-14	The project keeps a contract for CelesTrak, FCC, NPS trails and elevation, and the contract diff (C8) compares each source's schema with its contract before loading	F-1A.5	ADR 0018
FR-15	The publish gate (C9, using Great Expectations) checks freshness, empty results and value ranges; new data lands in staging, passes the checks, and only then goes live	F-1A.5	Engineering standard 4

ID	Requirement	Feature	Decision
FR-16	The publisher (C10) publishes the gold tables (the finished tables the page reads) as Parquet	F-1A.6	Platform serving template
FR-17	The trail page (C12) lets the visitor pick a trail and shows a strip map (bars where the carrier has coverage, no bars where it does not) and the next satellite windows along the trail, with credits	F-1A.6	Build plan 1A.6
FR-18	The trail page (C12) shows "Predictions valid until [time]" in the hiker's local time. When the data is late or the run failed, the page still shows the old data with a visible late notice that includes the time of the last good run, and does not hide the data	F-1A.6	ADR 0006 (amended)
FR-19	The trail page (C12) states that Direct-to-Cell texting needs T-Mobile service (in the United States) and a supported phone, and that T-Mobile is the Phase A carrier	F-1A.6	Build plan project description
FR-20	The trail page (C12) shows the cold-start notice: "The first request after a period of inactivity may take up to 10 seconds while the service initializes."	F-1A.6	Build plan 0.4, Decisions "Cold starts"
FR-21	Every pipeline stores timestamps in UTC, and only the trail page converts them to local time	all	Engineering standard 15
FR-22	Every pipeline task is idempotent (safe to run again with the same result) and atomic (all or nothing): it overwrites a partition instead of appending blindly, and uses the run's data interval instead of the current time	all	Engineering standard 3
FR-23	Every pipeline writes event time and a source key on every record	all	Engineering standard 2
FR-24	Each DAG writes freshness and row counts to the metrics table	all	Platform step 0.5

4.2 Phase B

ID	Requirement	Feature	Decision
FR-25	The all-carrier loader (C13) loads FCC coverage for every carrier for the park into Iceberg	F-1B.1	Build plan 1B.1
FR-26	The Ookla loader (C13) loads Ookla open mobile speed-test data for Arizona (quarterly, about 600-meter tiles, GPS-accurate tests only) into Iceberg	F-1B.1	Build plan 1B.1
FR-27	The trail page and the README show the exact Ookla license notice (see AC-1B.1)	F-1B.1	ADR 0013
FR-28	The nightly run publishes each new element set to Kafka. A Structured Streaming job (C14), run with the availableNow trigger, finds changed satellites and triggers re-propagation, using an explicit schema, checkpoints on S3, a watermark and dropDuplicatesWithinWatermark	F-1B.2	ADR 0016
FR-29	A refresh step at the end of each pipeline run makes Snowflake see new Iceberg data through the Glue catalog	F-1B.3	ADR 0015
FR-30	The dbt marts run on Snowflake, with SCD Type 2 rows (history kept as rows with validity ranges) for carrier coverage changes between FCC releases	F-1B.3	ADR 0015
FR-31	The project defines its metrics in MetricFlow, provides a Superset dashboard, and publishes snapshot charts to the site each night	F-1B.3	Build plan 1B.3
FR-32	A Snowflake resource monitor suspends the warehouse at 3 credits a month, and every query has a time limit	F-1B.3	Decisions "Snowflake"
FR-33	The platform runs OpenLineage with Marquez (an ARM build of one pinned version) and publishes the dbt docs catalog page	F-1B.3	Platform
FR-34	The confidence model (C17) predicts how likely the FCC map is to be right, using terrain, distance to the nearest tower (OpenCelliD, ADR 0024) and remoteness (distance to the nearest TIGER/Line road or NPS trail line, ADR 0024, ADR 0035)	F-1B.4	ADR 0008
FR-35	The confidence model (C17) gives a confidence level only where enough speed tests exist nearby; elsewhere the page says "Not enough measurements here; the coverage map's claim is unverified."	F-1B.4	ADR 0008
FR-36	John trains the model in Databricks (the data is uploaded through its Files API) and MLflow tracks every version	F-1B.4	ADR 0014

ID	Requirement	Feature	Decision
FR-37	After each training run, John exports the results and the chosen model by hand (downloads them and commits them to GitHub; whether this can be automated is open, OQ-19). A small monthly job keeps the free Databricks account active. The nightly pipeline always scores with the copy in the repo. Check: after each training run, a results file that carries the run id is present in the repo	F-1B.4	ADR 0014
FR-38	Every model call is logged to S3 (model, prompt version, tokens, latency, cost), and the nightly run loads the log into the `llm_calls` table	F-1B.5	Build plan 1B.5
FR-39	The assistant traces every agent run with OpenTelemetry into AWS X-Ray	F-1B.5	Decisions "Tracing"
FR-40	An MCP server (a tool interface for AI models) exposes the metrics as tools; raw SQL is a read-only, allowlisted fallback that is logged separately	F-1B.5	ADR 0017
FR-41	The project keeps prompts versioned in git, requires structured output on every model call, and uses only the cheapest model for now	F-1B.5	Decisions "AI"
FR-42	The project sets a daily question cap and the $10 monthly spend limit before the first model call	F-1B.5	Decisions "AI spending"
FR-43	The assistant answers a plain-language question with a check-in plan: where each carrier works, when satellite windows open, and how confident the map is, citing the trail points and times it used	F-1B.6	ADR 0017
FR-44	The assistant says so when the data cannot answer the question	F-1B.6	ADR 0017
FR-45	The eval suite has 30 or more questions with known answers, including ones the data cannot answer, and grades them exactly on tool choice, numbers and refusal	F-1B.7	Build plan 1B.7
FR-46	An LLM judge (a second model) scores answer clarity, calibrated on 50 answers that John graded by hand	F-1B.7	Build plan 1B.7
FR-47	The red-team set has 20 or more prompts covering instructions hidden in source text, write requests, off-topic requests and cost abuse	F-1B.7	Build plan 1B.7
FR-48	The assistant stores thumbs-up and thumbs-down votes in DynamoDB, and the project A/B tests two prompt versions	F-1B.7	Build plan 1B.7
FR-49	The trail page has "I had signal here" and "I had no signal here" buttons	F-1B.8	Build plan 1B.8
FR-50	When the hiker taps a button, the phone reads GPS, converts it to an H3 resolution 9 cell with the h3-js library, and discards the exact coordinates on the phone	F-1B.8	ADR 0009
FR-51	A report contains only the cell, the carrier, signal yes or no, and the time rounded to the hour	F-1B.8	ADR 0009
FR-52	The phone saves a report and sends it automatically when signal returns (in the background on Android; on iPhone, the next time the page is opened, with a "1 report waiting to send" note)	F-1B.8	ADR 0010
FR-53	The report API (Lambda, C22) writes each accepted report to DynamoDB	F-1B.8	ADR 0012
FR-54	The phone creates a random visitor ID, and the report API (C22) identifies a visitor only by that ID; no internet address is used	F-1B.8	ADR 0011 (amended)
FR-70	The report API (C22) refuses a visitor's 11th report in a day (the limit is 10 reports a day per visitor)	F-1B.8	ADR 0011 (amended)
FR-71	The report API (C22) refuses a second report from the same visitor for the same cell within 24 hours	F-1B.8	ADR 0011 (amended)
FR-72	The report API (C22) counts each report toward the limits by the hour the report was made on the phone, not by its arrival time	F-1B.8	ADR 0011 (amended)
FR-73	The report API (C22) refuses reports beyond 50 a day for the whole system	F-1B.8	ADR 0011 (amended)
FR-74	The report API (C22) stores limit counters apart from the reports and lets them expire (lifetime: 72 hours after the UTC day, ADR 0026), so a report never carries an ID that links it to a visitor	F-1B.8	ADR 0011 (amended)
FR-55	DynamoDB deletes each raw report row 30 days after the nightly copy into the lake and Snowflake. The lake and Snowflake copies follow the same 30-day rule and keep daily per-cell counts (OQ-43, ADR 0034)	F-1B.8	ADR 0012

ID	Requirement	Feature	Decision
FR-56	The nightly copy (C23) sets aside impossible patterns (reports outside any mapped area, bursts that contradict nearby reports) for review	F-1B.8	ADR 0012
FR-57	The nightly copy (C23) turns reports into labels for the confidence model, including the missing no-signal cases, only where at least N reports in the same cell on the same day agree. Each of those reports is from a different visitor, because one report per cell per visitor is allowed every 24 hours (N is open, OQ-12). The model retrains once there are enough labels	F-1B.8	ADR 0008, ADR 0012
FR-58	The trail page shows report counts per area	F-1B.8	Build plan 1B.8
FR-59	The project applies these S3 clean-up rules (each rule deletes by age): dev data after 14 days; temporary files and Spark checkpoints after 7 days; failed partial uploads after 7 days; pipeline logs after 90 days; security logs (CloudTrail, including Parameter Store reads) after 1 year; old file versions after 30 days. Raw source files and published data are kept. An alarm fires if storage passes a set size	F-1B.9	Decisions "Storage clean-up"
FR-60	The project measures Lambda cold starts and reports the median warm latency and the 95th-percentile cold-start time in the README	F-1B.9	Decisions "Cold starts"
FR-61	A weekly smoke test (a small trial pull) pulls one small sample from every source and alerts on any format change	F-1B.9	ADR 0018
FR-62	A failed run or a missed freshness target opens one GitHub issue, which is de-duplicated and closed on recovery	F-1B.9	Build plan 1B.9
FR-66	The platform removes internet addresses from CloudFront and API Gateway logs before anything is stored beyond a short log retention period (kept 7 days, OQ-29, ADR 0029)	F-1B.8	ADR 0023
FR-67	The OpenCelliD loader (C13) loads tower locations for the distance to the nearest tower; derived data carries the share-alike license, which is recorded in the license register and on the page	F-1B.4	ADR 0024
FR-68	The assistant answers only within the prediction span and says so when asked about later days	F-1B.6	ADR 0021
FR-69	The reports table has point-in-time recovery switched off, so deletion is final at 30 days; other tables keep it on	F-1B.8	ADR 0022

4.3 Phase C

ID	Requirement	Feature
FR-63	The project covers every national park and accepts a GPX route upload	F-1C.1
FR-64	The project offers an offline trip pack (map, satellite windows and check-in plan saved to the phone)	F-1C.2
FR-65	The project compares the cheapest model with a stronger one in the eval suite and switches only if the scores justify the cost	F-1C.3

5. Quality requirements (arc42 section 10)

Every quality requirement has a measure and a verification method. Where the build plan or Decisions document records a number, it is used; where it does not, the number is an open question and the requirement says so. No number below is invented.

5.1 Quality scenarios

ID	Quality	Scenario (stimulus, environment, response)	Measure	Verified by	Basis
QR-1	Freshness	The nightly run executes every day at 10:00 UTC (3 a.m. Arizona, 6 a.m. US Eastern)	A run every day; usually about 45 minutes; hard maximum 2 hours; hitting the maximum is recorded as an incident	Status page shows each pipeline's last good run; Airflow run history; incident record	Decisions "Data freshness", "Nightly shutdown"; defaults table

ID	Quality	Scenario (stimulus, environment, response)	Measure	Verified by	Basis
QR-2	Freshness	CelesTrak is requested	No more than once per 2-hour update	Unit test of the pull guard; DAG review	Build plan 1A.1
QR-3	Correctness (late data is labeled)	The nightly run fails and the page is opened after the previous predictions have expired	The page shows the old data with a visible late notice and the time of the last good run, and states "Predictions valid until [time]"; it does not hide the data. Definition of "late" is decided in ADR 0027 (OQ-23)	Front-end test with a controlled clock (late notice appears; data stays visible); manual check on the live page (AC-1A.6)	ADR 0006
QR-4	Correctness (data quality)	A source file arrives empty, stale or out of range	Publishing stops; nothing reaches the site	Great Expectations checks tested with small sample files in CI (AC-1A.5)	Build plan 1A.5
QR-5	Availability	A visitor opens the trail page while the session box is off	The page answers for every main trail	Test with the box off (AC-1A.6); Lambda runs DuckDB over published Parquet	Build plan 1A.6; platform serving template
QR-6	Performance (cold start)	First request after a period of inactivity	The page says it may take up to 10 seconds; median warm latency and 95th-percentile cold-start time are measured and published in Phase B	Measurement in F-1B.9; README	Decisions "Cold starts"
QR-7	Performance (targets)	Page latency, satellite window job duration	No numeric target is recorded. Satellite window job rows and runtime are measured on the session box (OQ-1)	Test run on the session box; limits set from the measurement	Documentation Plan "Questions still open"
QR-8	Cost	A normal month	Under $50 a month, browser only; skeleton and Phase A months usually under $20	Monthly cost review against the operating-model table; $40 AWS budget as backup	Decisions "Budget"; operating model
QR-9	Cost (hard limits)	Any paid item	A Spark job over its maximum rows or runtime stops before publishing and opens an incident; Snowflake limited to 3 credits a month and a time limit on every query; AI at $10 a month with a daily question cap and a token limit per answer; DynamoDB provisioned within the free tier; session limit 4 hours; nightly maximum 2 hours. The row and runtime values per job, the daily cap and the token limit have no recorded values (OQ-15, OQ-17)	Job-level checks tested in CI with sample data; Terraform resources inspected; resource monitor inspected	Decisions "Surprise costs"; build plan "Hard limits"
QR-10	Privacy	A hiker submits a report	The server receives no exact coordinates, only the H3 resolution 9 cell, carrier, signal yes or no and the time rounded to the hour; no internet address is used for identity; counters never link to stored reports	Request-capture test shows the payload; DynamoDB table inspection (AC-1B.8)	ADR 0009, ADR 0011
QR-11	Privacy (retention)	A raw report is 30 days past its nightly copy into the lake and Snowflake	The raw row is gone	DynamoDB expiry test (AC-1B.8). Point-in-time recovery is off for the reports table, so deletion is final (ADR 0022)	ADR 0012
QR-12	Licensing	Any page or README that shows derived data	Credits are shown; the Ookla notice is exact and present on the page and in the README; derived Ookla data carries the same license	Review checklist in the pull request; page text test for the Ookla notice (AC-1B.1)	ADR 0013; build plan 1B.1

ID	Quality	Scenario (stimulus, environment, response)	Measure	Verified by	Basis
QR-13	Security	Any key, role or deploy	Every key is in Parameter Store (the AWS key store) and none is in code or GitHub; one least-privilege IAM role (permission set) per pipeline; dev roles cannot write live data; gitleaks and Trivy scans in CI; source text treated as data, never as instructions; agents get read-only, allowlisted access	CI scans; role-refusal test from the platform skeleton; red-team set (FR-47)	Engineering standard 13; platform step 0.2
QR-14	Reliability (rerun safety)	A task is retried, a day is rerun, or AWS takes back a spot server (a cheaper server that can be reclaimed)	Rerunning a day replaces only that day; replaying a day of orbit updates writes the same rows	Re-run tests on sample data: running the same task twice gives the same result (AC-1A.3, AC-1B.2)	Engineering standard 3
QR-15	Reliability (spot)	Spot capacity unavailable at 10:00 UTC	The night is skipped; the next run catches up; the page shows the old data with the late notice	Platform skeleton checks; QR-3	Decisions "Nightly shutdown"; operating model
QR-16	Observability	A run fails or misses its freshness target	An email is sent and an incident record is written; in Phase B one de-duplicated GitHub issue opens and closes on recovery	First real failure (platform step 0.5); smoke test in F-1B.9	Platform step 0.5; build plan 1B.9
QR-17	Honesty of coverage claims	A trail point has too few nearby speed tests	The page says "Not enough measurements here; the coverage map's claim is unverified." and shows no confidence level	Test with a sparse-area fixture (AC-1B.4)	ADR 0008
QR-18	Honesty of AI answers	The assistant states a number	The number appears in the query result; otherwise the harness fails	Eval harness (AC-1B.5)	Build plan 1B.5
QR-19	Assistant quality	CI runs the eval suite	CI fails below its accuracy and red-team thresholds. The threshold values are not recorded (OQ-16)	Evals in CI (AC-1B.7)	Build plan 1B.7
QR-20	Data-contract integrity	An upstream source changes format	The change fails loudly and early: schema diff before loading; weekly smoke test in Phase B	Contract diff test in CI with sample files; weekly smoke test	Engineering standard 5; ADR 0018
QR-21	Time handling	Any timestamp	Stored in UTC; converted to local time only on the page	Unit test of the conversion; schema check that stored columns are UTC	Engineering standard 15
QR-22	Lineage and docs	A reviewer inspects the models	The lineage graph shows every step from raw to gold tables; the dbt docs are published	Marquez graph and docs page (AC-1B.3)	Build plan 1B.3

5.2 Constraints that shape the requirements

Constraint	Value	Basis
Monthly budget	Under $50, browser only	Decisions "Budget"
Phase A scope	Bright Angel, South Kaibab and North Kaibab, T-Mobile, next 24 hours (26 predicted)	ADR 0001, 0019, 0020
Non-commercial licenses	Ookla (CC BY-NC-SA 4.0) and Databricks Free Edition are non-commercial; OpenCelliD is share-alike; licensing must be checked before charging	Build plan "Assumptions"
Test method	Small sample files in CI; no staged production failures	Decisions "Failure testing"

6. Acceptance criteria per feature

These are the build plan's "Done when" checks. They are reproduced without change; the mapping to functional requirements is the right-hand column.

6.1 Phase A

Feature	Acceptance criterion (Done when)	Requirements
F-1A.1	AC-1A.1: Raw files land with hash and load time, and the selection is unit-tested, with disagreements listed.	FR-1 to FR-4
F-1A.2	AC-1A.2: Every trail point has a horizon profile, and a canyon-floor point shows a much higher horizon than a rim point.	FR-5, FR-6, FR-75, FR-76
F-1A.3	AC-1A.3: Every trail point has its satellite windows for the next 24 hours with a valid until time, and rerunning a day replaces only that day. (The run predicts 26 hours, ADR 0019; the criterion text is the build plan's.)	FR-7 to FR-11
F-1A.4	AC-1A.4: Every trail point has a coverage flag for the carrier.	FR-12
F-1A.5	AC-1A.5: dbt tests and check tests pass in CI, and publishing stops when a check fails.	FR-13 to FR-15
F-1A.6	AC-1A.6: The public link answers for every main trail with the box off, late data shows a visible late notice with the time of the last good run, and the README states exactly what is deployed. (Decided by John, Oct 9, 2026: the build plan's wording "expired satellite windows disappear on their own" is superseded; the late notice replaces it, ADR 0006.)	FR-16 to FR-20

Notes on measurability (open, not guessed): "much higher" in AC-1A.2 has no numeric threshold in the build plan; the test fixture compares a canyon-floor point and a rim point, and the numeric margin is 5 degrees between the medians of the canyon and rim fixtures (ADR 0036, OQ-24). The main trails are Bright Angel, South Kaibab and North Kaibab (ADR 0020); their names are checked against the NPS data in the first-run download (OQ-36).

6.2 Phase B

Feature	Acceptance criterion (Done when)	Requirements
F-1B.1	AC-1B.1: Coverage for every carrier and measured tiles are in Iceberg, and the license notice is on the page and in the README. The notice reads: "Includes data derived from Speedtest by Ookla® open data, licensed under CC BY-NC-SA 4.0. Derived data published here is shared under the same license."	FR-25 to FR-27
F-1B.2	AC-1B.2: Replaying a day of updates writes the same rows.	FR-28
F-1B.3	AC-1B.3: The marts pass their tests on Snowflake, one metric gives the same value from Snowflake and from DuckDB, and the lineage graph shows raw to gold.	FR-29 to FR-33
F-1B.4	AC-1B.4: MLflow shows the compared versions with their accuracy, the chosen model and its results are in the repo, and the trail page shows a confidence level where tests are dense enough and the "not enough measurements" notice elsewhere.	FR-34 to FR-37
F-1B.5	AC-1B.5: A test question runs end to end through the metrics tool, shows as a trace with its cost, and the harness fails when an answer states a number that is not in the query result.	FR-38 to FR-42
F-1B.6	AC-1B.6: Ten sample trip questions get correct, cited plans, checked by hand.	FR-43, FR-44
F-1B.7	AC-1B.7: CI fails below its accuracy and red-team thresholds, (the threshold values are open, OQ-16), and the README shows the scores, the judge's agreement with your grades and the A/B result.	FR-45 to FR-48
F-1B.8	AC-1B.8: A report made in airplane mode is saved and sends once the phone is back online, the server never receives exact coordinates, an 11th report in a day from one visitor is refused, the system refuses reports beyond 50 a day, (Decided by John, Oct 9, 2026: the build plan's wording "an 11th report in an hour is refused" is superseded by the per-day limits of ADR 0011) raw rows expire on schedule, and report counts per area are on the page.	FR-49 to FR-58, FR-66, FR-69 to FR-74
F-1B.9	AC-1B.9: The clean-up rules show on every bucket, warm and cold-start latency are in the README, the weekly smoke test runs, and the next real failure opens one issue.	FR-59 to FR-62

6.3 Phase C

No "Done when" checks are recorded for Phase C. The platform rule for the second round is that each project's design note is updated with measured numbers (build plan, "Build order and checkpoints"; the design note is this project's design document). Detailed criteria are a recorded gap: they are written when Phase C is designed (Decided by John, Oct 9, 2026).

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7. Traceability

Requirement group	Design doc section	ADR	Interface spec
Orbit ingest and selection (FR-1 to FR-4)	7 (C1, C2), 8.1	0002, 0018	contracts/celestrak_gp.yaml
Trails and terrain (FR-5, FR-6)	7 (C3, C4), 7.A	0004	contracts/nps_trails.yaml, copernicus_glo30.yaml, contracts/trail_point_horizon.yaml
Windows (FR-7 to FR-11)	7 (C5), 7.B	0003, 0005, 0006	contracts/fct_satellite_window.yaml
Coverage (FR-12, FR-25, FR-26)	7 (C6, C13)	0007, 0013	contracts/fcc_mobile_coverage.yaml, ookla_open_data.yaml, contracts/fct_trail_point_coverage.yaml
Checks and publish (FR-13 to FR-16)	7 (C7 to C10)	0018	all contracts
Page and API (FR-17 to FR-20)	7 (C11, C12), 10	0006, 0019, 0020	openapi.yaml
Streaming (FR-28)	7 (C14)	0016	none (internal topic)
Warehouse (FR-29 to FR-33)	7 (C15, C16)	0015	contracts/*
Confidence (FR-34 to FR-37, FR-67)	7 (C17)	0008, 0014, 0024	contracts/fct_coverage_confidence.yaml
AI and assistant (FR-38 to FR-48)	7 (C18 to C20)	0017	../Platform/contracts/llm_calls.yaml, openapi.yaml
Reports (FR-49 to FR-58)	7 (C21 to C23)	0009 to 0012	contracts/hiker_signal_reports.yaml, contracts/fct_report_counts_area.yaml, openapi.yaml
Operations (FR-59 to FR-62)	7 (C24), 12	0018	none

Platform requirements that this project relies on are in ../Platform/platform-requirements.md. They are not repeated here.

Part 3: Design document

`Will I have signal out there/design-doc.md`

Project 1 Design Doc: Will I have signal out there?

1. Status, owner, reviewers, last updated

Field	Value
Status	Draft for review. Coding of this project does not begin until John approves this document, the requirements, the ADRs and the interface specs
Owner	John
Reviewers	John
Last updated	Oct 9, 2026
Template	Documentation Plan "Design doc template" (Google design doc, arc42, C4 model)
Companion documents	requirements.md, adrs/, contracts/, openapi.yaml, ../Platform/platform-requirements.md, operations.md, security-and-privacy.md, model-card.md, README.md, changelog.md
Platform requirements	../Platform/platform-requirements.md: the platform parts this project depends on, specified in full. The separate platform document set is not written yet

How to read the values in this document:

- A value is either a recorded decision or a confirmed fact. A decision is cited as "ADR 00NN" (architecture decision record, a short file in `adrs/`) or as "Decisions" followed by a row name from the Build Plan Decisions document. A fact comes from the build plan, the Build Plan Decisions document or the Documentation Plan.
- Everything else is an open question in section 16 (OQ-n, where OQ means open question), each with a GitHub issue placeholder.
- "Engineering standard N" means rule number N in the build plan's "Engineering standards" table. "Platform step 0.N" means step 0.N of the platform skeleton (the first build stage, before Project 1). "FR", "QR" and "AC" are functional requirement, quality requirement and acceptance criterion identifiers in requirements.md.
- Terms are explained in the glossary (section 17).
- Rules cited by number, with their content:
- Engineering standard 1: keep raw files unchanged with source URL, fetch time and hash (for public archives that never change, keep a manifest of URL, size and hash instead of a copy).
- Engineering standard 2: carry event time and a source key on every record.
- Engineering standard 3: make tasks idempotent and atomic (overwrite a partition, never blind-append, use the run's data interval instead of the current time).
- Engineering standard 4: write, audit, publish (new data lands in staging, passes checks, then goes live).
- Engineering standard 5: every source has a contract, and the schema is diffed before loading.
- Engineering standard 6: dbt layers are staging, intermediate and marts, with `source()` only in staging and unique and not_null tests on keys.
- Engineering standard 8: keep business history in rows (SCD Type 2), not in Iceberg snapshots.
- Engineering standard 9: pick the H3 resolution from the source and join with cell_to_parent.
- Engineering standard 13: least privilege (read-only allowlisted tools for agents, write actions only with human approval, every key in Parameter Store, source text treated as data).
- Engineering standard 14: measure and publish the numbers (input bytes, rows, runtime, cost per run, problems caught, eval scores, users).
- Engineering standard 15: store every timestamp in UTC and convert only on the page.
- Engineering standard 16: anything that runs around the clock (Lambda) writes only to always-on storage.

- Platform step 0.1: repository and CI, including gitleaks and Trivy scans. Step 0.2: cloud base, with separate dev and live and one least-privilege role per pipeline. Step 0.4: serving template, including the cold-start notice. Step 0.5: status page, metrics table, failure email and incident records.
- Phases: Phase A is the first, smallest live version; Phase B adds depth (more sources, history, the warehouse, machine learning and the AI features); Phase C is the second-round scale phase and starts only after every project has reached Phase B. The platform skeleton is the first build stage, before Project 1: a tiny end-to-end version of the whole system.
- Three different things were called "satellite window" and now have separate names: "satellite window" (a period when a satellite is reachable), "prediction span" (the 26 hours the pipeline predicts) and "log retention period" (how long logs may be kept with internet addresses).
- "Valid until" in prose and `valid_until` in code mean the same thing: the time after which a run's predictions are no longer current.
- Names of components, functions, classes, tables, columns and endpoints other than those given in the Documentation Plan example (`compute_horizon`, `find_windows`) or in a source file (for example `TRLNAME`, `UNITCODE`, `[DTC]`) are proposals made by this document. John approves them with the document (OQ-21). They are marked "proposed" where they first appear in section 7.

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2. Summary

Project 1 is a trip planner for hikers, campers and backpackers called "Will I have signal out there?". The visitor picks a trail and sees where each mobile carrier (phone provider) has coverage and where it drops out, when a Starlink Direct-to-Cell satellite text could get through once canyon walls and mountains that block the sky are accounted for, and how much to trust the coverage map at each point. Phase A puts a live page up for Grand Canyon National Park's Bright Angel, South Kaibab and North Kaibab trails with one carrier (T-Mobile) and the next 24 hours of satellite passes (26 hours are predicted so there is no daily gap). Phase B adds every carrier, measured speed tests, a warehouse, a coverage-confidence model, a trip-planning assistant, evals (automated tests of the assistant's answers) and hiker signal reports sent from the trail. Phase C (second round) extends to every national park, adds route upload and an offline trip pack, and compares a stronger model. The outcome is a public page that answers while the session box (the rented AWS server that runs the nightly pipelines) is switched off, never presents late data as current (old data stays visible with a late notice and the time of the last good run), and says "not enough measurements" instead of guessing where speed-test data is thin.

---

3. Context, scope and constraints

3.1 Background facts

Fact	Source
CelesTrak's Starlink list carries a [DTC] tag in satellite names; John confirmed it	Decisions "Satellite selection"; build plan 1A first-run checks (confirmed)
Direct-to-Cell satellites fly at about 330 to 370 km with inclination 43 or 53 degrees; used as a cross-check	Build plan 1A.1
Direct-to-Cell texting also needs T-Mobile service (in the United States) and a supported phone; T-Mobile is the Phase A carrier	Build plan, Project 1 description; ADR 0020
FCC coverage files are downloaded per provider, state and technology as a Shapefile or GeoPackage; most datasets include H3 resolution 9 hexagons alongside the raw modeled coverage	Documentation Plan "Answered by research"
NPS Public Trails ArcGIS service: polylines, GeoJSON output supported, 2,000 records per request, trail name in `TRLNAME`, park in `UNITCODE`	Documentation Plan "Answered by research"
Copernicus GLO-30: 30 m, heights on the EGM2008 datum, a surface model that includes vegetation and buildings, water flattened	Documentation Plan "Answered by research"
Ookla open data: quarterly, about 600-meter tiles; license CC BY-NC-SA 4.0	Build plan 1B.1; "Measured coverage (Project 1)"

Fact	Source
H3 resolution 9 is the FCC coverage cell size, about 0.1 km2; a report cell is about 350 m across	Build plan 1B.8; Decisions "Report location"
H3 resolutions to use: 9 for FCC files and hiker reports, 8 for 15 arc-second rasters and Ookla tiles, 4 for 0.25 degree weather grids	Engineering standard 9

3.2 System boundary

Inside the boundary: the pipelines that ingest CelesTrak, NPS, Copernicus, FCC (Phase A) and Ookla (Phase B) data; the terrain and satellite window computations; the dbt models; the publish gate; the serving Lambda; the trail page; and in Phase B the report endpoint, the assistant, the model and the warehouse marts.

Outside the boundary, owned by the platform (the shared system that all five projects run on) and linked rather than repeated: the session box, k3s (a small Kubernetes), Airflow, Spark, Terraform, CI, Parameter Store, the metrics table and status page, S3 buckets and Glue catalog, the serving template, and the nightly schedule. See ../Platform/platform-requirements.md.

3.3 Budget and hard limits

Item	Limit	Source
Whole platform	Under $50 a month, working from a browser only; a normal month costs about $23 to $46 on spot pricing (cheaper interruptible servers); skeleton and Phase A months usually cost under $20	Decisions "Budget"; operating model
Session box	4-hour session limit; idle shutdown; nightly run hard maximum 2 hours	Decisions "Cost guards", "Nightly shutdown"
Backup guard	$40 AWS budget with an action	Decisions "Cost guards"
Spark jobs (Spark is the batch data-processing engine)	Each job has a maximum number of rows read and written and a maximum runtime. A job over either limit stops before publishing and opens an incident. The values are open (OQ-15)	Build plan "Hard limits"
Satellite windows	Stored as start and end times, never one row per minute; a size estimate goes in this design document before build step 1A.3 and the row limit is set from it (OQ-1)	Build plan "Hard limits"
Snowflake	Resource monitor at 3 credits a month (about $6); a time limit on every query	Decisions "Snowflake"
AI model	$10 a month spend limit set in the API account before the first call; a daily question cap; a token limit per answer; cheapest model only (20 questions a day and token limits, ADR 0038; OQ-17)	Decisions "AI spending"; build plan
DynamoDB	Provisioned within the free tier; heavy traffic is throttled, not billed	Build plan "Hard limits"
S3	Clean-up rules (step 1B.9) plus an alarm if storage passes a set size (size open, OQ-15)	Build plan "Hard limits"

3.4 Licenses and attribution

Source	Terms	Page and README obligation
Ookla open data	CC BY-NC-SA 4.0; non-commercial; derived data shared under the same license	Exact notice on the page and in the README (ADR 0013)
Copernicus elevation	Free under the Copernicus licence, with credit	Credit on the page
NPS trail files	Public, from data.gov	Credit on the page
TIGER/Line roads (U.S. Census Bureau)	Declared CC0 (public domain) on data.gov; some files carry a rights flag, "otherRestrictions". Checked on the Census page before the first load (ADR 0035)	Derived distances only; credit on the page

Source	Terms	Page and README obligation
CelesTrak, FCC	Credit wording and licence status in ADR 0037 (OQ-22)	Credit on the page ("with credits", build plan 1A.6)
Databricks Free Edition	Non-commercial	Licensing checked before charging for any project
OpenCelliD	Share-alike; used for tower locations (Decided by John, Oct 9, 2026, ADR 0024); derived data carries the same license	Recorded in the license register and on the page; credit wording in ADR 0037 (OQ-22; OpenCelliD terms to verify)

Licensing must be checked before charging for any of the projects (build plan "Assumptions").

3.5 Privacy rules

- Exact coordinates never leave the phone; the phone converts GPS to an H3 resolution 9 cell and discards the coordinates (ADR 0009).
- Visitor identity is a random ID created on the phone; no internet addresses are used (ADR 0011).
- Limit counters are stored apart from reports and expire; reports carry no ID that links them (ADR 0011). Limits: 10 reports a day per visitor, one per cell per visitor every 24 hours, 50 a day for the whole system, applied by the hour the report was made on the phone.
- Internet addresses in CloudFront and API Gateway logs are removed before anything is stored beyond a short log retention period (ADR 0023).
- Point-in-time recovery is off for the hiker reports table, so deletion is final at 30 days (ADR 0022).
- Raw report rows delete themselves 30 days after the nightly copy into the lake and Snowflake (ADR 0012).
- Email addresses are not collected by this project.
- Timestamps are stored in UTC (engineering standard 15).

---

4. Goals and non-goals

4.1 Goals

1. Phase A: a public trail page for Bright Angel, South Kaibab and North Kaibab showing, for one carrier (T-Mobile, ADR 0020), covered or not along the trail, and the next 24 hours of satellite windows (26 hours predicted, ADR 0019) with terrain accounted for (ADR 0001, 0003, 0004).
2. Never present late data as current: show old data with a visible late notice and the time of the last good run (ADR 0006, amended).
3. Show how much to trust the coverage map, and say "not enough measurements" where there are too few speed tests (ADR 0008).
4. Phase B: every carrier, measured speeds, a warehouse, a trip-planning assistant with cited plans, evals in CI, and hiker reports that stay private (ADR 0009 to 0012, 0017).
5. Keep every paid item under a hard limit (section 3.3).

4.2 Non-goals

Non-goal	Basis
Other parks, other carriers and speed tests in Phase A	ADR 0001; those arrive in Phase B and C
Guessing coverage in dead zones	Speed tests miss dead zones; the model must not guess there (ADR 0008)
Receiving exact coordinates from any visitor	ADR 0009
Identifying visitors by internet address, account or email	ADR 0011
Live model calls that depend on Databricks being up	The site never depends on Databricks (ADR 0014)

Non-goal	Basis
Staged production failures	Decisions "Failure testing"
Always-on servers	Everything on AWS at about $0 idle; public pages are static files plus Lambda
Commercial use	Non-commercial licenses (section 3.4)
Phase C features in the first two phases	Phase C starts only after every project has reached Phase B. Phase C done-when checks are a recorded gap, written when Phase C is designed (Decided by John, Oct 9, 2026)

---

5. Requirements

The full set is in requirements.md. Summary:

Functional (76 requirements, FR-1 to FR-76; numbers are never reused and a few are out of order, see the note in requirements.md section 4)

- Phase A (FR-1 to FR-24, FR-75, FR-76): orbit ingest and Direct-to-Cell selection; trail sampling at 50 m; terrain horizon to about 15 km; SGP4 at 1-minute steps for 26 hours; reachability above 10 degrees and the terrain horizon; satellite windows as start and end times with a valid until stamp; FCC coverage flag per trail point for one carrier; dbt on DuckDB; contracts; Great Expectations publish gate; gold Parquet; React trail page with the late notice and credits.
- Phase B (FR-25 to FR-62, FR-66 to FR-74): every carrier; Ookla tiles with license notice; orbit update stream; Snowflake marts with SCD Type 2; MetricFlow, Superset, Marquez; confidence model with a repo copy; AI foundations; trip-planning assistant; evals in CI; hiker reports with offline queue and limits; operations hardening.
- Phase C (FR-63 to FR-65): national scope and route upload, offline trip pack, model comparison.

Quality (QR-1 to QR-22; "quality" means speed, freshness, cost limits, privacy and availability)

Freshness (daily 10:00 UTC run, hard maximum 2 hours); late data never shown as current; publish gate; availability with the box off; cold-start notice and measurement; cost under $50 with hard limits; privacy of reports and retention; licensing credit; security; rerun safety; observability; honest confidence; honest AI answers; contract integrity; UTC handling. Each has a verification method in requirements.md section 5.

Acceptance criteria

Every feature's build plan "Done when" check is reproduced as AC-1A.1 to AC-1A.6 and AC-1B.1 to AC-1B.9 in requirements.md section 6.

---

6. Architecture

6.1 C4 level 1: system context

```
flowchart LR
  hiker(["Hiker or camper: reads page, sends reports"]):::person
  owner(["Owner (John): runs and reviews"]):::person
  subgraph SYS["Signal planner (Project 1)"]
    sp["Trail page, nightly pipelines, report endpoint, assistant"]:::cont
  end
  subgraph SRC["Public data sources"]
    celestrak["CelesTrak: Starlink GP JSON"]:::ext
    fcc["FCC: mobile coverage files"]:::ext
    nps["NPS Public Trails: ArcGIS polylines"]:::ext
    cop["Copernicus GLO-30: 30 m elevation"]:::ext
    ookla["Ookla open data: speed tiles (B)"]:::phb
  end
  subgraph SVC["Services"]
    models["Model API: cheapest model (B)"]:::phb
    dbx["Databricks Free Edition: training, MLflow (B)"]:::phb
    plat["Platform: box, Airflow, Spark, Glue, CI"]:::ext
  end
  hiker -->|"reads page, sends reports"| sp
  owner -->|"operates, reviews incidents"| sp
  sp -->|"pulls GP JSON"| celestrak
```

```
  sp -->|"downloads coverage"| fcc
  sp -->|"pulls trail lines"| nps
  sp -->|"reads elevation tiles"| cop
  sp -->|"loads tiles"| ookla
  sp -->|"calls for assistant"| models
  sp -->|"uploads data, trains"| dbx
  sp -->|"runs on"| plat
  classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0b1f4d
  classDef ext fill:#f3f4f6,stroke:#6b7280,color:#111827
  classDef cont fill:#dcfce7,stroke:#15803d,color:#052e16
  classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
  classDef phb fill:#ede9fe,stroke:#6d28d9,color:#2e1065
```

6.2 C4 level 2: containers

Phase A containers are unmarked; Phase B containers are marked (B). The view is split in two: what a visitor touches, and what runs each night.

Visitor-facing containers

```
flowchart TB
  hiker(["Hiker: browser or phone"]):::person
  page["Trail page: React on CloudFront and S3"]:::cont
  api["Serving API: FastAPI on Lambda, DuckDB"]:::cont
  rep["Report API (B): Lambda"]:::phb
  asst["Assistant API (B): Lambda"]:::phb
  models["Model API: cheapest model"]:::ext
  ddb[("Visitor tables (B): DynamoDB")]:::store
  lake[("Lake and gold Parquet: S3, Iceberg, Glue")]:::store
  hiker -->|uses| page
  page -->|GET data| api
  page -->|"POST reports (B)"| rep
  page -->|"POST questions (B)"| asst
  rep -->|writes| ddb
  asst -->|calls| models
  asst -->|reads metrics| lake
  api -->|reads gold Parquet| lake
  classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0b1f4d
  classDef ext fill:#f3f4f6,stroke:#6b7280,color:#111827
  classDef cont fill:#dcfce7,stroke:#15803d,color:#052e16
  classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
  classDef phb fill:#ede9fe,stroke:#6d28d9,color:#2e1065
```

Nightly containers

```
flowchart LR
  src["Public sources: CelesTrak, FCC, NPS, Copernicus, Ookla"]:::ext
  pipe["Nightly pipelines: Airflow and Spark on the session box"]:::cont
  dbt["dbt project: DuckDB, Snowflake in B"]:::cont
  gate["Publish gate: Great Expectations"]:::cont
  lake[("Lake and raw files: S3, Iceberg, Glue")]:::store
  ddb[("Visitor tables (B): DynamoDB")]:::store
  stream["Orbit update stream (B): Kafka"]:::phb
  train["Training (B): Databricks, MLflow"]:::phb
  wh[("Warehouse (B): Snowflake over Glue")]:::phb
  src -->|pulled by| pipe
  pipe -->|writes| lake
  pipe -->|runs| dbt
  pipe -->|calls before publish| gate
  ddb -->|nightly copy| pipe
  pipe -->|refresh| wh
  pipe -->|element sets| stream
  pipe -->|uploads data| train
  classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0b1f4d
  classDef ext fill:#f3f4f6,stroke:#6b7280,color:#111827
  classDef cont fill:#dcfce7,stroke:#15803d,color:#052e16
  classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
  classDef phb fill:#ede9fe,stroke:#6d28d9,color:#2e1065
```

6.3 C4 level 3: components of the nightly pipelines container

Component identifiers C1 to C24 are defined in section 7. The diagram is split in two: first the pipeline components, then the serving components and the Phase B additions.

Pipeline components (Phase A)

```
flowchart LR
  subgraph PIPE["Nightly pipelines"]
    c1["C1 Orbit ingest: pull, hash, store"]:::cont
    c8["C8 Contract diff: schema check"]:::cont
    c2["C2 DTC selector: tag and cross-check"]:::cont
    c3["C3 Trail loader: NPS lines, 50 m points"]:::cont
    c4["C4 Terrain horizon: 36 angles per point"]:::cont
    c5["C5 Satellite window engine: satellite windows with valid until"]:::cont
    c6["C6 FCC coverage: covered or not"]:::cont
    c7["C7 dbt project: staging, marts"]:::cont
    c9["C9 Publish gate: write, audit, publish"]:::cont
    c10["C10 Publisher: gold Parquet"]:::cont
  end
  c8 -->|checks schema| c1
  c1 -->|GP records| c2
  c2 -->|selected satellites| c5
  c3 -->|trail points| c4
  c4 -->|horizons| c5
  c3 -->|trail points| c6
  c5 -->|satellite windows| c7
  c6 -->|coverage| c7
  c7 -->|staged tables| c9
  c9 -->|passed| c10
  classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0b1f4d
  classDef ext fill:#f3f4f6,stroke:#6b7280,color:#111827
  classDef cont fill:#dcfce7,stroke:#15803d,color:#052e16
  classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
  classDef phb fill:#ede9fe,stroke:#6d28d9,color:#2e1065
```

Serving components and Phase B additions

```
flowchart TB
  c10["C10 Publisher: gold Parquet"]:::cont
  c1["C1 Orbit ingest"]:::cont
  c5["C5 Satellite window engine"]:::cont
  subgraph SERVE["Serving"]
    c11["C11 Serving API: FastAPI, DuckDB, Lambda"]:::cont
    c12["C12 Trail page: React, strip map, expiry"]:::cont
  end
  subgraph PB["Phase B additions"]
    c14["C14 Orbit stream (B): Kafka, Structured Streaming"]:::phb
    c23["C23 Report copy (B): DynamoDB to lake, set-aside"]:::phb
  end
  ddb[("DynamoDB reports (B)")]:::store
  lake[("Lake tables")]:::store
  c10 -->|Parquet| c11
  c11 -->|JSON| c12
  c1 -->|element sets| c14
  c14 -->|re-propagate| c5
  ddb -->|nightly read| c23
  c23 -->|copies and set-aside rows| lake
  classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0b1f4d
  classDef ext fill:#f3f4f6,stroke:#6b7280,color:#111827
  classDef cont fill:#dcfce7,stroke:#15803d,color:#052e16
  classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
  classDef phb fill:#ede9fe,stroke:#6d28d9,color:#2e1065
```

6.4 Architecture decisions that shape the diagrams

- Everything the public page needs is published as Parquet, so the page works with the box off (ADR 0006, amended: late data stays visible with a late notice; platform serving template).
- The nightly run writes; Lambda reads (and in Phase B writes only to DynamoDB), following engineering standard 16.
- Dev and live are separate copies built from the same code and containers (platform).

6.5 Data flow

The path every dataset takes, from source to the page. Phase B additions are purple.

```
flowchart TB
  subgraph S["Sources"]
    s1["CelesTrak GP JSON"]:::ext
    s2["NPS trail GeoJSON"]:::ext
    s3["Copernicus GLO-30 tiles"]:::ext
    s4["FCC coverage files"]:::ext
    s5["Ookla tile files (B)"]:::phb
    s6["Hiker reports (B)"]:::phb
  end
  subgraph R["Raw files (S3)"]
    r1["Raw files with hash and fetch time"]:::store
  end
  subgraph L["Lake tables (Iceberg)"]
    l1["celestrak_gp, dtc_satellite"]:::store
    l2["nps_trail_line, trail_point"]:::store
    l3["trail_point_horizon"]:::store
    l4["fcc_coverage_cell, trail_point_coverage"]:::store
    l5["satellite_window"]:::store
    l6["ookla_tile (B)"]:::phb
    l7["signal_report (B)"]:::phb
  end
  g{{"dbt build, then publish gate"}}:::cont
  subgraph P["Published tables (gold Parquet)"]
    p1["dim_trail_point, dim_time"]:::cont
    p2["fct_trail_point_coverage"]:::cont
    p3["fct_satellite_window"]:::cont
    p4["fct_coverage_confidence (B)"]:::phb
    p5["fct_report_counts_area (B)"]:::phb
  end
  subgraph V["Serving"]
    api["Serving API (Lambda, DuckDB)"]:::cont
    site["Trail page"]:::cont
  end
  s1 --> r1
  s2 --> r1
  s3 --> r1
  s4 --> r1
  s5 --> r1
  r1 --> l1
  r1 --> l2
  r1 --> l3
  r1 --> l4
  r1 --> l6
  s6 -->|"DynamoDB, nightly copy"| l7
  l1 --> l5
  l3 --> l5
  l2 --> l3
  l2 --> l4
  l5 --> g
  l4 --> g
  l2 --> g
  l6 --> g
  l7 --> g
  g --> p1
  g --> p2
  g --> p3
  g --> p4
  g --> p5
  p1 --> api
  p2 --> api
  p3 --> api
  p4 --> api
  p5 --> api
  api --> site
  classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0b1f4d
  classDef ext fill:#f3f4f6,stroke:#6b7280,color:#111827
  classDef cont fill:#dcfce7,stroke:#15803d,color:#052e16
  classDef store fill:#fef3c7,stroke:#b45309,color:#451a03
  classDef phb fill:#ede9fe,stroke:#6d28d9,color:#2e1065
```

7. Component breakdown

Naming status: component identifiers, function names, class names, table names and column names in this section are proposals of this document, except `compute_horizon(point, elevation_tiles)` and `find_windows(positions, horizon, min_elev)`, which come from the Documentation Plan, and source names such as `TRLNAME` and `UNITCODE` (OQ-21).

Every component runs on the platform listed in section 11. Re-run safety rules (engineering standard 3) apply to every component that writes data: it overwrites a partition (one slice of a table, here one run date), uses the run's data interval (the date the run is responsible for) instead of the current time, and never appends blindly.

7.1 Phase A components

C1 Orbit ingest (F-1A.1)

Item	Detail
Job	Pull CelesTrak GP data (orbital elements) as JSON, keep the raw file unchanged with its hash and load time, and pass the records to the selector (C2)
Inputs	CelesTrak Starlink list (GP, JSON). Exact URL and field list: OQ-20
Outputs	Raw file in the raw bucket with URL, fetch time and hash; Iceberg table `celestrak_gp` (partition per pull); a metrics-table row (freshness, row count)
Main functions and classes (proposed)	`fetch_gp_json(url, last_fetch_time)`: calls CelesTrak at most once per 2-hour update and stops on any response other than HTTP 200; `store_raw(payload, url, fetched_at)`: writes the file and manifest with hash; `load_gp_partition(spark, raw_path, logical_date)`: overwrites the partition; Airflow DAG `orbit_ingest`
Contract	contracts/celestrak_gp.yaml
ADRs	0002, 0018

C2 Direct-to-Cell selector (F-1A.1)

Item	Detail
Job	Select Direct-to-Cell satellites by the [DTC] tag in their names; cross-check altitude (about 330 to 370 km) and inclination (43 or 53 degrees); log any disagreement
Inputs	Records of `celestrak_gp` for the run
Outputs	Selected satellites (Iceberg `dtc_satellite`), disagreement list (logged and listed in the table `dtc_disagreement`)
Main functions and classes (proposed)	`select_dtc_satellites(gp_records)` returns selected and disagreements; `has_dtc_tag(name)`; `altitude_km_from_elements(record)` (derived from mean motion and eccentricity); `check_orbit(record)` returns the list of disagreements for one record; class `OrbitCheckResult`
Contract	None. These are internal tables (neither a source nor a published table); their schema is defined in code and tests
ADRs	0002

C3 Trail loader and sampler (F-1A.2)

Item	Detail
Job	Load NPS trail lines for Grand Canyon and sample a point every 50 meters
Inputs	NPS Public Trails ArcGIS service (polylines; GeoJSON output; 2,000 records per request; `TRLNAME`, `UNITCODE`). Park filter: Grand Canyon by `UNITCODE` (the code value is not recorded, OQ-20)
Outputs	Raw GeoJSON pages with hash; Iceberg `nps_trail_line`; Iceberg `trail_point` (50 m spacing)
Main functions and classes (proposed)	`fetch_trail_pages(service_url, unit_code)`: pages of up to 2,000 records; `parse_trail_geojson(payload)`; `sample_points_along_line(line, spacing_m)`; `build_trail_points(spark, trails_df, spacing_m)`; class `TrailPoint`
Contract	contracts/nps_trails.yaml
ADRs	0004

C4 Terrain horizon engine (F-1A.2)

Item	Detail
Job	For every trail point, compute how high terrain rises in each compass direction out to about 15 km; Spark computes many points at the same time
Inputs	`trail_point`; Copernicus GLO-30 tiles (30 m, EGM2008 datum, surface model) covering the park plus the 15 km margin
Outputs	Iceberg `trail_point_horizon` (36 horizon angles per point)
Main functions and classes	See table 7.A (full list)
Contract	contracts/copernicus_glo30.yaml (source); contracts/trail_point_horizon.yaml (output)
ADRs	0004

C5 Satellite window engine (F-1A.3)

Item	Detail
Job	Propagate selected satellites with SGP4 at 1-minute steps for 26 hours (ADR 0019; the page says the next 24 hours); find when each satellite is above 10 degrees and above the terrain horizon; write satellite windows with a valid until stamp; record how satellite windows change at 5 and 20 degrees
Inputs	`dtc_satellite`, `trail_point`, `trail_point_horizon`, run logical date
Outputs	Iceberg `satellite_window` (one row per point, satellite and satellite window; partition per run date); sensitivity figures for 5, 10 and 20 degrees
Main functions and classes	See table 7.B (full list)
Contract	contracts/fct_satellite_window.yaml (mart); internal Iceberg table follows the same columns
ADRs	0003, 0005, 0006, 0019

C6 FCC coverage loader (F-1A.4)

Item	Detail
Job	Load FCC mobile coverage (H3 resolution 9 files) for one carrier (T-Mobile, ADR 0020) in Arizona; mark each trail point covered or not
Inputs	FCC coverage file for one provider, Arizona, per technology (Shapefile or GeoPackage; includes H3 resolution 9 hexagons). The T-Mobile provider name and technology as named in the file: OQ-20 (Phase B carrier list: OQ-11)
Outputs	Raw file with hash; Iceberg `fcc_coverage_cell` (H3 resolution 9 cell, carrier, technology, release); Iceberg `trail_point_coverage` (point, carrier, covered flag)
Main functions and classes (proposed)	`download_fcc_file(provider, state, technology)`; `read_fcc_h3_cells(path)`; `point_to_cell(lat, lon, res=9)` using H3; `flag_points_covered(points_df, coverage_df, carrier)`
Contract	contracts/fcc_mobile_coverage.yaml
ADRs	0007

C7 dbt project (F-1A.5)

Item	Detail
Job	Build staging, intermediate and a small star schema (a central fact table joined to descriptive tables; here trail point, time, coverage and satellite windows) on DuckDB; `source()` only in staging; unique and not_null tests on keys
Inputs	Iceberg tables from C1 to C6 (how dbt on DuckDB reads Iceberg is a platform matter, OQ-40)
Outputs	Marts `dim_trail_point`, `dim_time`, `fct_trail_point_coverage`, `fct_satellite_window`
Models (proposed)	Staging: `stg_celestrak_gp`, `stg_nps_trail`, `stg_trail_point_horizon`, `stg_fcc_coverage`, `stg_satellite_window`. Intermediate: `int_trail_point_coverage`, `int_trail_point_windows`. Marts as above

Item	Detail
Contracts	contracts/*.yaml
ADRs	0018

C8 Contract and schema diff (F-1A.5)

Item	Detail
Job	Diff each source's schema against its contract before loading so an upstream change fails loudly and early
Inputs	Contract YAML (ODCS) and a sample or actual source file
Outputs	Pass, or a failure naming the changed field; written to the run log and the metrics table
Main functions and classes (proposed)	`load_contract(path)`; `diff_schema(contract, observed_schema)` returns added, removed and changed fields; `assert_no_breaking_change(diff)`
ADRs	0018

C9 Publish gate (F-1A.5)

Item	Detail
Job	Write, audit, publish: new data lands in staging, passes Great Expectations checks (freshness, empty results, value ranges), then goes live; publishing stops when a check fails
Inputs	Staged mart tables
Outputs	Pass or fail result; on pass, promotion to the publish location
Main functions and classes (proposed)	`run_publish_gate(staged_tables, suite)`; expectation suites `suite_satellite_window`, `suite_trail_point_coverage`, `suite_trail_point`; `promote_to_published(staged_path, published_path)`
Value-range limits	Two checks need no recorded number: a satellite window never starts after it ends, and elevation angles are between 0 and 90 degrees. Freshness thresholds and all other value ranges are open (OQ-15)
ADRs	0006, 0018

C10 Publisher (F-1A.6)

Item	Detail
Job	Write gold tables as Parquet to the published location read by the serving Lambda
Inputs	Mart tables that passed the gate
Outputs	Parquet files for `dim_trail_point`, `dim_time`, `fct_trail_point_coverage`, `fct_satellite_window`
Main functions and classes (proposed)	`publish_gold_parquet(tables, destination, logical_date)`
ADRs	0006

C11 Serving API (F-1A.6)

Item	Detail
Job	Answer page requests from published Parquet using DuckDB inside a FastAPI Lambda behind API Gateway; works with the box off
Inputs	HTTP requests (section 10); published Parquet
Outputs	JSON responses
Main functions and classes (proposed)	FastAPI app `app`; `list_trails()`; `get_trail_signal(trail_id)`; `query_gold(sql, params)` (read-only DuckDB connection); response models `TrailSummary`, `TrailSignal`, `PointSignal`, `WindowOut`, `ErrorOut`
Contract	openapi.yaml
ADRs	0006

C12 Trail page (F-1A.6)

Item	Detail
Job	Let the visitor pick a trail and show a strip map (a line along the trail with bars where the carrier has coverage and no bars where it does not) and the next satellite windows; show "Predictions valid until [time]" in local time; when the data is late or the run failed, keep showing the old data with a visible late notice that includes the time of the last good run (ADR 0006, amended); show credits, the T-Mobile and supported-phone statement and the cold-start notice
Inputs	Serving API responses
Outputs	The page in the browser
Main functions and classes (proposed)	React components `TrailPicker`, `StripMap`, `WindowList`, `ValidUntilBanner`, `Credits`, `ColdStartNotice`, `DtcRequirementsNote`; `LateNotice` (the visible late notice); functions `isRunLate(validUntilUtc, now)` (definition of late: OQ-23), `formatLocalTime(utcIso)`, `windowsWithLateFlag(windows, now)` (marks each satellite window as late or current and never hides data); a timer that re-evaluates lateness while the page stays open
ADRs	0006

7.2 Phase B components

C13 Ookla and all-carrier loader (F-1B.1)

Item	Detail
Job	Add every carrier's FCC coverage for the park; load Ookla's open mobile speed-test data for Arizona (GPS-accurate tests only); load OpenCelliD tower locations (ADR 0024); put the license notices on the page and in the README
Inputs	FCC files (all providers), Ookla quarterly tiles (about 600 m; a manifest of URL, size and hash instead of a full copy, per standard 1)
Outputs	Iceberg `fcc_coverage_cell` (all carriers), Iceberg `ookla_tile` (H3 resolution 8, per engineering standard 9)
Main functions and classes (proposed)	`load_ookla_arizona(spark, quarter)`; `filter_gps_accurate(df)`; `tile_to_h3_res8(tile)`; `load_all_carriers(spark, park)`; `load_opencellid_towers(spark)`
Contract	contracts/ookla_open_data.yaml, fcc_mobile_coverage.yaml
ADRs	0007, 0013

C14 Orbit update stream (F-1B.2)

Item	Detail
Job	Publish each new element set to Kafka; a Structured Streaming job (availableNow trigger) finds changed satellites and triggers re-propagation. This document states plainly that the stream is low-rate (few messages)
Inputs	New `celestrak_gp` records from C1
Outputs	Kafka topic of element sets; Iceberg table of changed satellites; trigger for C5
Main functions and classes (proposed)	`publish_element_sets(records, topic)`; `build_orbit_stream(spark, schema)` with an explicit schema, S3 checkpoints, a watermark and `dropDuplicatesWithinWatermark`; `detect_changed_satellites(df)`; `trigger_repropagation(satellite_ids)`
Parameters	Watermark value, topic name, definition of "changed": OQ-30
Contract	contracts/orbit_update_stream.yaml
ADRs	0016

C15 Warehouse sync (F-1B.3)

Item	Detail
Job	Connect Snowflake to the Glue catalog; refresh after each pipeline run; run dbt marts on Snowflake with SCD Type 2 carrier coverage rows; MetricFlow metrics; Superset dashboard with nightly snapshot charts; 3-credit resource monitor
Inputs	Iceberg tables via Glue; FCC release history

Item	Detail
Outputs	Snowflake marts; `dim_carrier_coverage_history` (SCD Type 2); metric definitions; snapshot chart files published to the site
Main functions and classes (proposed)	`refresh_snowflake_iceberg_tables(tables)`; dbt snapshots/models `dim_carrier_coverage_history`; MetricFlow files under `metrics/`; `publish_snapshot_charts(dashboard_id, destination)`
Open	Whether the connection works and the refresh shows new data: OQ-4
ADRs	0015

C16 Lineage (F-1B.3)

Item	Detail
Job	OpenLineage events from Airflow and Spark to Marquez (ARM build of one pinned version, made once in GitHub Actions); dbt docs catalog page
Open	Whether the ARM build (for the ARM processors of the session box) succeeds: OQ-5. The platform owns this question; if the build fails, the lineage tool choice is revisited
Platform link	../Platform/platform-requirements.md, section 6 (lineage)

C17 Coverage confidence model (F-1B.4)

Item	Detail
Job	Predict how likely the FCC map is to be right where speed tests exist; give a confidence level only where enough speed tests exist nearby; elsewhere output the "not enough measurements" notice. Train in Databricks, track in MLflow, export the results and model into the GitHub repository (repo), and score in the nightly run with the repo copy
Inputs	FCC coverage claims, Ookla tiles, terrain features, distance to nearest tower (OpenCelliD, ADR 0024), remoteness (distance in metres from each trail point to the nearest TIGER/Line road or NPS trail line; source decided in ADR 0035), report labels (from C23)
Outputs	Published `fct_coverage_confidence`; MLflow runs; repo files with results and the chosen model
Main functions and classes (proposed)	`build_training_frame(spark)`; `upload_to_databricks(frame_path)` (Files API); `train_and_log(frame)` (MLflow); `export_run_to_repo(run_id)` (a manual procedure: download results and model, commit to GitHub; automation is an open test, OQ-19); `score_cells(model_path, frame)`; `has_enough_tests(cell, tests_df, min_tests, radius)` (thresholds OQ-14); `keep_alive_ping()` (monthly job)
Model family and features	scikit-learn (a Python machine-learning library) is the planned library, from the build plan's technology table; the model type is logistic regression (ADR 0031; the remoteness source is still open, OQ-18)
ADRs	0008, 0014

C18 AI foundations (F-1B.5)

Item	Detail
Job	Log every model call to S3 and load it into `llm_calls`; trace every agent run to X-Ray; expose metrics as MCP tools with a read-only allowlisted SQL fallback logged separately; version prompts in git; enforce structured output; an eval harness (the program that runs the answer tests); a daily question cap; the $10 monthly spend limit
Main functions and classes (proposed)	`log_llm_call(model, prompt_version, tokens, latency, cost)`; `load_llm_calls(spark, date)`; MCP server `metrics_mcp` with example tools `get_signal_summary` and `get_windows` (the real tool list and names are open, OQ-21); `run_readonly_sql(sql)` (allowlist, separate log); `check_daily_cap(day)`; `with_trace(name)`
Contract	../Platform/contracts/llm_calls.yaml (platform table; Project 1 adds no columns)
ADRs	0017

C19 Trip-planning assistant (F-1B.6)

Item	Detail
Job	Take a plain-language question and return a check-in plan: where each carrier works, when satellite windows open, how confident the map is, citing the trail points and times used; say so when the data cannot answer
Main functions and classes (proposed)	`answer_trip_question(question)`; `plan_from_tool_results(results)`; `cite(points, times)`; `refuse_unanswerable(reason)`; structured output model `CheckInPlan`
Decided	The assistant answers only within the prediction span and says so for later days (ADR 0021)
ADRs	0017

C20 Evals in CI (F-1B.7)

Item	Detail
Job	30 or more questions with known answers, graded exactly on tool choice, numbers and refusal; an LLM judge (a second model that scores answers) for clarity, calibrated on 50 answers graded by hand; red-team set of 20 or more prompts; votes in DynamoDB; A/B test of two prompt versions
Main functions and classes (proposed)	`run_eval_suite(cases)`; `grade_exact(case, result)`; `judge_clarity(answer)`; `calibrate_judge(hand_grades)`; `run_red_team(prompts)`; `record_vote(vote)`; `assign_prompt_variant(visitor_id)`
Thresholds	Not recorded: OQ-16
ADRs	0017

C21 Report client on the phone (F-1B.8)

Item	Detail
Job	Two buttons; read GPS; convert to an H3 resolution 9 cell with h3-js (the H3 JavaScript library); discard coordinates; build the report; queue it on the phone; send when signal returns; show "1 report waiting to send" on iPhone
Main functions and classes (proposed)	React component `ReportButtons`; `captureCell()` (GPS to cell, then drops coordinates); `buildReport(cell, carrier, signal, now)` (time rounded to the hour, UTC); `enqueueReport(report)` (persistent local store); `flushQueue()`; background send on Android; `PendingReportsNote`; `getOrCreateVisitorId()` (random ID, stored on the phone only)
Answered	Visitor ID format and transport: ADR 0025 (OQ-38). Report time zone: UTC per standard 15
ADRs	0009, 0010, 0011

C22 Report API and limiter (F-1B.8)

Item	Detail
Job	Validate a report; apply the limits by the hour the report was made on the phone (not arrival time): 10 reports a day per visitor, one per cell per visitor every 24 hours, and at most 50 a day for the whole system (ADR 0011, amended); write the report to DynamoDB without any ID; store counters apart from reports with expiry (lifetime: OQ-26); a system-wide daily counter holds no visitor ID
Main functions and classes (proposed)	Lambda handler `post_report`; `validate_report(body)`; `check_and_increment_limits(visitor_id, cell, now)`; `write_report(report)`; DynamoDB tables `signal_reports`, `report_limit_counters`; models `ReportIn`, `ReportAccepted`, `LimitExceeded`
Answered	Day boundary, accepted report hours and counter lifetime: ADR 0026 (OQ-26). Log retention: 7 days, ADR 0029 (OQ-29). Counters recovery: off, ADR 0025 (OQ-41)
Contract	contracts/hiker_signal_reports.yaml; openapi.yaml
ADRs	0011, 0012

C23 Nightly report copy and set-aside (F-1B.8)

Item	Detail
Job	Copy the day's raw reports into the lake and Snowflake; set each copied row to expire 30 days after the copy; set aside impossible patterns for review; produce label candidates (reports that may train the model) and per-area counts
Main functions and classes (proposed)	`copy_reports_to_lake(spark, logical_date)`; `set_expiry_after_copy(report_keys)` (sets the DynamoDB expiry attribute to 30 days after the copy); `find_impossible_reports(reports_df, mapped_area_df)`; `find_contradicting_bursts(reports_df)`; `build_label_candidates(reports_df, n_per_cell_day)` (a cell qualifies at N reports in one day; N: OQ-12); `build_report_counts_area(reports_df)`
Outputs	Iceberg `signal_report`, `signal_report_set_aside`; published `fct_report_counts_area`
Open	OQ-12, OQ-13. Area and minimum count answered in ADR 0030 (OQ-35)
ADRs	0012, 0008

C24 Operations hardening (F-1B.9)

Item	Detail
Job	S3 lifecycle rules (automatic deletion by age); storage-size alarm; cold-start measurement; weekly smoke test of every source; one de-duplicated GitHub issue per failure, closed on recovery
Main functions and classes (proposed)	Terraform lifecycle rules (platform module); `measure_cold_starts(endpoint, n)`; `smoke_pull(source)` per source; `open_or_update_issue(incident_key)`; `close_issue_on_recovery(incident_key)`
ADRs	0018

7.A Terrain horizon (1A.2): every function and class

Convention proposed here: 36 horizon angles, one per 10-degree compass sector (36 sectors is the figure in the Documentation Plan's example; the 10-degree width follows from 360 divided by 36). Sector 0 is centered on north and sectors run clockwise (proposed convention; OQ-24).

Function or class	Purpose	Takes	Returns
`TrailPoint` (class)	One sampled point on a trail	`point_id`, `trail_id`, `seq`, `lon`, `lat`, `elevation_m`	Immutable record
`ElevationTiles` (class)	The loaded set of GLO-30 tiles for the park plus margin, with fast lookup	Tile keys or a bounding box	Object with `elevation_at(lon, lat)` and `profile(lon, lat, azimuth_deg, max_range_m, step_m)`
`HorizonProfile` (class)	The 36 angles for one point	`point_id`, `angles_deg` (36 values)	Immutable record
`sample_points_along_line(line, spacing_m)`	Place a point every `spacing_m` (50) along a trail polyline	A line geometry, spacing in meters	Ordered list of `(lon, lat)`
`build_trail_points(spark, trails_df, spacing_m)`	Spark wrapper that numbers points per trail and adds elevation	Spark DataFrame of trail lines, spacing	DataFrame of `TrailPoint` rows
`load_elevation_tiles(bbox)`	Find and open the GLO-30 tiles covering the bounding box plus the horizon range	Bounding box in lon/lat	`ElevationTiles`
`compute_horizon(point, elevation_tiles)`	How high terrain rises in each compass direction around a trail point	A trail point, elevation tiles	36 horizon angles
`ray_elevation_profile(point, azimuth_deg, elevation_tiles, max_range_m)`	Terrain heights along one compass direction out to about 15 km	Point, azimuth, tiles, range	Arrays of distance and height

Function or class	Purpose	Takes	Returns
`max_elevation_angle(observer_height_m, distances_m, heights_m)`	The highest angle above the horizontal along one profile (what blocks the sky in that direction)	Observer height, profile arrays	One angle in degrees
`compute_horizons(spark, trail_points_df, tiles_index)`	Run `compute_horizon` for every point in parallel in Spark	Points DataFrame, tile index	DataFrame of `HorizonProfile` rows
`horizon_at(horizon, azimuth_deg)`	Horizon angle for any azimuth, used by the satellite window engine	A `HorizonProfile`, an azimuth	One angle in degrees (selection of sector or interpolation: OQ-24)
`validate_horizon(horizon)`	Value-range check used by tests and the publish gate	A `HorizonProfile`	Pass or a list of violations

Decided in ADR 0036 (OQ-24): range 15 km; ray step 30 m; observer height 1.7 m; Earth curvature applied; atmospheric refraction not applied (conservative); surface model kept; 36 sectors, final; `horizon_at` interpolates linearly; the AC-1A.2 margin is 5 degrees between the medians of the canyon and rim fixtures.

7.B Satellite windows (1A.3): every function and class

Function or class	Purpose	Takes	Returns
`SatelliteElements` (class)	One selected satellite's orbital elements from a GP record	GP record fields	Immutable record with NORAD id, name, epoch, elements
`SatellitePosition` (class)	A satellite's position at one minute	Satellite id, time (UTC), position in an earth-fixed frame	Immutable record
`Window` (class)	One reachable interval for one point and satellite	`point_id`, `satellite_id`, `start_utc`, `end_utc`	Immutable record
`WindowRun` (class)	One run's metadata	Run id, logical date, span start, span length, `min_elev_deg`, `valid_until_utc`	Immutable record
`propagate_satellites(elements, start, minutes, step_minutes)`	Position of each selected satellite at 1-minute steps for 26 hours using SGP4	Elements list, start time, 1,560 minutes (26 hours), 1-minute step	`SatellitePosition` list per satellite
`look_angles(position, point)`	Azimuth and elevation of a satellite seen from a trail point	A `SatellitePosition`, a `TrailPoint`	`(azimuth_deg, elevation_deg)`
`find_windows(positions, horizon, min_elev)`	When a satellite clears both the minimum elevation and the terrain	Satellite positions, a horizon profile, the minimum elevation	Windows as start and end times
`reachable_flags(positions, horizon, min_elev)`	One true or false value per minute: above `min_elev` and above `horizon_at(horizon, azimuth)`	Same as `find_windows`	Boolean list aligned to the minutes
`flags_to_windows(flags, times)`	Collapse consecutive true minutes into start and end times, so no row per minute is ever stored	Flags, times	List of `(start_utc, end_utc)`
`compute_valid_until(span_start, span_minutes)`	The time the run's predictions stop being valid	Span start, span length	A UTC time: the end of the prediction span (ADR 0019); the late test built on it is OQ-23
`run_window_job(spark, run, satellites_df, points_df, horizons_df)`	Spark job: propagate once per satellite, evaluate against every trail point in parallel, build satellite windows	Run metadata and the three DataFrames	DataFrame of `Window` rows plus the `WindowRun`
`enforce_row_limit(df, max_rows)`	Stop before publishing if the job would write more rows than its limit	DataFrame, limit	The same DataFrame, or raises and opens an incident (limit value: OQ-1, OQ-15)

Function or class	Purpose	Takes	Returns
`write_window_partition(df, logical_date)`	Overwrite only the run date's partition so a rerun replaces only that day	DataFrame, logical date	Iceberg commit
`sensitivity_report(positions, horizons, elevations)`	Count and total length of satellite windows at 5, 10 and 20 degrees for the design note	Positions, horizons, `[5, 10, 20]`	A summary table

Edge convention proposed for `flags_to_windows`: a satellite window starts at the first minute whose flag is true and ends at the last minute whose flag is true. Because the steps are 1 minute apart, satellite window edges are accurate to 1 minute. This convention is a proposal that John approves with the document (OQ-21).

Partition rule: the partition key is the run's logical date in UTC (engineering standard 3: use the run's data interval, not the current time).

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8. Runtime view

8.1 Nightly run, Phase A

The platform starts the session box at 10:00 UTC (3 a.m. Arizona, 6 a.m. US Eastern) on spot only, runs every pipeline, publishes, and shuts the box down when the pipelines finish, with a hard maximum of 2 hours (about 45 minutes expected). The Project 1 tasks are shown below (C1 to C10 are the components of section 7; "box" means the session box):

```
flowchart TD
  A[Box starts 10:00 UTC] --> B[Restore Airflow database from S3]
  B --> C[C1 pull CelesTrak GP JSON, hash, store raw]
  C --> C8a[C8 schema diff against contract]
  C8a --> D[C2 select DTC satellites and cross-check]
  B --> E[C3 NPS trails and 50 m points]
  E --> F[C4 terrain horizon per point]
  B --> G[C6 FCC coverage flag per point]
  D --> H[C5 propagate 26 hours and find satellite windows]
  F --> H
  H --> I[C7 dbt build on DuckDB, staged]
  G --> I
  I --> J[C9 publish gate]
  J -->|pass| K[C10 publish gold Parquet]
  J -->|fail| X[Stop, keep previous publish, incident, email]
  K --> L[Write freshness and row counts, status.json]
  L --> M[Iceberg maintenance: last task]
  M --> N[Save Airflow database to S3, shut down]
```

Step by step, with failure handling:

1. Restore Airflow's database (its record of past runs) from S3. If the latest copy is lost, the pipelines can safely run again (Decisions "Airflow's database").
2. C1 calls CelesTrak at most once per 2-hour update. On any non-200 response the task stops and the DAG fails; the raw store is untouched. The previous publish stays live and the page shows it with the late notice once it is late (QR-3, ADR 0006).
3. C8 diffs the new file's schema against the contract. A mismatch stops the run before anything is loaded.
4. C2 selects satellites and logs disagreements with the altitude and inclination cross-check. Disagreements do not fail the run; they are listed (AC-1A.1).
5. C3, C4 and C6 run on the trail points. The cadence of these reference-data steps is decided in ADR 0028 (OQ-25). Either way, each of them overwrites its own partitions, so running again is safe.
6. C5 propagates once per satellite and evaluates all trail points in parallel. If the row count or runtime exceeds its limit the job stops before publishing and opens an incident (the limit values come from OQ-1).
7. C7 runs dbt on DuckDB into staging. A failed dbt test fails the run.
8. C9 runs the Great Expectations checks. On failure, publishing stops; the previously published Parquet stays (QR-4).
9. C10 publishes gold Parquet; each run is stamped with its valid until time.

10. The metrics table gets freshness and row counts; `status.json` is regenerated; a missed freshness target or failed run sends an email.
11. Iceberg maintenance (compact, expire snapshots, remove orphan files) is the last task, so it never competes with the pipelines.
12. Airflow's database is saved to S3 and the box shuts down. If the 2-hour maximum is hit, it is recorded as an incident and the next run catches up. If spot is unavailable, the night is skipped and the next run catches up.

No daily gap: the run predicts 26 hours (24 plus the 2-hour nightly maximum), so the previous predictions are still valid when the next run publishes (ADR 0019). The 24 hours is the promise to the visitor; the extra 2 hours are a safety margin.

8.2 A visitor opens the trail page (Phase A)

```
sequenceDiagram
  participant H as Hiker browser
  participant CF as CloudFront and S3
  participant API as Serving API (Lambda, DuckDB)
  participant P as Published Parquet (S3)
  H->>CF: GET page (static React build)
  CF-->>H: HTML, JS
  H->>API: GET /trails
  API->>P: read dim_trail_point
  API-->>H: trail list
  H->>API: GET /trails/{trail_id}/signal
  API->>P: read coverage and satellite windows
  API-->>H: points, coverage flags, satellite windows, valid_until
  Note over H: Convert UTC to local time. Compare valid_until with now and show the late notice if late, and never hide
data. Show "Predictions valid until [time]"
```

Failure handling:

- Cold start: the first request after inactivity may take up to 10 seconds; the page says so (FR-20).
- Unknown trail: 404 error body (openapi.yaml).
- Published data missing or unreadable: 503 error body; the page shows an error and no satellite windows.
- When the run is late (the valid until time has passed, or the publish failed), the page keeps showing the old data with a visible late notice and the time of the last good run. It does not hide data (ADR 0006, amended). How satellite windows already in the past are drawn, and the exact lateness rule, are decided in ADR 0027 (OQ-23).

8.3 A hiker makes an offline report (Phase B)

```
sequenceDiagram
  participant H as Hiker phone
  participant Q as Phone local queue
  participant R as Report API (Lambda)
  participant D as DynamoDB
  H->>H: Tap button, read GPS
  H->>H: h3-js converts GPS to res 9 cell, GPS discarded
  H->>Q: Save report (cell, carrier, signal, hour) and keep visitor ID
  Note over H,Q: No signal. Report waits.
  H->>R: Send when signal returns (Android background, iPhone on next page open)
  R->>R: Validate, check limits by visitor ID
  alt over a limit
    R-->>H: 429 for a limit counted by the report's own hour
  else accepted
    R->>D: Put report (no visitor ID)
    R->>D: Update counters (own table, with expiry)
    R-->>H: 202 accepted
  end
```

Failure handling: a send that fails for network reasons stays in the queue; a 400 (invalid report) is dropped from the queue; limits are counted by the hour the report was made, so a batch sent after being offline is not refused for arriving together (ADR 0011). What happens to a report refused with 429 is decided in ADR 0026 (OQ-26); the day boundary is UTC and reports older than 48 hours are refused. On iPhone, the page shows "1 report waiting to send".

8.4 Nightly report copy (Phase B)

1. Read reports not yet copied from DynamoDB.
2. Write them to the lake (Iceberg) and let Snowflake see them through the refresh step.
3. Set each copied row's expiry to 30 days after the copy.
4. Set aside impossible patterns (reports outside any mapped area, bursts that contradict nearby reports) in `signal_report_set_aside` for review.
5. Produce label candidates only where at least N reports in the same cell on the same day agree (each is a different visitor because of the one-per-cell rule; N: OQ-12) and per-area counts for the page.
6. If the copy fails, no expiry is set, so no row is deleted before it is copied.

8.5 Assistant question (Phase B)

1. The page sends the question to the assistant endpoint.
2. The endpoint checks the daily question cap; over the cap it returns a refusal (429).
3. The model (cheapest only) chooses metric tools from the MCP server; raw SQL is a read-only allowlisted fallback logged separately.
4. The answer is produced with structured output, citing trail points and times.
5. Every call is logged to S3 (model, prompt version, tokens, latency, cost) and traced to X-Ray.
6. If the tools return nothing that answers the question, the answer says so.
7. Source text is treated as data, never as instructions.

8.6 Training and scoring (Phase B)

1. Upload training data to Databricks through the Files API (OQ-3).
2. Train and compare versions in MLflow.
3. The results and the chosen model are exported manually (downloaded and committed to GitHub); whether this can be automated is an open test (OQ-19). A monthly job keeps the free account active.
4. The nightly run scores with the repo copy; the site never depends on Databricks being up.

---

9. Data model

Layers (a layer is a stage in the data flow): raw files (S3, never changed, kept with hash and fetch time), Iceberg tables in the lake (Glue catalog), dbt staging and marts, published Parquet, and in Phase B DynamoDB and Snowflake. Dev and live have separate buckets, Glue databases and roles.

Every row carries event time and a source key (engineering standard 2). All timestamps are UTC (standard 15).

Table or object	Layer	Phase	What one row is (grain) and key (proposed)	Contract
CelesTrak raw JSON files	Raw	A	One file per pull; URL, fetch time, hash	contracts/celestrak_gp.yaml
`celestrak_gp`	Iceberg	A	One row per satellite element set per pull; partition per pull	same
`dtc_satellite`	Iceberg	A	One row per selected satellite per run	internal, no standalone contract
`dtc_disagreement`	Iceberg	A	One row per disagreement per run	internal
NPS raw GeoJSON pages	Raw	A	Per request page	contracts/nps_trails.yaml
`nps_trail_line`	Iceberg	A	One row per trail line (`TRLNAME`, `UNITCODE`)	same
`trail_point`	Iceberg	A	One row per sampled point (trail id plus sequence)	internal; published form is `dim_trail_point`

Table or object	Layer	Phase	What one row is (grain) and key (proposed)	Contract
Copernicus GLO-30 tiles	Raw (manifest only: URL, size, hash, per standard 1 for public archives that never change)	A	Per tile	contracts/copernicus_glo30.yaml
`trail_point_horizon`	Iceberg	A	One row per trail point, with 36 angles	contracts/trail_point_horizon.yaml
FCC raw files	Raw	A	One file per provider, state, technology and release	contracts/fcc_mobile_coverage.yaml
`fcc_coverage_cell`	Iceberg	A (one carrier), B (all)	One row per H3 res 9 cell, carrier, technology, release	same
`trail_point_coverage`	Iceberg	A	One row per trail point and carrier	internal; published form is `fct_trail_point_coverage`
`satellite_window`	Iceberg	A	One row per trail point, satellite and satellite window; partition per run date	published form is `fct_satellite_window`
`dim_trail_point`	Mart, published Parquet	A	One row per trail point	contracts/dim_trail_point.yaml
`dim_time`	Mart, published Parquet	A	One row per UTC minute (OQ-33, ADR 0028)	contracts/dim_time.yaml
`fct_trail_point_coverage`	Mart, published Parquet	A	Trail point by carrier	contracts/fct_trail_point_coverage.yaml
`fct_satellite_window`	Mart, published Parquet	A	Trail point by satellite by satellite window; run date and valid until	contracts/fct_satellite_window.yaml
Ookla raw tile files	Raw (manifest)	B	Per quarterly file	contracts/ookla_open_data.yaml
`ookla_tile`	Iceberg	B	One row per tile and quarter, with H3 res 8	same
`dim_carrier_coverage_history`	Snowflake mart (SCD Type 2)	B	One row per cell, carrier and validity range	contracts/dim_carrier_coverage_history.yaml
`orbit_update_stream`	Kafka topic	B	One message per element set	contracts/orbit_update_stream.yaml
`llm_calls`	Iceberg	B	One row per model call	../Platform/contracts/llm_calls.yaml
`signal_reports`	DynamoDB	B	One item per report; no visitor ID	contracts/hiker_signal_reports.yaml
`report_limit_counters`	DynamoDB	B	Counter per visitor ID and day (and per cell), plus one system-wide daily counter with no ID; expires (lifetime: OQ-26)	same file
`opencellid_tower`	Iceberg	B	One row per tower (share-alike license)	contracts/opencellid_towers.yaml
`tiger_road`	Iceberg	B	One row per road segment and vintage	contracts/tiger_roads.yaml
`trail_point_remoteness`	Iceberg	B	One row per trail point: distance in metres to the nearest road segment and to the nearest NPS trail line	internal (ADR 0035)

Table or object	Layer	Phase	What one row is (grain) and key (proposed)	Contract
`feedback_votes`	DynamoDB	B	One item per thumbs-up or thumbs-down	contracts/feedback_votes.yaml
`signal_report`	Iceberg	B	Lake copy of `signal_reports`	contracts/hiker_signal_reports.yaml
`signal_report_set_aside`	Iceberg	B	Reports set aside for review. Deleted 30 days after the copy (ADR 0034)	contracts/signal_report_set_aside.yaml
`report_cell_day`	Iceberg	B	One row per H3 resolution 9 cell and UTC day: report count and count showing signal. Kept after the raw rows are deleted (ADR 0034)	Contract to be written before Phase B
`fct_coverage_confidence`	Mart, published Parquet	B	Cell (or trail point) by carrier with confidence or "not enough measurements"	contracts/fct_coverage_confidence.yaml
`fct_report_counts_area`	Mart, published Parquet	B	Area by carrier by signal	contracts/fct_report_counts_area.yaml
Model results and chosen model	Repo files	B	Per training run	not a table; governed by ADR 0014

Published tables and their keys (the Phase A star schema, plus the Phase B confidence and report-count tables; `fct_report_counts_area` has its own area key and no join to the trail dimension, OQ-35):

```
erDiagram
  dim_trail_point ||--o{ fct_trail_point_coverage : "trail_point_id"
  dim_trail_point ||--o{ fct_satellite_window : "trail_point_id"
  dim_time ||--o{ fct_satellite_window : "time key"
  dim_trail_point ||--o{ fct_coverage_confidence : "trail_point_id"
  dim_trail_point {
    string trail_point_id
    string trail_id
    string trail_name
    string unit_code
  }
  fct_trail_point_coverage {
    string trail_point_id
    string carrier
    boolean covered
  }
  fct_satellite_window {
    string trail_point_id
    string satellite_id
    timestamp window_start_utc
    timestamp window_end_utc
    timestamp valid_until_utc
  }
  dim_time {
    string time_key
    timestamp time_utc
  }
  fct_coverage_confidence {
    string trail_point_id
    string carrier
  }
  fct_report_counts_area {
    string area_id
    string carrier
  }
```

Retention (from Decisions "Storage clean-up"): raw and published data kept; dev data 14 days; temporary files and Spark checkpoints 7 days; failed uploads 7 days; pipeline logs 90 days; security logs 1 year; old versions 30 days. Live raw files are locked against deletion or overwrite for 30 days; dev files are not locked. The 30-day deletion rule for

reports applies to DynamoDB, the lake and Snowflake copies alike, and the lake and Snowflake keep daily per-cell counts instead of raw rows (OQ-43, ADR 0034).

9.1 Physical locations (proposed names)

Every dataset in its contract has a location. The names below are proposals that John approves with this document (OQ-21). `{env}` stands for `dev` or `live`. Dev and live never share a bucket, database, table or role.

Store	Name pattern	What it holds
Raw bucket	`s3://{env}-signal-raw/<source>/`	Raw source files with URL, fetch time and hash; manifests for public archives
Lake bucket	`s3://{env}-signal-lake/`	Iceberg tables, read through the Glue catalog database `{env}_signal`
Published Parquet	`s3://{env}-signal-lake/published/<table>/`	Gold tables the serving Lambda reads; replaced in one step by the publisher (C10)
LLM call log	`s3://{env}-signal-lake/llm-calls/`	One JSON record per model call, loaded nightly into `llm_calls` (Phase B)
Visitor tables	DynamoDB `{env}-signal-reports`, `{env}-signal-report-limits`, `{env}-signal-feedback`	Reports, limit counters and votes (Phase B)
Orbit topic	Kafka topic `orbit-element-sets` on the session box	Element sets for the low-rate stream (Phase B, OQ-30)
Warehouse	Snowflake database `SIGNAL_{ENV}`	Marts built by dbt, reading Iceberg through Glue (Phase B)
Model copy	Repository path `models/confidence/<mlflow-run-id>/`	Results file and chosen model, copied by hand after each training run (Phase B, ADR 0014)
Public sources	Source URLs	Fixed by the first-run checks (OQ-20); not stored here

---

10. API

The full specification is openapi.yaml (OpenAPI 3.0.3, pinned in ADR 0032). All endpoints sit behind API Gateway; the public pages use the HTTP type of API Gateway. The REST type is needed only where API keys and per-key limits are used, and this project issues no API keys. Paths and field names are proposals (OQ-21).

Operation id	Method and path	Phase	Purpose
`listTrails`	GET /trails	A	The main trails the page can show
`getTrailSignal`	GET /trails/{trail_id}/signal	A (confidence fields added in B)	Points along the trail with coverage flags, satellite windows, the run's valid until time and in B the confidence level or the "not enough measurements" notice
`submitReport`	POST /reports	B	Accept one hiker signal report
`getReportCounts`	GET /report-counts	B	Report counts per area
`askTripQuestion`	POST /assistant/questions	B	Trip-planning question; returns a cited check-in plan or a refusal
`submitFeedback`	POST /feedback	B	Thumbs-up or thumbs-down vote

Errors: 400 for invalid input, 404 for an unknown trail, 429 for a limit (report limits; assistant daily cap), 503 when published data is unavailable. Limits: reports 10 a day per random visitor ID, one per cell per visitor every 24 hours, 50 a day for the whole system, counted by the report's own hour; the assistant has a daily question cap and a token limit per answer (values: OQ-17). Lambda memory, timeout, concurrency and API throttling values: OQ-28.

The first request after a period of inactivity may take up to 10 seconds (Decisions "Cold starts").

Error codes

Every error body has `error.code` and `error.message`. The codes below are proposals (OQ-21).

Code	HTTP status	Used by	Meaning
`invalid_request`	400	every POST operation	The body or a parameter fails the schema
`trail_not_found`	404	getTrailSignal	No trail has this `trail_id`
`report_limit_exceeded`	429	submitReport	A report limit is reached: 10 a day per visitor, one per cell per visitor every 24 hours, or 50 a day for the whole system. The message names which one
`assistant_limit_exceeded`	429	askTripQuestion	The daily question cap or the token limit is reached
`vote_limit_exceeded`	429	submitFeedback	Reserved. No vote limit is recorded yet (OQ-39)
`data_unavailable`	503	listTrails, getTrailSignal, getReportCounts	Published data is missing or cannot be read
`dependency_unavailable`	503	submitReport, askTripQuestion, submitFeedback	A store or the model API cannot be reached. A phone keeps an unsent report in its queue and retries

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11. Deployment view

Platform-wide deployment is described in ../Platform/platform-requirements.md, sections 2 to 5. This section lists what Project 1 puts where.

Component	Runs on	Environments	Schedule
C1 to C10 (pipelines, dbt, gate, publish)	Session box on k3s (Airflow, Spark)	dev and live, same code and containers, separate buckets, Glue databases and roles	Nightly at 10:00 UTC, spot only, hard maximum 2 hours
C14 stream (B)	Session box (Kafka one broker, Structured Streaming with availableNow)	dev and live	Inside the nightly run
C15, C16 (B)	Session box and Snowflake; Marquez on the box (ARM image built once in GitHub Actions); dbt docs on the site	dev and live	After each pipeline run (refresh); Superset and Grafana during sessions only; snapshot charts published nightly
C11 Serving API	Lambda (FastAPI, DuckDB) behind API Gateway	dev and live	Always available, box off
C12 Trail page	S3 and CloudFront, preview link per pull request	dev preview and live	Always available
C21 (B)	The hiker's phone (browser, local storage, background send on Android)	live	No schedule; runs when the hiker taps a button or signal returns
C22 Report API (B)	Lambda writing DynamoDB	dev and live	Always available
C17 training (B)	Databricks Free Edition	single (free account)	After a model retrain; a monthly keep-alive job
C18 to C20 (B)	Lambda (agents), X-Ray, S3 log, Grafana reads X-Ray during sessions	dev and live	Always available
C24 smoke test (B)	GitHub Actions or the box (platform decision)	live	Weekly

Merge flow (platform): short-lived branches merge to main; CI runs Spark tests on sample data on GitHub's free machines (the AWS server stays off); only if they pass does the change deploy to dev and then to live; every website change gets a preview link.

Terraform: every resource for this project is defined as code and destroyed with `terraform destroy`.

---

12. Cross-cutting concerns

12.1 Security

- Every key (including the model API key, Snowflake and Databricks credentials) lives in AWS Parameter Store; none in code or GitHub; GitHub Actions reads them through OIDC (Decisions "Key storage").
- One least-privilege IAM role per pipeline; dev roles cannot write live data (platform step 0.2).
- The report and assistant Lambda functions each get the narrowest IAM role (permission set) that still lets them reach their own tables and tools. The assistant gets read-only tools from a fixed allowlist. Engineering standard 13 allows write actions only through human approval, and this project's assistant has no write tool.
- Source text (CelesTrak names, trail names, user questions) is treated as data, never as instructions. The red-team set tests this (FR-47).
- gitleaks and Trivy scans run in CI (platform step 0.1).
- Security logs (CloudTrail, including Parameter Store reads) are kept 1 year.
- The threat model, personal-data inventory and licence register are in security-and-privacy.md.

12.2 Privacy

Data	Where	Retention	Basis
Report cell, carrier, signal, hour	DynamoDB; lake and Snowflake after the nightly copy	Deleted 30 days after the copy in DynamoDB, the lake and Snowflake. The lake and Snowflake keep daily per-cell counts (OQ-43)	ADR 0012, ADR 0034
Visitor random ID	Phone; sent for limiting; stored only in counters	Counters expire (lifetime: OQ-26)	ADR 0011
Exact GPS	Phone memory only; discarded after the cell is computed	Never sent	ADR 0009
Internet addresses	Not used for identity or limits; logs may be written but addresses are removed before anything is stored beyond a short log retention period (length: OQ-29)	Short log retention period	ADR 0023
Votes	DynamoDB; whether a vote carries any identity is not recorded (OQ-39)	Not recorded (OQ-39)	Build plan 1B.7

DynamoDB point-in-time recovery is OFF for the hiker reports table so deletion is final at 30 days (ADR 0022); other tables keep it on. The counters table holds keyed hashes of visitor IDs for a short time, and its point-in-time recovery is off (OQ-41, ADR 0025).

Proposed control, for approval with this document (OQ-21): access logs must not record the `X-Visitor-Id` header of `POST /reports` or the body of a report. Without this, a log could link a visitor ID to a report cell and hour.

12.3 Observability

- Every DAG writes freshness and row counts to the metrics table; `status.json` feeds the public status page showing each pipeline's last good run.
- A missed freshness target or failed run sends an email; each real incident gets a short record (what broke, the cause, the fix). In Phase B one de-duplicated GitHub issue is opened and closed on recovery.
- Data lineage: OpenLineage with Marquez (Phase B).
- Model calls are logged to S3 and loaded into `llm_calls`; agent runs are traced to X-Ray; Grafana reads X-Ray during sessions.
- The numbers published (standard 14): input bytes, rows, runtime, cost per run, problems caught, eval scores, users.

12.4 Cost limits

See section 3.3. The satellite window job's row limit comes from the measured size (OQ-1). Every limit that has a recorded value is stated; limits without a recorded value are open questions, not guesses.

12.5 Reliability

- Every task is idempotent (safe to run again with the same result) and atomic (all or nothing); it overwrites a partition and uses the run's data interval instead of the current time (engineering standard 3).
- Write, audit, publish (standard 4): bad data never reaches the site.
- The page degrades safely: late data stays visible with a late notice and the time of the last good run (ADR 0006, amended).
- Spot interruption: every job is safe to rerun; a skipped night is caught up by the next run.
- Raw files are immutable and, for live, locked for 30 days (Decisions "Raw-file lock").
- Always-on parts (Lambda) write only to DynamoDB, S3, CloudWatch and X-Ray (standard 16).
- Business history is kept in rows (SCD Type 2, one partition per pull), not in Iceberg snapshots (standard 8).

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13. Testing strategy

Tests use small sample files in CI like ordinary unit tests; there are no staged production failures. Each phase starts with first-run checks.

13.1 Unit tests

Component	Tests
C2	Tagged and untagged names; each disagreement type (altitude outside about 330 to 370 km, inclination not 43 or 53) is listed; AC-1A.1
C3	`sample_points_along_line` spacing of 50 m on synthetic lines; pagination of 2,000 records
C4	`compute_horizon` on a synthetic elevation tile (flat ground gives low angles; a wall to the east gives a high angle in the east sectors); canyon-floor versus rim fixture for AC-1A.2
C5	`find_windows` with synthetic positions that cross and do not cross a horizon; `flags_to_windows` edge cases; `compute_valid_until`; rerun of a day leaves other days unchanged (AC-1A.3); `enforce_row_limit` raises over the limit
C6	`flag_points_covered` on a small hexagon set
C8	`diff_schema` with a renamed, removed and added field
C12	`isRunLate` and `windowsWithLateFlag` with a controlled clock (late notice shown, data never hidden); local time formatting from UTC
C21 (B)	`buildReport` rounds time to the hour; no coordinates in the payload; queue persists across a reload
C22 (B)	The 11th report in a day from one visitor is refused; the system refuses reports beyond 50 a day; a second report for the same cell within 24 hours is refused; a batch of reports made earlier and sent together is judged by the report hours; counters expire

13.2 Data tests

- dbt tests: unique and not_null on keys in staging and marts; relationships between facts and dimensions.
- Great Expectations: freshness, empty results and value ranges on sample files; publishing stops on failure (AC-1A.5).
- Contract diff: each source's sample file against its contract.
- Idempotence: running the same pull twice leaves the row count unchanged; replaying a day of orbit updates writes the same rows (AC-1B.2).
- One metric gives the same value from Snowflake and from DuckDB (AC-1B.3).

13.3 Evals (Phase B)

- 30 or more questions with known answers, including ones the data cannot answer, graded exactly on tool choice, numbers and refusal.
- LLM judge for clarity, calibrated on 50 answers graded by hand.
- Red-team set of 20 or more prompts: instructions hidden in source text, write requests, off-topic requests, cost abuse.
- A/B test of two prompt versions with votes from DynamoDB.
- CI fails below accuracy and red-team thresholds (values: OQ-16).
- A test question runs end to end through the metrics tool and shows as a trace with its cost; the harness fails when an answer states a number that is not in the query result (AC-1B.5).
- Ten sample trip questions are checked by hand for correct, cited plans (AC-1B.6).

13.4 First-run checks

Phase A (build plan): the [DTC] tag appears in CelesTrak's Starlink list (confirmed); the tag-based selection agrees with the altitude and inclination check (OQ-6); the size of the FCC file for Arizona is measured (OQ-7); the NPS trail file and the Copernicus elevation tiles for the park download and open correctly (OQ-8).

Phase B (build plan): the Ookla open-data files for Arizona read correctly from S3 (OQ-10); an upload to Databricks Free Edition works (OQ-3); Snowflake's refresh command shows new Iceberg data after a pipeline run (OQ-4); the ARM build of the pinned Marquez version and its web page succeeds (OQ-5).

Each first-run check has a fallback if it fails. The fallbacks recorded in the build plan are: if the Marquez build fails, the lineage tool choice is revisited; and the site never depends on Databricks, so a Databricks problem does not stop the site.

13.5 Acceptance

The build plan's "Done when" checks are the acceptance tests (requirements.md section 6).

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14. Alternatives considered

Only alternatives recorded in the Decisions document or the build plan are listed. Details are in the ADRs.

Decision	Chosen	Alternative(s) recorded	ADR
Phase A scope	Bright Angel, South Kaibab, North Kaibab; T-Mobile; 24 hours shown, 26 predicted	Wider scope deferred to Phase B and C	0001, 0019, 0020
Direct-to-Cell selection	[DTC] tag with altitude and inclination cross-check	Altitude and inclination alone (kept as the cross-check)	0002
Reachability rule	Above 10 degrees and above the terrain horizon	5 and 20 degrees compared in the design note	0003
Terrain model	Horizon from Copernicus GLO-30	None recorded	0004
Window storage	Start and end times	One row per minute (rejected as a cost risk)	0005
Late predictions	Valid until shown; old data kept with a late notice and the last good run time	Hiding expired satellite windows (superseded)	0006
Grid	H3 resolutions 9 and 8 with cell_to_parent joins	Trail-spot lists (do not scale)	0007
Confidence	Only where enough tests exist; reports as labels	Guessing in dead zones (rejected)	0008
Report location	H3 cell on the phone	Trail-spot lists; exact coordinates (rejected)	0009
Offline reports	Saved on the phone, sent on return	Requiring signal (rejected)	0010
Visitor identity	Random ID on phone, no internet addresses	Internet-address based limits (not chosen; most private option picked)	0011
Report storage	DynamoDB with 30-day expiry after copy	Earlier options did not handle writes at any hour	0012

Decision	Chosen	Alternative(s) recorded	ADR
Ookla license	Notice on page and README	None	0013
ML training	Databricks Free Edition with repo copy	Self-hosted MLflow (removed as duplicate)	0014
Warehouse	Snowflake over Glue; SCD Type 2; refresh step	BigQuery (removed)	0015
Orbit updates	Low-rate Kafka with availableNow	None recorded	0016
Assistant	Defined metrics first; raw SQL as logged fallback	Raw SQL first (more wrong numbers)	0017
Source changes	Contracts plus weekly smoke test	None recorded	0018

Decisions of Oct 9, 2026 added after the first draft (decided by John): 0019 26-hour span (alternative: exactly 24 hours, rejected, daily gap); 0020 T-Mobile and three trails; 0021 assistant limited to the span; 0022 recovery off for reports (alternative: on, rejected); 0023 addresses removed from logs; 0024 OpenCelliD and remoteness.

Changes from the build plan

Where a decision changed the build plan, this table records the change. The build plan is the schedule; where it and this document differ, this document wins.

Topic	Build plan says	This document says	Record
Prediction span	Predict and show 24 hours	Predict 26 hours; the page says "the next 24 hours"	ADR 0019
Late data	Hide expired windows	Keep old data visible with a late notice and the time of the last good run	ADR 0006 (amended; the superseded 0006a was removed as a duplicate)
ML export	Automatic copy of results and model to GitHub	Manual export by John; automation is an open test	ADR 0014, OQ-19
Report limits	10 an hour and 50 a day per visitor	10 a day per visitor; 50 a day for the whole system; counted by the report's hour	ADR 0011
Acceptance check 1B.8	"An 11th report in an hour is refused"	"An 11th report in a day from one visitor is refused"	ADR 0011
Recovery on reports table	Point-in-time recovery on	Off for the reports table	ADR 0022
Internet addresses in logs	Not addressed	Removed before anything is kept beyond a short log retention period	ADR 0023
Carrier and trails	"One carrier" and "main trails" not named	T-Mobile; Bright Angel, South Kaibab and North Kaibab	ADR 0020
Tower distance and remoteness	No source or definition	OpenCelliD towers; remoteness is distance to the nearest road or trailhead	ADR 0024
Assistant and later days	Example question about a Saturday hike	The assistant answers only within the prediction span	ADR 0021
Report location phase	Step 1B.8, but "(A)" in the Decisions document	Phase B; the "(A)" was a typo	ADR 0009
Phase C acceptance checks	Not recorded	A recorded gap, written when Phase C is designed	Section 4.2

Platform-level alternatives (k3s vs Docker Compose, Terraform vs OpenTofu, DuckDB vs Trino, Spark vs Flink, and others) are in ../Platform/platform-requirements.md, section 8.

Minor decisions folded from ADRs

These were separate ADRs and are recorded here in full because each is a setting, a wording rule, a format or a repeat of a platform rule. The old numbers are kept so references still resolve.

Old ADR	Decision	Detail
ADR-0013	Show the Ookla license notice on the page and in the README	The trail page and the README carry this exact notice: "Includes data derived from Speedtest by Ookla® open data, licensed under CC BY-NC-SA 4.0. Derived data published here is shared under the same license."
ADR-0019	Predict 26 hours of satellite windows so there is no daily gap	Propagate 26 hours (24 plus the 2-hour nightly maximum, 1,560 one-minute steps). The page still says "the next 24 hours": the 24 hours is the promise to the visitor, and the extra 2 hours are a safety margin. The run's valid until time is the end of the prediction span.
ADR-0020	Phase A carrier is T-Mobile; Phase A trails are Bright Angel, South Kaibab and North Kaibab	The Phase A carrier is T-Mobile, the only carrier with a confirmed satellite service (T-Satellite, which runs over Starlink). The build plan's [DTC] first-run check (a test that the tag exists in CelesTrak's list) confirms it. The Phase A trails are Bright Angel, South Kaibab and North Kaibab. Their names are checked against the NPS data in the first-run download.
ADR-0021	The trip-planning assistant is limited to the prediction span	The assistant answers only within the prediction span (ADR 0019) and says so when asked about later days. Coverage and confidence questions that do not need satellite windows are still answered.
ADR-0022	Point-in-time recovery is off for the hiker reports table	Point-in-time recovery is turned off for the hiker reports table, so deletion is final at 30 days. Other tables keep it on.
ADR-0023	Internet addresses are removed from logs before anything is kept beyond a short log retention period	Logs may be written, but addresses are removed before anything is stored beyond a short log retention period.
ADR-0025	Visitor ID: a 128-bit random value on the phone, stored on the server only as a keyed hash	Format. 16 random bytes from the phone's cryptographic random source (`crypto.getRandomValues`), written as 32 hexadecimal characters. Created once and kept in the phone's local storage. It is never shown to the hiker. Transport. Only in the `X-Visitor-Id` header of `POST /reports` , and only over HTTPS. Server. The counters table stores HMAC-SHA256 of the ID, keyed with a secret held in Parameter Store. This is the keyed-hash pattern the build plan already uses for email addresses. The raw ID is never written to DynamoDB or to any log. Recovery. Point-in-time recovery is off for the counters table (OQ-41). Recovery would keep hashed IDs past their expiry, which breaks ADR 0026.
ADR-0026	Report limits count by UTC day; reports are accepted only within 48 hours; counters expire 72 hours after their day	Day boundary. Midnight UTC. Standard 15 stores all times in UTC, and a UTC day is the same for every visitor. Accepted hours. A report's `report_hour_utc` must be no more than 48 hours before the time it arrives, and no more than 1 hour after it (clock tolerance). Outside that window the report is refused as `invalid_request` . Counter lifetime. A counter expires 72 hours after the end of its UTC day.
ADR-0027	Data is late when its validity time has passed or the last scheduled run did not publish	Late. A trail's data is late when either the current time is after `valid_until_utc` , or the most recent scheduled nightly run did not publish. The alerts use the same rule. Notice. "Data is late. Last good run: {local time}. Predictions valid until: {local time}." It is shown above the strip map, in the visitor's local time. Windows that have passed. Drawn in grey and labelled "passed". They are never hidden (ADR 0006).
ADR-0028	Reference data refreshes only when its source changes; the orbit stream and dim_time have fixed parameters	Reference data (OQ-25). NPS trails, trail points, horizons and FCC release files are recomputed only when the source's manifest hash changes. The nightly run checks the hash and writes only on a change. This costs less, and every rerun reproduces the same output. Orbit update stream (OQ-30). Topic `orbit-element-sets` . Watermark 2 hours, matching the nightly maximum. A satellite counts as changed when a new element set has a later epoch than the last stored element set for its catalog number. Checkpoints go to S3. Time grain (OQ-33). `dim_time` has one row per UTC minute, covering each run's prediction span. Columns: `time_key` (the minute), `date_utc` and `hour_utc` .

Old ADR	Decision	Detail
ADR-0029	Starting limits for Lambda, API throttling, reserved concurrency, and a 7-day log window	Lambda memory and timeout (starting values). Serving API 512 MB and 15 seconds. Report API 256 MB and 5 seconds. Assistant 1024 MB and 60 seconds. Reserved concurrency (caps cost and abuse). Report API 10. Vote API 5. Assistant 5. Serving API 50. API Gateway throttling. POST routes 10 requests per second with a burst of 20. GET routes 50 per second with a burst of 100. Access logs (OQ-29). Kept for 7 days, then deleted. Internet addresses are removed by a processing step before the log is kept (ADR 0023). If that step cannot be built for a log source, that log is not stored. The mechanism is confirmed at the first deploy. Logged fields. The `X-Visitor-Id` header and POST request bodies are excluded from every access log (security and privacy, section 8).
ADR-0030	Report counts are shown per H3 resolution 6 area, and only where at least 5 reports exist	Area. The H3 resolution 6 parent of each report's cell, using `cell_to_parent`. A resolution 6 cell is about 36 km² on average. Minimum count. A count is shown for an area, carrier and answer only when the area has at least 5 reports in the last 7 days. Below that the page says "Too few reports to show."
ADR-0031	Start the confidence model with logistic regression; Phase B carriers are every carrier in the park's FCC files	Model family. Logistic regression, from scikit-learn, as the first model. It is small, its coefficients can be read, and it is easy to compare with the baseline (the FCC claim alone). A tree-based model can be compared in Phase C. Carriers and technologies (OQ-11). Every carrier and every technology that has an FCC coverage file for the park's area. The file decides the set, not a list in the plan. Remoteness source (OQ-18). Decided in ADR 0035: TIGER/Line roads and NPS trail lines.
ADR-0032	Pin ODCS v3.0.2 and OpenAPI 3.0.3	Pin the Open Data Contract Standard at v3.0.2 and OpenAPI at 3.0.3.
ADR-0033	Votes carry no visitor identity and are kept 90 days; assistant question text is not stored by the project	Votes. Each vote item holds the answer ID, the vote, and the UTC time. It holds no visitor ID. Votes are deleted 90 days after they are recorded. Vote limit. Votes are rate-limited with the same HMAC-keyed counters as reports (ADR 0025), at 50 votes a day per visitor. The 429 response on `POST /feedback` is therefore defined. Question text. The project does not store it, in the call log or elsewhere (platform contract `llm_calls`, ../../Platform/contracts/llm_calls.yaml).
ADR-0038	The assistant's monthly spend limit is $10; 20 questions a day; 2,500 input and 800 output tokens per answer	Monthly spend limit: $10, set in the Anthropic account before the first call. Lowered from $15 (ADR 0017). Daily question cap: 20 questions per UTC day. Derived: $10 over about 30 days is about $0.33 a day, and 20 questions a day uses that budget. Token limit per answer: 2,500 input tokens and 800 output tokens. A question longer than the input limit is refused with `invalid_request`. An answer that reaches the output limit is cut and marked as incomplete. Price check. Before the first call, the cost of one answer at these limits is worked out from the price list in force. The daily cap holds only if one answer costs no more than about $0.017 ($0.33 divided by 20). If the price is higher, the daily cap is lowered in a new ADR before the assistant is turned on. Stop. When the monthly spend reaches the limit, the provider stops the calls. The assistant then returns `assistant_limit_exceeded` until the next month.

15. Risks

Risk	Impact	How it is handled
Late data read as current after a failed or skipped run	Hiker relies on a satellite pass that is not valid	Valid until stamp; visible late notice with the time of the last good run (ADR 0006, amended); lateness rule OQ-23
Daily gap between expiry and the next publish	Page would be late for part of each day	Handled: 26 hours predicted (ADR 0019)
Surface model counts trees and buildings as terrain	Horizons may be higher than rock alone	Stated as a limitation; effect to be measured (OQ-24)
FCC map overstates coverage	Hiker trusts coverage that is not there	Confidence model; "not enough measurements" notice (ADR 0008)
Speed tests miss dead zones	Model would guess where it has no data	Confidence only where enough tests exist; reports as labels (ADR 0008)
Report floods or fake reports	Poisoned labels and counts	Limits per random ID; counters; filtering; set-aside; label only where N reports in a cell on a day agree (ADR 0008, 0011, 0012; OQ-12)
Offline sync burst hits the rate limit	Honest reports dropped	Handled: limits counted by the report's own hour (ADR 0011); a phone could state a false hour, sub-parts in OQ-26

Risk	Impact	How it is handled
iPhone cannot send in the background	Delayed reports	Sent at next page open with a "1 report waiting to send" note (ADR 0010)
Satellite window job explodes in rows or runtime	Cost or overrun	Interval storage; row and runtime limits; incident (ADR 0005; OQ-1, OQ-15)
Upstream format change	Run fails	Contracts, schema diff, weekly smoke test (ADR 0018)
Spot unavailable or reclaimed	Skipped night	Rerun-safe jobs; next run catches up; the page shows the old data with the late notice
Databricks Free Edition limits or lapses	Training blocked	Site never depends on it; repo copy; monthly keep-alive (ADR 0014; OQ-3, OQ-19)
Snowflake cannot read Glue or refresh	Warehouse stale	Refresh step; first-run check (ADR 0015; OQ-4)
Marquez ARM build fails	No lineage graph	Platform first-run check; lineage choice revisited (OQ-5)
Non-commercial licenses	Cannot charge	Licensing checked before charging (section 3.4)
Model cost	Surprise spend	$10 limit, daily cap, token limit, cheapest model (FR-42)
DTC tag changes or is dropped	Selection breaks	Altitude and inclination cross-check logs disagreements (ADR 0002)
Coverage claims taken as guarantees	Misleading page	Page states what DTC needs and that coverage is a map claim; wording set in ADR 0037 (OQ-22)
Assistant asked about a date outside the prediction span	Wrong or missing plan	Handled: limited to the prediction span and says so (ADR 0021)

16. Open questions

Each question has a GitHub issue to be created. A question is removed once answered and its answer moves to the right section. Open questions do not block coding of the parts they do not affect.

A row that says "Plan conflict" marks a place where the build plan or the Decisions document seems contradictory, impossible or incomplete, so John must decide a change before the affected feature is built.

ID	Question	What answering it needs	Affects	Issue
OQ-1	How many rows does a satellite-window run for the park write, and how long does it take? The row limit and runtime limit for the satellite window job are set from the answer	A test run on the session box	C5, QR-7, QR-9	issue: to be created
OQ-2	Are there enough Ookla speed tests along Grand Canyon trails?	Downloading Ookla's open data	C17, F-1B.4	issue: to be created
OQ-3	Does an upload to Databricks Free Edition work (Files API)?	John's Databricks account	C17, F-1B.4	issue: to be created
OQ-4	Does Snowflake connect to the Glue catalog and refresh? Snowflake's docs list no edition requirement; check in 1B.3	John's Snowflake trial, run just before Phase B	C15, F-1B.3	issue: to be created
OQ-5	Does the ARM build of the pinned Marquez version and its web page succeed? (Owned by the platform; first-run check in Phase B)	Running the build in GitHub Actions	C16	issue: to be created
OQ-6	Does the tag-based selection agree with the altitude (about 330 to 370 km) and inclination (43 or 53 degrees) check for all tagged satellites?	First-run check on a real CelesTrak file	C2, ADR 0002	issue: to be created
OQ-7	What is the size of the FCC file for Arizona (and of the Ookla and OpenCelliD files)? The build plan says to measure sizes first	Downloading the files	C6, C13, QR-9	issue: to be created

ID	Question	What answering it needs	Affects	Issue
OQ-8	Do the NPS trail file and the Copernicus elevation tiles for the park download and open correctly?	First-run check	C3, C4	issue: to be created
OQ-9	How do the satellite windows change at 5 and 20 degrees compared with 10? (Decide in 1A.3, starting at 10)	The satellite window job output	C5, ADR 0003	issue: to be created
OQ-10	Do the Ookla open-data files for Arizona read correctly from S3?	First-run check	C13	issue: to be created
OQ-11	Which FCC technologies are used, and what is the carrier list for Phase B? (Phase A carrier is T-Mobile, Decided by John, Oct 9, 2026, ADR 0020) **Answered (Oct 9, 2026):** every carrier and every technology with an FCC file for the park's area; the file decides the set (ADR 0031).	John's decision	C6, C13	closed
OQ-12	What is N, the number of reports in the same cell on the same day that lets reports become labels? (Method decided: N reports in one cell on one day, each a different visitor because of the one-per-cell rule, ADR 0008)	John's decision after seeing traffic; the 50 a day system cap bounds it	C23, ADR 0008	issue: to be created
OQ-13	How many labeled reports are enough to retrain the model?	John's decision, set in the design note	C17, C23	issue: to be created
OQ-14	How many speed tests, within what distance, count as "enough tests nearby" for a confidence level?	Decision from the measured Ookla density (OQ-2)	C17, ADR 0008	issue: to be created
OQ-15	What are the maximum rows and runtime for each Spark job, the freshness thresholds and value-range limits for the publish gate, and the S3 storage alarm size?	Measured sizes from test runs (OQ-1, OQ-7)	C5, C9, C24, QR-9	issue: to be created
OQ-16	What are the CI thresholds for eval accuracy, the red-team pass rate and the judge's agreement?	John's decision after a first scoring run	C20, QR-19	issue: to be created
OQ-17	What are the assistant's daily question cap and the token limit per answer? **Answered by John (Oct 9, 2026):** daily cap of 20 questions per UTC day, 2,500 input and 800 output tokens per answer, and a monthly spend limit of $10 (ADR 0038).	John's decision	C18, C19, QR-9	closed
OQ-18	What is the remoteness data source (decided: distance to the nearest road or trailhead from published data, ADR 0024; tower source decided: OpenCelliD) and which model family is used? **Answered by John (Oct 9, 2026):** the remoteness source is the U.S. Census TIGER/Line roads and the NPS trail lines; the model family is logistic regression (ADR 0035, ADR 0031). The licence check of the exact Census vintage is a first-run check.	John's decision; a contract for the remoteness source	C17, ADR 0024	closed
OQ-19	Can the export of MLflow results and the model from Databricks to GitHub be automated, given Databricks Free Edition has limited outbound internet? (Decided: manual export for now, ADR 0014)	A test in the Databricks account (with OQ-3)	C17, ADR 0014	issue: to be created
OQ-20	What are the exact source URLs, field lists, the `UNITCODE` value for Grand Canyon and the FCC file names for each source? The source contracts are drafts until this is answered	First-run checks (OQ-6, OQ-7, OQ-8, OQ-10)	contracts/*	issue: to be created
OQ-21	Approval of proposed names: components, functions, classes, tables, columns, endpoint paths, DAG ids, MCP tool names **Answered (Oct 9, 2026):** the proposed names are accepted as working names; John may rename them before coding (no ADR; recorded in the changelog).	John's review	all	closed

ID	Question	What answering it needs	Affects	Issue
OQ-22	What are the terms and credit text for CelesTrak, FCC, NPS and OpenCelliD data (OpenCelliD is share-alike, ADR 0024; the attribution text is not recorded), and what wording does the page use to state that coverage is a map claim and that satellite texting is not guaranteed? **Answered by John (Oct 9, 2026):** credit wording for each source, the status of each licence check, and the page wording that coverage is a claim and satellite texts are not guaranteed. Two credits are unverified: OpenCelliD and FCC mobile data (ADR 0037).	Reading each provider's terms; John's wording	C12, section 3.4	closed
OQ-23	What exactly makes data "late" (the valid until time passing, a failed run, or both), what is the late notice wording, and how are individual satellite windows already in the past drawn while the run is current? (Decided: old data stays visible with a late notice and the last good run time, ADR 0006; 26 hours predicted, valid until is the end of the span, ADR 0019. Anchor of the span start: the run's start time is assumed) **Answered (Oct 9, 2026):** data is late when its validity time has passed or the last scheduled run did not publish; passed windows are greyed, never hidden (ADR 0027).	John's decision	C5, C12, ADR 0006	closed
OQ-24	Open details of the terrain horizon computation: range cutoff ("about 15 km"), ray step, Earth curvature and refraction, observer height above the surface model, vegetation in the surface model, whether 36 sectors is final, direction convention, interpolation in `horizon_at`, and the numeric margin that makes a canyon-floor horizon "much higher" than a rim horizon **Answered by John (Oct 9, 2026):** the horizon defaults: 15 km range, 30 m ray step, 1.7 m eye height, Earth curvature applied, refraction not applied, surface model kept, 36 sectors final, linear interpolation, 5-degree AC-1A.2 margin (ADR 0036).	A spike on the park's tiles	C4, AC-1A.2	closed
OQ-25	How often are the reference datasets (NPS trails, FCC files, horizons) refreshed: every night or only when the source changes? **Answered (Oct 9, 2026):** reference data is recomputed only when the source's manifest hash changes (ADR 0028).	A measured run time (OQ-1) and release cadence of each source	C3, C4, C6	closed
OQ-26	Remaining details of the report limits (decided: counted by the report's own hour, 10 a day per visitor, 50 a day system-wide, ADR 0011): the day boundary (UTC or local), the oldest report hour still accepted, how long counters live given offline delay, and handling of a refused report on the phone **Answered (Oct 9, 2026):** day boundary midnight UTC; reports accepted if no more than 48 hours old; counters expire 72 hours after their UTC day (ADR 0026).	John's decision	C21, C22, ADR 0011	closed
OQ-28	Lambda memory, timeout, reserved concurrency, API throttling values and any page-latency target **Answered (Oct 9, 2026):** starting Lambda, reserved-concurrency and throttling values, to be measured in F-1B.9 (ADR 0029).	A measurement in F-1B.9 and a decision	C11, C22, QR-6	closed
OQ-29	How long is the short log retention period for logs, and what mechanism removes internet addresses? (Decided: addresses are removed before anything is stored beyond that period, ADR 0023) **Answered (Oct 9, 2026):** access logs kept 7 days; internet addresses removed before storage (ADR 0029).	Platform configuration	C22, QR-10	closed
OQ-30	Orbit stream parameters: watermark value, topic name, checkpoint path, and what counts as a "changed" satellite **Answered (Oct 9, 2026):** topic orbit-element-sets; watermark 2 hours; a satellite is changed when its element set has a later epoch (ADR 0028).	A first low-rate run	C14, ADR 0016	closed

ID	Question	What answering it needs	Affects	Issue
OQ-32	Which versions of the Open Data Contract Standard and OpenAPI 3 are pinned? (The contracts use ODCS v3.0.2 and OpenAPI 3.0.3 as working placeholders; API Gateway import compatibility is to be confirmed) **Answered (Oct 9, 2026):** ODCS v3.0.2 and OpenAPI 3.0.3 are pinned (ADR 0032).	John's decision after validating the files	contracts, openapi.yaml	closed
OQ-33	What does one row of `dim_time` represent (for example a minute, an hour or a day)? **Answered (Oct 9, 2026):** one row per UTC minute of each run's prediction span (ADR 0028).	John's decision	dbt star schema	closed
OQ-35	What is an "area" for report counts per area (a parent H3 cell at which resolution, or a trail)? **Answered (Oct 9, 2026):** area is the H3 resolution 6 parent of the report cell; counts are shown only from 5 reports in 7 days (ADR 0030).	John's decision	C23, fct_report_counts_area	closed
OQ-36	Do the NPS TRLNAME values match Bright Angel, South Kaibab and North Kaibab (decided trails, ADR 0020), and is each one line or several segments?	First-run download of the NPS file	C3, AC-1A.6	issue: to be created
OQ-38	What is the visitor ID format (type, length) and how is it transported? **Answered (Oct 9, 2026):** 16 random bytes written as 32 hexadecimal characters; sent only in the header, over HTTPS; stored on the server as a keyed HMAC (ADR 0025).	John's decision	C21, C22	closed
OQ-39	Which model (the cheapest) and which API account are used, how long are votes kept, does a vote carry any visitor identity, and are votes rate-limited (the 429 response on POST /feedback depends on it)? **Partly answered (Oct 9, 2026):** votes carry no visitor ID and are kept 90 days; question text is not stored by the project (ADR 0033). Provider chosen by John: the Anthropic API, on the cheapest Claude model (ADR 0034). Still open: recording the provider's terms before the first call.	John's decision	C18, C20, openapi.yaml	issue: to be created
OQ-40	How does dbt on DuckDB read Iceberg tables (platform-owned)?	Platform design	C7	issue: to be created
OQ-41	Does the limit-counters table also have point-in-time recovery off? It holds visitor IDs for a short time, and the decision covers only the reports table (ADR 0022) **Answered (Oct 9, 2026):** point-in-time recovery is off for the counters table (ADR 0025).	John's decision	C22	closed
OQ-42	Which AWS region and accounts hold the dev and live environments? The build plan does not name a region **Answered by John (Oct 9, 2026):** us-east-1 (N. Virginia) for dev and live (ADR 0034).	John's decision	platform, section 9.1, every contract	closed
OQ-43	How long are the lake and Snowflake copies of reports (`signal_report`, `signal_report_set_aside`) kept? The 30-day rule covers DynamoDB rows only **Answered by John (Oct 9, 2026):** raw report rows deleted 30 days after the copy in DynamoDB, the lake and Snowflake; daily per-cell counts are kept; set-aside rows follow the same rule (ADR 0034).	John's decision (privacy)	C23, hiker_signal_reports, security-and-privacy.md	closed

Closed by John's answers on Oct 9, 2026 (moved into the sections and ADRs named; IDs are not reused): OQ-27 (assistant limited to the prediction span, ADR 0021), OQ-31 (recovery off for the reports table, ADR 0022), OQ-34 (Phase C done-when checks are a recorded gap, written when Phase C is designed, section 4.2), OQ-37 (report location is step 1B.8, Phase B; the "(A)" label is a typo, ADR 0009). Narrowed to their open sub-parts: OQ-11, OQ-12, OQ-18, OQ-19, OQ-23, OQ-26, OQ-29, OQ-36.

Answered on Oct 9, 2026 under the security-first default rule for minor questions, each recorded as an ADR and marked in the table above: OQ-11, OQ-21, OQ-23, OQ-25, OQ-26, OQ-28, OQ-29, OQ-30, OQ-32, OQ-33, OQ-35, OQ-38 and OQ-41 (ADRs 0025 to 0032). Partly answered: OQ-18 (model family answered; remoteness source open) and OQ-39 (votes and question text answered; provider and account open). These defaults can be changed by John.

Answered by John on Oct 9, 2026: OQ-42 (region, ADR 0034), OQ-43 (report copies, ADR 0034), the provider part of OQ-39 (ADR 0034), OQ-17 (spend $10 a month, ADR 0038), OQ-22 (credit wording and page wording, ADR 0037, two credits to verify) and OQ-24 (horizon defaults, ADR 0036).

A platform question that Project 1 depends on: whether the whole stack fits in 32 GB of memory. The platform owns it, and it is answered by running the stack on the session box.

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17. Glossary

Term	Meaning
[DTC]	Name tag on Starlink Direct-to-Cell satellites in CelesTrak
ADR	Architecture decision record: a short file that records one decision, its context, alternatives and consequences
Airflow	The scheduler that runs the nightly pipelines as DAGs (directed acyclic graphs: ordered task lists)
API Gateway	The AWS service that receives web requests and passes them to Lambda functions
availableNow	A Spark trigger that processes everything new since the last run and then stops
Azimuth	Compass direction in degrees, measured clockwise from north
C4	A way of drawing software architecture at four levels (context, containers, components, code)
Carrier	A mobile phone provider
CelesTrak	A public source of satellite orbit data (GP data)
Checkpoint	Saved progress of a Spark streaming job, kept on S3, so the job can continue where it stopped
CI	Continuous integration: the automatic tests that GitHub Actions runs on every change
CloudFront	The AWS service that delivers the static website files
Cold start	The delay on the first request to a Lambda function after inactivity
Contract (data contract)	A written description of a table's schema, rules, freshness target, owner and version (ODCS)
Copernicus GLO-30	Global 30 m elevation model; surface model including vegetation and buildings
Databricks	The service where the confidence model is trained, in its free edition
dbt	Tool that builds tables from SQL models and tests them
Direct-to-Cell	Starlink satellites that connect to ordinary phones for texting, through T-Mobile in the United States
Done when	The build plan's check that shows a step is finished; reproduced as an acceptance criterion
dropDuplicatesWithinWatermark	A Spark function that drops repeated records that arrive inside the watermark time
DuckDB	The embedded query engine used in Lambda and on the box
DynamoDB	The AWS database that is always on and stores visitor data
EGM2008	The vertical datum of Copernicus heights
Elevation (satellite)	Angle of a satellite above the horizontal seen from a point
Eval	An automated test of the assistant's answers against known answers
FastAPI	The Python web framework used for the serving API
FCC	U.S. Federal Communications Commission; publishes mobile coverage data
Glue catalog	AWS catalog that registers Iceberg tables
Gold	The last layer of tables, ready for the page to read
GP data	General perturbations orbital elements, the format CelesTrak serves
Grafana	The tool that shows platform health charts during work sessions; it reads X-Ray traces
Great Expectations	Tool that checks data before it is published

Term	Meaning
H3	A grid of hexagon cells covering the Earth; resolution 9 cells are about 0.1 km2 (the FCC cell size) and resolution 8 is used for Ookla tiles
Horizon profile	The highest terrain angle in each of 36 compass sectors around a point
Iceberg	Table format on S3
Idempotent	Safe to run again with the same result
Intermediate	The dbt layer between staging and marts, where tables are combined
k3s	A small version of Kubernetes (software that runs and restarts programs on servers) used on the session box
Kafka	A message queue that carries element sets during the nightly run
Lambda	AWS code that runs on demand and costs nothing when idle
Late data	Data from a run whose valid until time has passed or whose publish failed; shown with a late notice (exact rule: OQ-23)
Late notice	The visible message on the page that says the data is late and gives the time of the last good run
LLM judge	A second language model that scores answers, for example for clarity
Logical date (data interval)	The date a run is responsible for, used instead of the current time
Manifest	A list of a file's URL, size and hash kept instead of a copy of the file
Marquez, OpenLineage	Lineage graph tool and its event standard
Mart	A finished table built by dbt for a specific use
MCP	Model Context Protocol; the way tools are exposed to an AI model
MetricFlow	The dbt tool that defines business metrics once so every report gives the same number
MLflow	Tool tracking model training runs
Nightly run	The run of every pipeline that starts at 10:00 UTC
OIDC	OpenID Connect: the way GitHub Actions proves its identity to AWS without stored keys
Ookla	Speed-test company whose open data is used for measured speeds
OpenAPI	A standard YAML format that describes a web API
Parameter Store	The AWS service that stores every key and password
Parquet	A compact file format for tables
Partition	One slice of a table, here the rows of one run date, that can be replaced on its own
Platform skeleton	The first build stage of the platform, before Project 1 starts
Prediction span	The period the satellite windows cover (26 hours predicted; the page says the next 24 hours)
Red-team set	Test prompts written to try to make the assistant misbehave
Satellite window	A period during which a satellite is reachable from a trail point
SCD Type 2	Keeping history as rows with validity ranges
Serving template	The platform's reusable web setup: Lambda with DuckDB behind API Gateway, plus a React site
Session box	The rented AWS server that runs the nightly pipelines and is switched off the rest of the time
SGP4	The standard model that predicts a satellite's position from its orbital elements
Snowflake	The cloud data warehouse used from Phase B
Spark	The batch data-processing engine used for heavy computations
Spot	AWS servers that cost less and can be taken back with two minutes of warning
Staging	The dbt layer and area where new data lands before it is checked
Star schema	A table design with a central fact table joined to descriptive dimension tables

Term	Meaning
Superset	The dashboard tool for data in Snowflake, used during work sessions
Remoteness	How far a trail point is from the nearest road or NPS trail line (ADR 0035)
TIGER/Line	The U.S. Census Bureau's road and boundary files
Trail point	A point sampled every 50 meters along a trail line
Valid until	The time after which a run's predictions must not be shown as current
Walking skeleton	A tiny end-to-end version of the whole system that is built first and then grows
Watermark	In streaming, how long late records are still accepted before results are final
X-Ray	The AWS service that records traces of agent runs
ODCS	Open Data Contract Standard: the YAML format used for the data contracts
Residual risk	A risk that remains after its controls, which John reviews and accepts
Runbook	Step-by-step response to one known failure
Threat model	A list of what could go wrong, who or what could cause it, and what stops it

Part 4: Operations

`Will I have signal out there/operations.md`

Project 1 operations

Field	Value
Status	Draft for review. Written before coding so the failure handling is designed in. Numbers from measured runs replace the estimates in section 7 when the first runs exist
Owner	John
Last updated	Oct 9, 2026
Related	design-doc.md (sections 8, 12 and 13), ../Platform/platform-requirements.md, security-and-privacy.md

This document says how the project runs day to day: when things happen, what is watched, what a person does when something breaks, and what the project costs. It is written to be followed by someone who did not build the system.

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1. Schedule

Job	When	Where it runs	Maximum run time
Nightly pipeline run (C1 to C10, plus Iceberg maintenance last)	Every day at 10:00 UTC (3 a.m. Arizona, 6 a.m. US Eastern)	Session box, spot only	2 hours. Usually about 45 minutes
Orbit element pull (C1)	At most once per 2-hour update, inside the nightly run	Session box	Stops on any response other than HTTP 200
Nightly report copy (C23, Phase B)	Inside the nightly run	Session box, reading DynamoDB	Part of the 2-hour maximum
Kafka orbit stream (C14, Phase B)	Inside the nightly run, with availableNow	Session box	Part of the 2-hour maximum
Snowflake refresh (C15, Phase B)	After each pipeline run	Snowflake	Query time limit on every query
Weekly source smoke test (C24, Phase B)	Once a week	GitHub Actions or the box (platform decision)	Not recorded
Databricks keep-alive (Phase B)	Once a month	Databricks Free Edition	Not recorded
Model retrain and export (Phase B)	After a training run, by hand	John's computer and Databricks	Not recorded

Hitting the 2-hour maximum is recorded as an incident. The next run catches up.

2. Freshness targets

A freshness target is the longest time a published dataset may be out of date before it is treated as late.

Published data	Target	What the page shows when the target is missed
Satellite windows (`fct_satellite_window`)	Published every night, inside the 2-hour maximum. Each run predicts 26 hours, so the previous run is still valid when the next one publishes (ADR 0019)	Old data with a late notice and the time of the last good run (ADR 0006)
Trail coverage (`fct_trail_point_coverage`)	Rebuilt every night. The FCC input changes only with a new release	Same as above
Trail points and horizons (`dim_trail_point`, `trail_point_horizon`)	Reference data. Recomputed when the inputs change, or nightly if OQ-25 decides so	Same as above

Published data	Target	What the page shows when the target is missed
Confidence values (`fct_coverage_confidence`, Phase B)	Scored every night with the model copy in the repo	Same as above
Report counts (`fct_report_counts_area`, Phase B)	Nightly, from the copy of reports	Same as above

"Late" is defined in ADR 0027 (OQ-23). A trail's data is late when the current time is past its validity time, or when the most recent scheduled run did not publish. The alerts use the same rule.

3. What is watched

Signal	Where it is shown	Who is told
Freshness and row count for every pipeline run	Metrics table, written by every DAG (platform step 0.5)	Read on the status page
Each pipeline's last good run	`status.json`, read by the status page	Read on the status page
A failed run or a missed freshness target	Email through Amazon SNS (proposed, OQ-21)	John, by email
A failed run or missed target (Phase B)	One GitHub issue. Duplicates are merged into it. It closes when the pipeline recovers	John, in GitHub
Storage size passes a set level	CloudWatch alarm on the lake bucket. Level is open (OQ-15)	John, by email
Session box idle	CloudWatch alarm stops the box	Logged; no email unless the box fails to stop
Session limit (4 hours)	The box is stopped by the limit	Logged
Spend	AWS Budget at $40 with an action (backup); Snowflake resource monitor at 3 credits; Model API limit at $10 a month and a daily question cap	Budget email; resource monitor suspends the warehouse
Model calls (Phase B)	Call log in S3, loaded nightly into `llm_calls`; traces in AWS X-Ray	Grafana reads X-Ray during work sessions
Cold-start time (Phase B)	Measured, reported as median warm latency and 95th-percentile cold-start time in the README	Read in the README
Source format changes (Phase B)	Weekly smoke test pulls one small sample from every source and alerts on a format change	Email and GitHub issue

4. Cost

Estimates are from public price pages as of Oct 8, 2026. They are rechecked before the first run and after every change of instance type.

Item	Spot	On-demand	Hard limit
Session box, `t4g.2xlarge`, about 70 hours a month	$5.40 to $9.70	$18.82	4-hour session limit; idle shutdown
Heavy batch on `m7g.2xlarge`, about 30 hours a month	$4.80 to $6.20	$9.79	Same limits
Disk, 50 GB gp3 while running	$0.55	$0.55	Disk size fixed in Terraform
S3 storage, up to about 150 GB by Phase C (far less early on)	Up to $3.45	Up to $3.45	Clean-up rules (section 6); size alarm
Fargate ingest tasks	About $1	About $1	Fixed schedule
Public IPv4 while running	$0.50	$0.50	Only while the box runs

Item	Spot	On-demand	Hard limit
Nightly pipeline run, about 22 hours a month	$2 to $3.35	$2 to $3.35	Always spot; 2-hour maximum
AWS total	$18 to $25	About $37	$40 AWS budget as backup
Snowflake after the trial	Up to $6	Up to $6	3 credits a month; query time limit
Model API, cheapest model only	$5 to $10	$5	$10 a month spend limit; daily question cap; token limit per answer
All in, normal month	About $23 to $46	About $45 to $48	Under $50 a month

The skeleton and Phase A months are usually under $20, because they need few box hours and no warehouse.

Each Spark job also has a limit on rows read and written and on runtime. A job over either limit stops before publishing and opens an incident. The values are open until measured (OQ-1 and OQ-15).

5. Incidents

An incident record is written for every real failure. A short record is enough for a small failure. A blameless postmortem is written for each real incident.

Record fields:

- **ID and date**, and the pipeline or component that failed
- **Detected by**: the alarm, the email, the status page or a person
- **What visitors saw**: the page as it was, including any late notice
- **Cause**
- **Fix**
- **Follow-up**: a change to a runbook, a contract, a limit or a design decision, with its own record
- **Runbook updated**: yes or no

Blameless means the record asks what allowed the failure, not who caused it.

6. Runbooks

Each runbook names the symptom, how it is detected, what the visitor sees, the first action, the fix, and what to record. Every failure here leaves the previously published data live, so a failed run is never shown as a new result.

RB-1: CelesTrak returns a status other than 200

- **Detected by:** the orbit ingest DAG (C1) stops and the run fails.
- **Visitor sees:** the previous satellite windows, with the late notice once their `valid_until_utc` passes.
- **First action:** check the CelesTrak status and the URL in the contract (OQ-20).
- **Fix:** none needed for a short outage. The next run retries. If the URL moved, update the contract through a reviewed change.
- **Record:** an incident if the outage lasts past one nightly run.

RB-2: Schema diff fails (C8)

- **Detected by:** the schema diff compares the new file with the contract and stops the run before anything loads.
- **Visitor sees:** the previous data, with the late notice once it is late.
- **First action:** read the diff. It names each added, removed or changed field.
- **Fix:** if the source changed on purpose, update the contract, rerun the tests with the new sample, and then rerun the pipeline. If not, report it to the source.
- **Record:** an incident, with the diff.

RB-3: The DTC cross-check lists disagreements (C2)

- **Detected by:** the selector lists satellites whose altitude or inclination disagrees with the `[DTC]` tag.

- **Visitor sees:** nothing new. The run does not fail on disagreements (design doc, section 8.1).
- **First action:** read the list. Decide whether the tag or the cross-check is wrong for each satellite.
- **Fix:** record the decision in an ADR. Change the selector only with a test that shows the change.
- **Record:** the list and the decision.

RB-4: A job exceeds its row or runtime limit (C5)

- **Detected by:** `enforce_row_limit` stops the satellite window job before publishing.
- **Visitor sees:** the previous data, with the late notice once it is late.
- **First action:** read the row count and runtime from the metrics table. Compare them with the size estimate in the design doc.
- **Fix:** if the estimate was wrong, record the measured figures and set the limit from them (OQ-1, OQ-15). Do not raise a limit to get past a failure without a recorded reason.
- **Record:** an incident with the measured figures.

RB-5: A dbt test fails (C7)

- **Detected by:** dbt test failure stops the run before the publish gate.
- **Visitor sees:** the previous data, with the late notice once it is late.
- **First action:** read the failing test name and its model.
- **Fix:** fix the model or the input. A test is never removed to get a run through; a change to a test is recorded in an ADR.
- **Record:** an incident.

RB-6: The publish gate fails (C9)

- **Detected by:** a Great Expectations check fails (freshness, empty results or value ranges). Publishing stops, and the previously published Parquet stays live.
- **Visitor sees:** the previous data, with the late notice once it is late.
- **First action:** read the failed expectation.
- **Fix:** if the source is at fault, hold the publish until the source is fixed. If the expectation is wrong, change it through a recorded decision.
- **Record:** an incident.

RB-7: The nightly run reaches the 2-hour maximum

- **Detected by:** the session box stops at the maximum. The run is marked incomplete.
- **Visitor sees:** the previous data. The next night's run catches up.
- **First action:** read the metrics table to see which step took the time.
- **Fix:** move the slow step out of the nightly run if it can run in a work session (backfills always run in sessions). Otherwise, change the step.
- **Record:** an incident with the step and its time.

RB-8: Spot capacity is unavailable at 10:00 UTC

- **Detected by:** the box does not start. The night is recorded as skipped.
- **Visitor sees:** the previous data, with the late notice once it is late.
- **First action:** none. The next run catches up.
- **Fix:** if spot is unavailable for several nights, reconsider the instance choice in the design doc. The build plan makes the nightly run spot only, so switching it to on-demand needs a recorded decision and a new cost check.
- **Record:** a note in the metrics table. An incident if two nights in a row are skipped.

RB-9: A spot server is taken back mid-run

- **Detected by:** the interruption handler records the two-minute warning and the run stops.
- **Visitor sees:** the previous data.

- **First action:** rerun the nightly pipeline on a new box.
- **Fix:** none. Every job overwrites its own partition, so a rerun replaces only that day.
- **Record:** a note in the metrics table.

RB-10: Airflow's saved database cannot be restored

- **Detected by:** the box starts without its run history.
- **Visitor sees:** nothing new. Pipelines can run from the start.
- **First action:** start the pipelines from the beginning. Every task is safe to rerun.
- **Fix:** check the S3 copy and the restore step. Keys come from Parameter Store, so the saved copy holds none.
- **Record:** an incident.

RB-11: Databricks is unavailable or its free account lapsed (Phase B)

- **Detected by:** training or export fails. The nightly scoring does not use Databricks, so it is unaffected.
- **Visitor sees:** nothing new. The page uses the model copy in the repo.
- **First action:** check the Databricks account and its quota.
- **Fix:** if the account lapsed, the monthly keep-alive job has failed. Restore the account and run the keep-alive. Training waits.
- **Record:** a note. An incident only if the copy in the repo is out of date.

RB-12: Snowflake does not see new data after the refresh (Phase B)

- **Detected by:** the refresh step shows no new rows, or a metric differs between Snowflake and DuckDB.
- **Visitor sees:** nothing. The site does not read Snowflake (ADR 0015).
- **First action:** check the Glue connection and the refresh command.
- **Fix:** rerun the refresh after the pipeline has finished. If the connection fails, see OQ-4.
- **Record:** an incident if a metric is wrong.

RB-13: The report copy fails (C23, Phase B)

- **Detected by:** the copy stops before setting any expiry.
- **Visitor sees:** nothing. No report is deleted before it is copied (ADR 0012).
- **First action:** rerun the copy. It is safe to rerun.
- **Fix:** read the failing step.
- **Record:** an incident if the copy has failed for more than one night.

RB-14: A flood or fake reports arrive (Phase B)

- **Detected by:** the per-day counts in `fct_report_counts_area` jump, or the set-aside table fills.
- **Visitor sees:** counts that may be wrong until the set-aside step catches the pattern.
- **First action:** read the set-aside rows (`signal_report_set_aside`).
- **Fix:** tighten a limit only with a recorded decision. The system-wide cap of 50 a day stops floods at the write step.
- **Record:** an incident with the pattern.

RB-15: The assistant's spend or question count rises (Phase B)

- **Detected by:** the daily question cap refuses questions, or the model API spend nears $10 in a month.
- **Visitor sees:** the assistant refuses questions over the cap, with the `assistant_limit_exceeded` error.
- **First action:** read `llm_calls` for the calls and their prompt versions.
- **Fix:** if the traffic is abusive, leave the cap in place. If it is real use, raise the cap by a recorded decision.
- **Record:** an incident.

RB-16: A source changes its format (weekly smoke test, Phase B)

- **Detected by:** the weekly smoke test finds a format change before the nightly run fails.
- **Visitor sees:** nothing yet.

- **First action:** read the change.
- **Fix:** update the contract and its tests, through a reviewed change, before the next nightly run.
- **Record:** an incident and a changelog line.

RB-17: The page shows the late notice when it should not

- **Detected by:** a visitor or John sees a late notice on fresh data.
- **Visitor sees:** a notice that is wrong about freshness.
- **First action:** compare the device clock with UTC. The page compares `valid_until_utc` with the browser clock (ADR 0006), so a wrong device clock changes the result.
- **Fix:** if the clock was right, check `last_good_run_utc` and the status page.
- **Record:** an incident if the data itself was stale.

7. Backups and recovery

What	Backup	Recovery
Raw source files	Never changed. Live raw files are locked for 30 days	Re-download from the source, or from the raw file
Lake tables (Iceberg)	Snapshots and the raw files they came from	Rerun the affected day. Each day's partition is replaced
Published Parquet	The previous publish stays until the next one passes	The previous publish stays live
Airflow's database	Saved to S3 after every run and at shutdown	Restored at startup. If lost, the pipelines run again
Reports (DynamoDB, Phase B)	Point-in-time recovery is off for this table, so deletion is final at 30 days (ADR 0022). The lake copy is the only other copy	A bad write cannot be undone from DynamoDB; the lake copy is the recovery source
Visitor votes (Phase B)	Point-in-time recovery is on	Restore to an earlier point in time
Visitor counters (Phase B)	Point-in-time recovery off (OQ-41, ADR 0025)	Not recoverable by design. Counters expire 72 hours after their UTC day (ADR 0026)
Terraform state	Versioned S3 bucket	Restore an earlier version

8. Clean-up and retention

Data	Rule
Dev data	Deleted after 14 days
Temporary files and Spark checkpoints	Deleted after 7 days
Failed partial uploads	Deleted after 7 days
Pipeline logs	Deleted after 90 days
Security logs (CloudTrail, including Parameter Store reads)	Kept 1 year
Old file versions	Deleted after 30 days
Raw source files and published data	Kept
Raw report rows in DynamoDB (Phase B)	Deleted 30 days after the nightly copy
Lake and Snowflake copies of reports	Deleted 30 days after the copy. Daily per-cell counts are kept (OQ-43, ADR 0034)
Access log addresses (CloudFront and API Gateway)	Removed before storage. Kept 7 days, then deleted (OQ-29, ADR 0029)

Iceberg maintenance (compact, expire snapshots, remove orphan files) runs as the last task of the nightly run so it never competes with the pipelines.

9. Routine maintenance

Task	When
Review the open questions list and the issues for each one	Each time a spike answers a question
Check the cost against section 4	Monthly
Check the weekly smoke test	Weekly (Phase B)
Keep the Databricks account active	Monthly (Phase B)
Check the budget, the resource monitor and the model API spend limit	Monthly
Review the incident records and close runbook gaps	Monthly
Recheck the Marquez ARM image	Only when an upgrade is chosen (Phase B)

10. Access during operations

- John owns the system and is the only person with production access.
- Each pipeline runs under its own least-privilege role. Dev roles cannot write live data (platform step 0.2).
- Agents, from Phase B, have read-only access through an allowlist (design doc, section 12.1).
- Keys are read from Parameter Store only. Each read is logged through CloudTrail.

11. Open operational questions

Question	Where it is tracked
Row and runtime limits; storage alarm size; freshness thresholds	OQ-1, OQ-15
Whether the Databricks export can be automated	OQ-19
Whether the platform stack fits 32 GB of memory	Design doc, section 16

Part 5: Model card

`Will I have signal out there/model-card.md`

Project 1 model card

Field	Value
Status	Draft for review. **No model has been trained and no evaluation has been run.** Every result field is empty on purpose until the first run
Owner	John
Last updated	Oct 9, 2026
Phase	B (both components). Phase A has no AI
Related	design-doc.md (components C17 to C20), requirements.md (FR-34 to FR-48), security-and-privacy.md

This card describes the two AI components of Project 1 and what is known about each. It states what the components are for, what they must not be used for, how they are tested, and where they fail. It is updated after every training run and every evaluation.

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1. Components

	Coverage confidence model	Trip-planning assistant
What it does	Predicts how likely the FCC coverage claim is to be right at a trail point and carrier. It gives a confidence level only where enough speed tests exist nearby. Elsewhere it gives "not enough measurements" (ADR 0008)	Answers a plain-language trip question with a check-in plan: where each carrier works, when satellite windows open, and how confident the map is. It cites the trail points and times it used. It says so when the data cannot answer (ADR 0017)
Kind	Supervised classifier. Scikit-learn is the planned library. The model family is logistic regression (ADR 0031)	Large language model: the cheapest Claude model from Anthropic (ADR 0034), calling a fixed set of metric tools through an MCP server
Where it is trained	Databricks Free Edition, tracked in MLflow (ADR 0014)	Not trained. Prompts are versioned in git
Where it runs	Nightly run scores each point with the copy of the chosen model in the repository. The site never depends on Databricks	Lambda behind API Gateway, from Phase B
Version	Recorded in each row of `fct_coverage_confidence` as `model_version`	Recorded in each answer as `prompt_version`

2. Intended use

- Helping a hiker judge how far to trust the FCC coverage map at a trail point.
- Helping a hiker see when a Direct-to-Cell satellite could pass, and plan where to check in.
- Showing the confidence of each claim, and saying plainly when there are not enough measurements to judge.

3. Out of scope

- **Emergency communication.** Nothing here may be used to decide whether a call or text will get through in an emergency, or any decision where a missed message is a safety risk. The page wording that says so is open (OQ-22).
- **Any area outside the scope of the phase in use.** Phase A covers three Grand Canyon trails with one carrier. Its results do not describe other areas or carriers.
- **Later days than the prediction span.** The assistant answers only within the prediction span and says so (ADR 0021).

- **Commercial use.** The project is non-commercial (ADR 0013; Databricks Free Edition terms).

4. Data

4.1 Confidence model

Input	Source	Notes
FCC coverage claim per cell and carrier	FCC mobile coverage files, H3 resolution 9	The claim is a model's output, not a measurement
Ookla speed-test tiles	Ookla open data, quarterly, about 600 m tiles, GPS-accurate tests only	Speed tests exist only where phones already had signal. Dead zones are missing from the data
Terrain features	Copernicus GLO-30	A surface model, so trees and buildings count as terrain (ADR 0004)
Distance to the nearest tower	OpenCelliD tower locations (ADR 0024)	Share-alike licence
Remoteness	Distance to the nearest road or trailhead, from published data. Source: TIGER/Line roads and NPS trail lines (ADR 0035)	Definition recorded in ADR 0024
Labels	Hiker signal reports (Phase B), where at least N reports in one cell on one day agree (ADR 0008; N open, OQ-12)	Each report is from a different visitor, because of the one-per-cell rule. Reports arrive only from hikers who choose to tap

Label volume is limited by the report caps: 10 reports a day per visitor and 50 a day for the whole system (ADR 0011). Labels accrue slowly.

4.2 Assistant

The assistant reads only through the metric tools, and the metric tools read the published tables. It never reads raw source files or the DynamoDB reports table. Raw SQL is a read-only, allowlisted fallback, and each use is logged separately (ADR 0017).

5. Evaluation

5.1 Confidence model

Measure	Plan	Threshold
Agreement with held-out Ookla measurements, where enough tests exist	Compare each model version in MLflow on the same held-out split (method proposed; OQ-16)	Not set (OQ-16)
Coverage of the "not enough measurements" rule	Test on a sparse-area fixture (AC-1B.4, QR-17)	The page must show the notice and no confidence level there
Baseline	The FCC claim alone, with no confidence model. This is the version without AI, against which the model must show a gain (proposal, OQ-21)	Not set

The exact split, the metric, and the accuracy threshold are set by John after the first scoring run (OQ-14, OQ-16). Until then, no accuracy figure is claimed.

5.2 Assistant

Measure	Plan	Threshold
Correct, cited plans on sample questions	Ten sample trip questions, checked by hand (AC-1B.6)	Not set
Graded eval suite	30 or more questions with known answers, including ones the data cannot answer. Graded exactly on tool choice, numbers and refusal (FR-45)	CI fails below the threshold (OQ-16)

Measure	Plan	Threshold
Clarity	An LLM judge, a second model, scores clarity. It is calibrated on 50 answers John grades by hand (FR-46)	Agreement with John's grades is reported; threshold not set
Red team	20 or more prompts: instructions hidden in source text, write requests, off-topic requests, cost abuse (FR-47)	CI fails below the threshold (OQ-16)
Numbers	The harness fails when an answer states a number that is not in the query result (AC-1B.5, QR-18)	Zero unsupported numbers
Later days	Questions about days beyond the span, where the correct answer is the statement of the limit (ADR 0021)	Must refuse the later day and say so
A/B	Two prompt versions compared through votes in DynamoDB (FR-48)	Result reported; no winner is assumed

5.3 Results

No results exist yet. The results section is filled in when the first run happens. It records the date, the model version or prompt version, the eval scores, the agreement with John's grades, and the A/B result. Nothing is copied from another project.

6. Known failure cases

These are known in advance from the design. Each one is tested or shown on the page where possible.

Failure	Why it happens	What the system does
Overconfident in dead zones	Speed tests are missing where phones had no signal	No confidence level where tests are too few. The page shows the "not enough measurements" notice (ADR 0008)
Claims the FCC map is right when it is not	The FCC coverage is a modeled claim	The page says the coverage is a map claim. The wording is set in ADR 0037
Trees and buildings counted as terrain	The elevation model is a surface model (ADR 0004)	Stated as a limitation. The effect on horizons is measured later (OQ-24)
Biased labels	Only hikers who choose to report, and only where they have a signal to report	Labels only where N reports agree. Reports that contradict nearby reports are set aside (ADR 0012)
Few labels, slow learning	Report caps (10 a day per visitor, 50 a day for the system)	Retraining waits until there are enough labels (OQ-13)
Remoteness is a rough proxy	Distance to a road or trail line is not a measure of signal. The trail network stands in for trailheads	Listed as an input. Its value is checked in evaluation (ADR 0035)
Wrong numbers in an answer	A model can state a figure it did not read	The harness fails an answer with a number that is not in the query result (AC-1B.5)
Answers about a later day	Windows exist only for the span	Refused, with a statement of the limit (ADR 0021)
Instructions hidden in text change the answer	Source text and user questions are untrusted	Source text is data. Tools are read-only and allowlisted. Red team tests this (FR-47)
Answer when the data cannot answer	The model may try to be helpful	The answer says so. The eval suite tests refusal (FR-44)

7. Safety and privacy

- The confidence model sees only cells and aggregated speed tests. It never sees a visitor ID or exact coordinates.
- Reports carry no visitor ID, so a label cannot be linked to a visitor (ADR 0009, ADR 0011).
- The assistant is given no write tool (ADR 0017).
- Each model call is logged with its model, prompt version, tokens, latency and cost. The question text is not stored in the log (platform contract `llm_calls`). It is not stored (ADR 0033).

- The question text goes to the model provider. The provider's terms are recorded before the first call (OQ-39, ADR 0034).
- Spending is capped: $10 a month, a daily question cap and a token limit per answer (FR-42).

8. Updates and monitoring

- **Confidence model:** retrained only after the labels reach the level set in OQ-13. Each training run is tracked in MLflow. The chosen model and its results are copied to the repository by hand (ADR 0014). The nightly run uses the copy in the repository. A results file that carries the run ID must be in the repository after each run (FR-37).
- **Assistant:** the prompt version is recorded in every answer. Changes to prompts are made in git and evaluated in CI before they reach the site. The model is changed only when the evals justify the cost (FR-65, Phase C).
- **Monitoring:** calls and costs are in the nightly `llm_calls` table. Traces are in AWS X-Ray. Grafana reads X-Ray during work sessions (design doc, section 12.3).

9. Licences

- Ookla open data: CC BY-NC-SA 4.0. Confidence values derived from it are shared under the same licence, and the exact notice is shown (ADR 0013).
- OpenCelliD: share-alike. Confidence values that use tower distance carry the same licence (ADR 0024).
- Databricks Free Edition: non-commercial.
- Model provider: Anthropic. Terms to be recorded before the first call (OQ-39).

10. Open items

Item	Tracked in
Remoteness source	OQ-18
Enough nearby tests for a confidence level	OQ-14
Labels needed to retrain; the N that makes a label	OQ-13, OQ-12
Eval, judge and red-team thresholds	OQ-16
Assistant daily cap and token limit	OQ-17
Baseline for the assistant (the version without the model)	Proposed here; approve with the design doc (OQ-21)

Part 6: Security and privacy

Will I have signal out there/security-and-privacy.md

Project 1 security and privacy

Field	Value
Status	Draft for review. Residual risks are accepted by John when he approves this document
Owner	John
Last updated	Oct 9, 2026
Related	design-doc.md (sections 3.5, 12.1 and 12.2), ../Platform/platform-requirements.md (sections 3, 9), operations.md, model-card.md

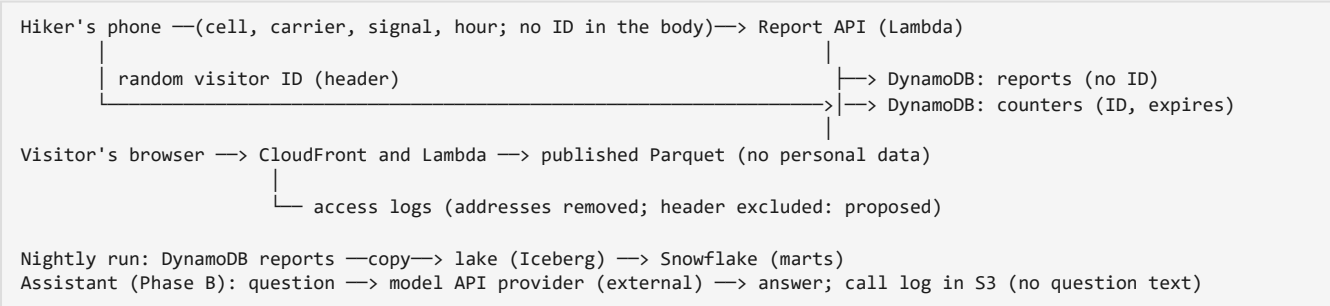
This document covers three things: what could go wrong (the threat model), what personal data the project holds and for how long (the inventory), and where each licensed dataset comes from and what it requires (the register). Each control named here is either a recorded decision (cited as an ADR), a rule in the design doc, or a proposal marked as one.

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1. Principles

1. **Exact locations never leave the phone.** A report carries only an H3 resolution 9 cell (about 350 m across), a carrier, a yes or no for signal, and the time rounded to the hour (ADR 0009).
2. **No identity from the network.** Limits use a random ID created on the phone. Internet addresses are not used for identity or limits (ADR 0011). They are removed from logs (ADR 0023).
3. **Data that is no longer needed is deleted on a fixed schedule.** Raw reports in DynamoDB are deleted 30 days after the nightly copy, and point-in-time recovery is off for that table so the deletion is final (ADR 0012, ADR 0022).
4. **Source text is data, never instructions.** Trail names, satellite names and user questions never change what the system does (design doc, section 12.1).
5. **Least privilege everywhere.** One role per pipeline, read-only roles for agents, and dev cannot write live data (platform step 0.2).
6. **Every key is in Parameter Store.** None is in code or GitHub (platform section 9).
7. **Nothing can run up a bill.** Every paid item has a hard limit (design doc, section 3.3).

2. What the system touches



3. Personal data inventory

Data	Where it lives	Who can read it	Retention	Basis
Exact GPS position	The hiker's phone, in memory, for a moment	The hiker's phone only	Discarded once the cell is computed. Never sent	ADR 0009

Data	Where it lives	Who can read it	Retention	Basis
Report: H3 cell, carrier, signal yes or no, hour rounded to the hour	DynamoDB `signal_reports` (Phase B)	The report API (write); the nightly copy (read)	Deleted 30 days after the nightly copy	ADR 0012, ADR 0022
Report copy in the lake and in Snowflake (Phase B)	Iceberg `signal_report`; Snowflake marts	Pipeline roles and Snowflake roles	Deleted 30 days after the copy; daily counts kept (OQ-43, ADR 0034)	Resolved, see section 8
Set-aside reports (Phase B)	Iceberg `signal_report_set_aside`	John and pipeline roles, for review	Deleted 30 days after the copy (OQ-43, ADR 0034)	Resolved, see section 8
Random visitor ID	The phone's local storage. Sent in the `X-Visitor-Id` header of `POST /reports`. Stored in `report_limit_counters` only	The report API; the counters table	Stored as an HMAC of the ID. Expires 72 hours after the UTC day (ADR 0025, ADR 0026)	ADR 0011
System-wide daily counter	DynamoDB counters table, with no visitor ID	The report API	Expires	ADR 0011
Internet addresses	CloudFront and API Gateway access logs	Platform logging only	Removed before storage. Kept 7 days (OQ-29, ADR 0029)	ADR 0023
Votes (thumbs up or down) (Phase B)	DynamoDB `feedback_votes`	The vote API (write); John	90 days, then deleted. No visitor ID (ADR 0033)	Build plan 1B.7
Assistant question text (Phase B)	Sent to the model API provider. Not stored in the call log	The provider, under its own terms	The provider's retention follows its own terms, recorded before the first call (OQ-39, ADR 0034)	Contract `llm_call_log`
Assistant call log (Phase B)	S3, then Iceberg `llm_calls`: model, prompt version, tokens, latency, cost	Pipeline roles; John	Not yet decided. The clean-up rules do not name this log (confirm before Phase B)	Build plan 1B.5
Email addresses	Not collected by this project	Not applicable	Not applicable	Design doc, section 3.5

What is deliberately not stored

- Exact coordinates, on the phone or on the server.
- A visitor ID in any report record or in any table with a report.
- Internet addresses for identity or limits.
- Email addresses.
- Question text in the call log (the contract lists the field set; any change needs a decision, OQ-39).

4. Threat model

Each threat has a control, a way the control is checked, and a residual risk. A residual risk is what remains after the controls. John reviews each one when he approves this document.

#	Threat	What it affects	Controls	How the control is checked	Residual risk
T1	Exact location is sent to the server	Hiker privacy	The phone computes the H3 cell and discards the coordinates. The payload holds only the cell, carrier, signal and hour (ADR 0009)	Request-capture test (AC-1B.8, QR-10)	Low. A cell about 350 m across still shows the area
T2	A report is linked to a hiker across reports	Hiker privacy	Reports carry no ID. Counters are apart from reports. Times are rounded to the hour. Raw rows are deleted 30 days after copy (ADR 0011, 0012)	DynamoDB inspection; expiry test (QR-11)	Medium. Two reports from the same small area in the same hour can point to one visit, although no ID links them

#	Threat	What it affects	Controls	How the control is checked	Residual risk
T3	A visitor ID is linked to reports through access logs	Hiker privacy	Counters hold the ID, not reports. Internet addresses are removed (ADR 0023). **Proposed:** access logs exclude the `X-Visitor-Id` header and the report body	Log configuration review. The proposed exclusion is not yet built	Medium until the proposed exclusion is built and reviewed
T4	A visitor clears storage to get a new ID and send more reports	Report integrity	Per-ID limits, plus the system-wide cap of 50 a day (ADR 0011). Set-aside step (ADR 0012)	Limit tests (AC-1B.8, section 13.1 of the design doc)	Medium. The system cap bounds the total, but one person can use up part of it
T5	Fake or flood reports distort labels and counts	Model and page accuracy	One report per cell per visitor every 24 hours. Labels only where N reports in one cell on one day agree (ADR 0008; N open, OQ-12). Set-aside. System cap	Fixtures in the nightly copy tests (C23); runbook RB-14	Medium. Low-volume fake reports within the limits remain possible
T6	A stale prediction is shown as current	Hiker safety	`valid_until_utc` and the late notice. The last good run time is shown. Old data is never hidden (ADR 0006)	Front-end test with a controlled clock (QR-3, AC-1A.6)	Low. A wrong device clock changes what the page thinks is late
T7	A key leaks (AWS, model API, Snowflake, Databricks)	All systems	Parameter Store only. OIDC for CI, so no AWS keys in GitHub. gitleaks and Trivy in CI. Least privilege	CI scans; role refusal test (QR-13)	Low to medium. A key pasted into a commit is exposed until it is rotated
T8	Instructions hidden in source text or in a user question change the assistant's behavior	Assistant integrity	Source text is data. Metric tools come from a fixed allowlist. Raw SQL is read-only, allowlisted and logged. No write tool. Structured output	Red-team set of 20 or more prompts in CI (FR-47, AC-1B.7)	Medium. No model is immune. Its effect is limited to what it can read and answer
T9	The assistant is used to run up model costs	Cost	$10 monthly spend limit set before the first call. Daily question cap. Token limit per answer. Cheapest model only (FR-42)	Limits set in the account; cost review (QR-9)	Low to medium. A burst inside one day can use up to the cap
T10	Dev code writes to live data	Live data	Separate buckets, Glue databases and roles. Dev roles cannot write live (platform step 0.2)	Refusal test in Terraform checks	Low
T11	A bad write to the reports table is lost	Report integrity	The lake copy is the only other copy. Point-in-time recovery is off by decision (ADR 0022)	None beyond the nightly copy	Accepted by John (ADR 0022). A bad write cannot be rolled back
T12	Licensed data is published without its required notice	Licence compliance	Exact Ookla notice on the page and in the README (AC-1B.1). Credits for every source. License register (section 6)	Page text test; pull request checklist (QR-12)	Low. Credit wording for CelesTrak, FCC, NPS, Copernicus and OpenCelliD is still to confirm (OQ-22)
T13	A source changes format and bad data is published	Data accuracy	Contract diff before loading (C8). Great Expectations gate (C9). Publish stops on failure (QR-4)	Tests on sample files in CI	Low
T14	A compromised image or library reaches production	All systems	Trivy scans. Image manifests are checked before first use. The ARM Marquez build is pinned to one version	CI scans	Medium. Dependencies change often

#	Threat	What it affects	Controls	How the control is checked	Residual risk
T15	Someone else gets into John's AWS account	All systems	Least privilege. Budget action. Session limit and idle shutdown limit exposure	**Not recorded:** whether MFA is on for the root and admin accounts. John confirms this before the first deploy	Not assessed until confirmed
T16	Small-area counts reveal one person's report	Hiker privacy	Counts are per area. Area is H3 resolution 6; minimum 5 reports (ADR 0030)	Review of the area definition	Medium until a minimum count is set (OQ-35)

5. Access

Who or what	Can read	Can write	Notes
John	All `dev` data; `live` data through the console and Terraform	Terraform applies after approval; Parameter Store	Account access uses MFA (to confirm, T15)
CI (GitHub Actions, through OIDC)	Terraform state; sample files in CI	Plan on pull requests; apply after approval; dev deploys	Stores no AWS keys
Each pipeline role	Its own raw source; its tables in its environment	Its own raw source; its tables in its environment	One role per pipeline. Dev roles cannot write live
Serving Lambda	The published Parquet prefix only	Nothing	Reads only what the page needs
Report Lambda (Phase B)	Nothing in the lake	`signal_reports` and `report_limit_counters`	No read of the lake
Nightly copy role (Phase B)	`signal_reports`	The lake; the expiry attribute on copied rows	Sets expiry only after a successful copy
Assistant Lambda (Phase B)	Published tables; metric tools on the allowlist; read-only SQL on allowlisted views	The call log in S3	No write tool (ADR 0017)
Vote endpoint (Phase B)	Nothing	`feedback_votes`	Runs behind API Gateway like the other endpoints
Hiker's phone	Its own queue	Sends reports to the report API only	Holds the visitor ID and the queue

Exact policies are written in Terraform, in the platform repository. This table is the requirement they must meet.

6. Licence and attribution register

Source	Terms	What is published from it	What must appear	Status
Ookla open data	CC BY-NC-SA 4.0. Non-commercial. Derived data is shared under the same licence	Speed tiles, confidence values (Phase B)	The exact notice: "Includes data derived from Speedtest by Ookla® open data, licensed under CC BY-NC-SA 4.0. Derived data published here is shared under the same license." On the page and in the README (ADR 0013, AC-1B.1)	Required from Phase B
OpenCelliD	Share-alike, CC BY-SA 4.0 according to secondary sources. Primary page not readable (HTTP 403)	Confidence values that use tower distance (ADR 0024)	Cell tower locations: OpenCelliD (opencellid.org), licensed CC BY-SA 4.0	Unverified: read the OpenCelliD terms before publishing (ADR 0037)

Source	Terms	What is published from it	What must appear	Status
Copernicus GLO-30 elevation	Free of charge under the Copernicus DEM licence (read from the licence text)	Horizons (derived, so the modified-data notice applies)	"produced using Copernicus WorldDEM-30 © DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved." Plus the liability statement when distributed	Verified (ADR 0037)
TIGER/Line roads (U.S. Census Bureau)	Declared CC0 (public domain) on data.gov. Some files carry a rights flag, "otherRestrictions"; check the Census page for the exact vintage before the first load	Derived distances only; road geometry is kept only if the check allows it	Credit the Census Bureau on the page. Check the flag (first-run check, ADR 0035)	Phase B
NPS Public Trails	Public domain under NPS policy; credit appreciated, not required	Trail lines, trail points, horizons	Trail data: National Park Service	Verified (ADR 0037)
CelesTrak	No terms found; courtesy credit	Satellite windows (derived, not raw sets)	Orbital data: CelesTrak (celestrak.org), maintained by Dr. T.S. Kelso	Courtesy credit. Ask the maintainer before republishing raw sets (ADR 0037)
FCC mobile coverage	Courtesy credit per secondary sources; carrier submissions may carry restrictions. Primary page not readable (HTTP 403)	Coverage flags, confidence values	Coverage data: FCC National Broadband Map (broadbandmap.fcc.gov)	Unverified: check before Phase B publishes carrier data (ADR 0037)
Databricks Free Edition	Service terms, non-commercial	Nothing published. Model copy in the repo is derived from training	Licensing is checked before any charge (design doc, section 3.4)	Phase B
Model API provider	Anthropic API (ADR 0034). Terms to record before the first call	Nothing published. Questions are sent to it	Recorded before the first call (OQ-39)	Phase B; open
Open-source libraries (for example h3-js, FastAPI, DuckDB)	Their own licences	Code	Licence text recorded in the repository when each library is added	Ongoing

The project is non-commercial. Licensing is checked before charging for any of it (design doc, section 3.4).

7. Logs and retention

Log or record	Kept	Deleted	Notes
Pipeline logs	90 days	Yes, after 90 days	Clean-up rule
Security logs (CloudTrail, Parameter Store reads)	1 year	Yes, after 1 year	Matches the PCI convention noted in the Build Plan Decisions
Access logs (CloudFront, API Gateway)	7 days	Addresses are removed before storage (ADR 0023)	Deleted after 7 days (ADR 0029)
Report rows in DynamoDB	Until 30 days after copy	Yes	ADR 0012

Log or record	Kept	Deleted	Notes
Report copy in the lake and Snowflake	30 days after the copy	Yes	ADR 0034
Counters	72 hours after the UTC day	Yes	ADR 0026
X-Ray traces	Per the X-Ray setting	Per the X-Ray setting	Traces hold agent runs. Their retention is not recorded; confirm before Phase B

8. Gaps this review found

These are places where the documents did not yet protect a privacy promise. Each is recorded so that John decides it.

1. **Resolved (ADR 0034): the lake and Snowflake copies of reports now follow the 30-day rule.** Before this decision they had no rule. The 30-day rule covers only the DynamoDB rows. Without a rule, the copies in the lake and Snowflake would be kept under the general "raw and published data kept" rule. Decision: OQ-43.
2. **Access logs could record the visitor ID.** The visitor ID travels in a header of `POST /reports`. Without an exclusion, an access log could link it to a report. Proposed control: exclude the header and the report body from logs (design doc, section 12.2). Decision: approve with the design doc (OQ-21).
3. **Assistant questions go to an outside provider.** A question can contain personal detail, such as who is hiking with the visitor. The call log does not store question text, but the provider receives it, and its terms are not recorded. Decision: OQ-39.
4. **Small counts can identify one person.** A count of one report for an area and hour can show one person's visit. Decided: counts are shown only from 5 reports, in an H3 resolution 6 area (ADR 0030).
5. **MFA is not recorded for John's AWS accounts.** Confirm before the first deploy (T15).

9. Incidents involving privacy

Privacy incidents follow the process in operations.md, section 5. The response is:

1. Stop the copy or the publish that causes it.
2. Delete the affected rows or files. Deletion of DynamoDB rows is final, so confirm the scope before deleting.
3. Record what was exposed, for how long, and to whom.
4. Change the control, or record why the risk is accepted, before the next run.

10. Open items

Item	Where it is tracked
Model provider's terms and question retention	OQ-39
Verify OpenCelliD terms, FCC mobile-data terms and the Census rights flag before publishing	ADR 0037
Approval of the log exclusion for the visitor ID header	OQ-21 (with the design doc)
MFA on John's AWS accounts	Confirm before the first deploy
Retention of X-Ray traces	Confirm before Phase B

Part 7: Architecture decision records

`Will I have signal out there/adrs/0001-phase-a-scope-grand-canyon.md`

0001. Phase A scope is Grand Canyon main trails, one carrier, the next 24 hours

- Status: Accepted (John, Build Plan Decisions, Oct 9, 2026)
- Date: Oct 9, 2026
- Source: Build Plan Decisions, Project 1 table, "Phase A scope"

Context

Phase A (the first build stage, "live") is the smallest version that works end to end, on a small area or sample, using the same technology as the full version, and it goes public as soon as it works. Project 1 needs terrain that blocks the sky strongly enough that the satellite-blocking effect is obvious.

Decision

Phase A covers Grand Canyon National Park's main trails (Bright Angel, South Kaibab and North Kaibab, ADR 0020), one carrier (T-Mobile, ADR 0020), and the next 24 hours of satellite passes (26 hours are predicted so there is no daily gap, ADR 0019). Reason recorded by John: canyon walls make the satellite-blocking effect obvious.

Alternatives considered

- Wider scope (every carrier, other parks, longer prediction span): recorded as later phases. Every carrier and measured speeds are Phase B (1B.1); every national park is Phase C.
- No other alternatives are recorded.

Consequences

- One carrier means a single coverage flag per trail point in Phase A.
- The span limits what the page and the Phase B assistant can say about later days (ADR 0021).
- Carrier and trails are decided in ADR 0020; FCC technologies are decided in ADR 0031 (OQ-11).

`Will I have signal out there/adrs/0002-select-dtc-satellites-by-name-tag.md`

0002. Select Direct-to-Cell satellites by the [DTC] name tag, cross-checked by altitude and inclination

- Status: Accepted (John, Build Plan Decisions, Oct 9, 2026)
- Date: Oct 9, 2026
- Source: Build Plan Decisions "Satellite selection"; build plan 1A.1 and "Decisions and open items"

Context

The page needs to know which Starlink satellites can send Direct-to-Cell texts. CelesTrak (a public source of satellite orbit data) lists the Starlink satellites, and its list carries a [DTC] tag in satellite names. John confirmed the tag exists.

Decision

Select satellites by the [DTC] tag in their names. Cross-check each selected satellite against altitude (about 330 to 370 km) and inclination (43 or 53 degrees). Any disagreement is logged and listed. The selection is unit-tested.

Alternatives considered

- Altitude and inclination alone: kept only as the cross-check, not as the selector.
- No other alternatives are recorded.

Consequences

- If CelesTrak renames or drops the tag, the cross-check log shows disagreements, so the selection does not change silently.
- The weekly source smoke test (ADR 0018, Phase B) and the contract diff catch format changes.
- Whether the tag-based selection agrees with the cross-check on a real file is a first-run check (OQ-6).
- "About 330 to 370 km" has no tighter tolerance in the sources. The developer sets the exact comparison rule when implementing the check and records it in the unit test.

`Will I have signal out there/adrs/0003-reachability-rule-and-minimum-elevation.md`

0003. A satellite is reachable only above the minimum elevation and the terrain horizon; start at 10 degrees

- Status: Accepted as the starting value (build plan "Decide in 1A.3"); the 5 and 20 degree comparison is pending
- Date: Oct 9, 2026
- Source: Build plan 1A.3 and "Decisions and open items" ("Satellite elevation rule (Project 1)")

Context

Canyon walls and mountains block the sky. A satellite counts as reachable only if it is high enough in the sky and clear of the terrain in its direction.

Decision

A satellite counts as reachable from a trail point only when it is above both 10 degrees and the terrain horizon in its direction. 10 degrees is the starting minimum elevation. The design note also records how the satellite windows change at 5 and 20 degrees.

Alternatives considered

- Minimum elevation of 5 degrees and of 20 degrees: compared in the design note after the first satellite window run (OQ-9). The choice stays at 10 unless that comparison and John's decision change it.

Consequences

- `find_windows(positions, horizon, min_elev)` takes the minimum elevation as a parameter so the comparison needs no code change.
- The satellite window run records `min_elev` so published rows say which value produced them.
- Windows are accurate to the 1-minute propagation step.
- The prediction span is 26 hours (ADR 0019).
- The horizon comes from a surface model that includes vegetation (ADR 0004), so the horizon is conservative where trees are tall (effect to be measured, OQ-24).

`Will I have signal out there/adrs/0004-terrain-horizon-from-copernicus-glo30.md`

0004. Compute the terrain horizon from Copernicus GLO-30 at trail points sampled every 50 meters

- Status: Accepted (build plan 1A.2; trail and terrain data "Answered")

- Date: Oct 9, 2026
- Source: Build plan 1A.2 and "Decisions and open items"; Documentation Plan "Answered by research"

Context

The satellite windows depend on how much sky the terrain blocks at each trail point. The build plan fixes the inputs and the procedure; the data formats were answered by research.

Decision

- Trail lines come from the National Park Service (NPS) Public Trails ArcGIS service for Grand Canyon, limited in Phase A to Bright Angel, South Kaibab and North Kaibab (ADR 0020) (polylines, GeoJSON output, 2,000 records per request, trail name in `TRLNAME`, park in `UNITCODE`).
- A point is sampled every 50 meters.
- Elevation comes from the Copernicus 30-meter model on AWS (GLO-30: heights on the EGM2008 datum, a surface model that includes vegetation and buildings, water flattened; free under the Copernicus licence, with credit).
- Each point's horizon is how high the terrain rises in each compass direction out to about 15 km. Spark (the batch data-processing engine) computes it for many points at the same time and stores it in Iceberg (a table format on S3).
- The horizon is stored as 36 horizon angles per point (the figure used in the Documentation Plan's example for `compute_horizon`).

Alternatives considered

- No other elevation source or sampling spacing is recorded.

Consequences

- Because GLO-30 is a surface model (it measures the top of trees and buildings, not bare ground), trees and buildings count as terrain, so horizons may be higher than rock alone.
- Public archives that never change store a manifest of URL, size and hash instead of a full copy (engineering standard 1: keep raw files unchanged, or a manifest for public archives that never change).
- The exact range cutoff, ray step, Earth curvature and refraction handling, observer height, sector convention and the numeric margin for "much higher" are open (OQ-24).
- 36 sectors is taken from the Documentation Plan's example; it is final only after John's review (OQ-24).
- The number of trail points is unknown until the NPS file is loaded; it drives the satellite window row count (OQ-1).

`Will I have signal out there/adrs/0005-store-windows-as-intervals.md`

0005. Store satellite windows as start and end times, never one row per minute

- Status: Accepted (build plan "Hard limits on everything that costs money")
- Date: Oct 9, 2026
- Source: Build plan "Hard limits" table, "Satellite windows (Project 1)"; Decisions "Surprise costs"

Context

Propagation runs at 1-minute steps for the 26-hour prediction span (ADR 0019). Storing one row per trail point, satellite and minute would multiply the row count, and nothing in the project is allowed to surprise with cost.

Decision

Windows are stored as start and end times. A size estimate goes in the design note before step 1A.3 and the row limit is set from it. A job that would exceed its row limit or runtime limit stops before publishing and opens an incident (a recorded failure with its cause and fix).

Alternatives considered

- One row per minute: rejected by the build plan as the cost and size risk.

Consequences

- `flags_to_windows` collapses consecutive reachable minutes into intervals.
- The row limit value cannot be set before the test run on the session box (OQ-1, OQ-15).
- The satellite window table partition is the run's logical date so rerunning a day replaces only that day (ADR 0006).

```
Will I have signal out there/adrs/0006-predictions-valid-until-and-late-notice.md
```

0006. Stamp predictions with a valid until time and show late data with a visible late notice

- Status: Accepted, amended (Decided by John, Oct 9, 2026). The original decision to hide expired satellite windows is superseded.
- Date: Oct 9, 2026
- Source: Build Plan Decisions "Stale predictions" (superseded); Decided by John, Oct 9, 2026 (all five projects show old data with a visible late notice instead of hiding it); build plan 1A.3, 1A.6; engineering standards 3 (idempotent tasks that use the run's data interval) and 15 (timestamps in UTC)

Context

The session box runs only for the nightly run. If a nightly run fails or is skipped (spot unavailable), the last published satellite windows stop being current while the page is still served. The public page works with the box off. The first version of this decision hid expired satellite windows. John then decided that every project shows old data with a visible late notice and does not hide it.

Decision

- Each satellite window run is stamped with the time its predictions stop being valid (the end of the prediction span, ADR 0019).
- The page shows "Predictions valid until [time]" in the hiker's local time.
- When the data is late or the run failed, the page still shows the old data, with a visible late notice that includes the time of the last good run. It does not hide it.
- Timestamps are stored in UTC and converted to local time only on the page.
- Rerunning a day replaces only that day: the satellite window table is partitioned by the run's logical date and a rerun overwrites only that partition.

Alternatives considered

- Hiding expired satellite windows (the original decision): superseded by the late notice rule.
- No other alternatives are recorded.

Consequences

- The late notice must be unmistakable so old satellite windows are not read as current. The exact definition of "late", the notice wording, and how individual satellite windows already in the past are shown are open (OQ-23).
- The page needs the browser clock to compare with the valid until time.

- The published data must carry the time of the last good run (column `run_started_utc`, field `last_good_run_utc`).
- Supersedes the "expired satellite windows disappear on their own" part of AC-1A.6; see requirements.md.

`Will I have signal out there/adrs/0007-h3-resolutions-and-joins.md`

0007. Use H3 resolution 9 for FCC files and hiker reports, 8 for Ookla tiles, joined with cell_to_parent

- Status: Accepted (engineering standard 9: pick the H3 resolution from the source; Decisions "Report location")
- Date: Oct 9, 2026
- Source: Build plan engineering standard 9; Build Plan Decisions "Report location"; build plan 1A.4 and 1B.1

Context

Sources come at different grains: FCC coverage files include H3 resolution 9 hexagons, Ookla tiles are about 600 meters, and trail points are sampled every 50 meters.

Decision

H3 is a grid of hexagon cells covering the Earth; a higher resolution number means smaller cells. Pick the H3 resolution from the source (the cell just larger than the pixel) and join across sources with cell_to_parent (the H3 function that finds the larger cell containing a smaller one): resolution 9 for FCC files and hiker reports, resolution 8 for 15 arc-second rasters and Ookla tiles, resolution 4 for 0.25 degree weather grids (the last is not used by Project 1). Each trail point is assigned an H3 resolution 9 cell to receive the coverage flag.

Reason recorded by John: trail-spot lists do not scale; H3 works anywhere and matches the FCC grid.

Alternatives considered

- Lists of trail spots: rejected, they do not scale.
- No other alternatives are recorded.

Consequences

- Using the source's own cell size avoids false precision and mismatched joins.
- Phase C (every national park, GPX routes) needs no new grid.
- The rule for converting an Ookla tile (about 600 meters) to an H3 resolution 8 cell is a detail the developer settles in the loader. The Ookla file's actual fields are open (OQ-20).

`Will I have signal out there/adrs/0008-coverage-confidence-only-where-tests-exist.md`

0008. Give a coverage confidence only where enough tests exist; use hiker reports as labels

- Status: Accepted (John, Build Plan Decisions, Oct 9, 2026); amended Oct 9, 2026 (Decided by John, Oct 9, 2026)
- Date: Oct 9, 2026
- Source: Build Plan Decisions "Coverage confidence"; build plan 1B.4 and 1B.8; "Decisions and open items"

Context

The FCC map is a model's claim. Ookla speed tests measure what phones actually got, but speed tests only exist where phones already had signal, so dead zones are missing from the data.

Decision

- A model predicts how likely the FCC map is to be right, using terrain, distance to the nearest tower and remoteness.
- It gives a confidence level only where enough tests exist nearby. Elsewhere the page says: "Not enough measurements here; the coverage map's claim is unverified."
- Hiker signal reports become labels over time, including the missing no-signal cases, only where enough different visitors agree. Amendment (Decided by John, Oct 9, 2026): "enough agree" means N reports in the same cell on the same day. Because a visitor can report once per cell per 24 hours (ADR 0011), each report in a cell is from a different visitor, so reports need no ID to be counted. N stays an open number. The model retrains once there are enough labels.
- Model inputs (amendment, Decided by John, Oct 9, 2026): terrain, distance to the nearest tower from OpenCelliD, and remoteness defined as distance to the nearest road or trailhead (ADR 0024).

Reason recorded by John: speed tests miss dead zones, so the model must not guess there.

Alternatives considered

- Giving a confidence level everywhere: rejected, it would guess in dead zones.
- No other alternatives are recorded.

Consequences

- The page has two states for a point: a confidence level or the notice.
- Open numbers: tests that count as enough and the nearby distance (OQ-14), N reports in a cell per day (OQ-12), labels needed to retrain (OQ-13).
- The model family is logistic regression (ADR 0031). The remoteness data source is still open (OQ-18).
- The day boundary for "same day" is UTC midnight (ADR 0026, OQ-26). The system accepts at most 50 reports a day (ADR 0011), which caps how fast labels can accrue.

`Will I have signal out there/adrs/0009-report-location-reduced-to-h3-cell-on-phone.md`

0009. Reduce a report's location to an H3 cell on the phone; exact coordinates never leave it

- Status: Accepted (John, Build Plan Decisions, Oct 9, 2026); phase confirmed as B (Decided by John, Oct 9, 2026)
- Date: Oct 9, 2026
- Source: Build Plan Decisions "Report location" (the label "(A)" there is a typo; Decided by John, Oct 9, 2026: the feature is step 1B.8, Phase B); build plan 1B.8; "Personal data"

Context

Hikers report where they did or did not have signal. Exact positions are personal data and add nothing the coverage grain needs.

Decision

When a hiker taps a report button, the phone reads its GPS, converts it to an H3 resolution 9 cell (the FCC coverage grain, about 0.1 km2, about 350 m across) with the h3-js library and discards the exact coordinates on the phone. The report holds only the cell, carrier, signal yes or no and the time rounded to the hour.

Reason recorded by John: trail-spot lists do not scale; H3 works anywhere and matches the FCC grid.

Alternatives considered

- Sending exact coordinates: not chosen; it conflicts with the personal-data decision.
- Trail-spot lists: rejected, they do not scale.

Consequences

- The server cannot recover exact positions; verified by a request-capture test (AC-1B.8).
- Reports and FCC cells join directly at resolution 9.
- The cell is still a location; the 30-day expiry (ADR 0012) and the absence of an ID (ADR 0011) limit what it reveals.

```
Will I have signal out there/adrs/0010-offline-report-queue.md
```

0010. Save reports on the phone and send them when signal returns

- Status: Accepted (John, Build Plan Decisions, Oct 9, 2026)
- Date: Oct 9, 2026
- Source: Build Plan Decisions "Offline reports"; build plan 1B.8

Context

The hikers with no signal are the ones who need to report it.

Decision

A report is saved on the phone and sent automatically when signal returns: in the background on Android; on iPhone, Safari sends it the next time the page is opened, with a "1 report waiting to send" note.

Alternatives considered

- Requiring signal at the moment of the report: rejected, since those hikers are the ones who need to report.
- No other alternatives are recorded.

Consequences

- The phone keeps each report in a local queue (storage that survives closing the page) until a send succeeds.
- On iPhone, reports can wait until the page is opened again.
- A burst of queued reports arriving at once is judged by the hour each report was made (ADR 0011); remaining details are in OQ-26.
- Verified by the airplane-mode test in AC-1B.8.

```
Will I have signal out there/adrs/0011-anonymous-visitor-id-and-report-limits.md
```

0011. Use a random ID created on the phone for report limits; no internet addresses

- Status: Accepted (John, Build Plan Decisions, Oct 9, 2026); limits amended Oct 9, 2026 (Decided by John, Oct 9, 2026)
- Date: Oct 9, 2026
- Source: Build Plan Decisions "Visitor identity" and "Report limits"; build plan 1B.8; "Hiker report limits (Project 1)"

Context

Reports over time must be limited to stop floods and fake reports, without identifying the hiker.

Decision

- Visitor identity is a random ID created on the phone only. No internet addresses are used (reason recorded: the most private option).

- Limits (amended, Decided by John, Oct 9, 2026): 10 reports a day per visitor (replaces 10 an hour and 50 a day per visitor), one per cell per visitor every 24 hours, and at most 50 reports a day accepted by the whole system, plus filtering. John raises the caps as traffic grows. Reason recorded earlier: a limit on reports over time stops floods and fake reports.
- Limits are applied by the hour the report was made on the phone (the report's rounded hour), not by arrival time, so a batch sent after being offline is not refused for arriving together (Decided by John, Oct 9, 2026).
- Limit counters are stored apart from the reports and expire, so reports never carry an ID that links them. The build plan sets expiry at 24 hours; the offline delay and the day boundary are decided in ADR 0026 (OQ-26).

Alternatives considered

- Limits by internet address: not chosen, because John picked the most private option.
- No other alternatives are recorded.

Consequences

- A visitor can reset their identity by clearing phone storage; filtering and the set-aside step (ADR 0012) cover that.
- Counting for label confirmation uses the one-per-cell rule: N reports in one cell on one day come from N visitors (ADR 0008).
- A phone could state a false report hour. The sub-parts are decided in ADR 0026: UTC day, reports up to 48 hours old, counters kept 72 hours after their day (OQ-26).
- A system-wide counter (no visitor ID) enforces the 50 a day cap.
- Internet addresses are removed from logs (ADR 0023).
- The ID format and transport are decided in ADR 0025 (OQ-38).

`Will I have signal out there/adrs/0012-report-storage-retention-and-labels.md`

0012. Store reports in DynamoDB, delete raw rows 30 days after the nightly copy, set aside impossible patterns

- Status: Accepted (John, Build Plan Decisions, Oct 9, 2026); recovery amended Oct 9, 2026 (Decided by John, Oct 9, 2026)
- Date: Oct 9, 2026
- Source: Build Plan Decisions "Visitor data", "DynamoDB", "Email addresses"; build plan 1B.8 and "Personal data"; engineering standard 16 (round-the-clock Lambda writes only to always-on storage)

Context

Reports arrive at any hour while the session box is off. Personal data should stay in one place and not accumulate.

Decision

- A Lambda function (AWS code that runs on demand) writes each report to DynamoDB, which is always on and free at this traffic (reason recorded: earlier options did not handle visitors writing at any hour). Point-in-time recovery is OFF for the hiker reports table (amended, ADR 0022), so deletion is final at 30 days. Other tables keep it on.
- The nightly run copies reports into the data lake (S3 storage with Iceberg tables) and into Snowflake. Raw report rows delete themselves 30 days after that copy.
- The copy sets aside impossible patterns (reports outside any mapped area, bursts that contradict nearby reports) for review.
- Report counts per area are shown on the page.

Alternatives considered

- PostgreSQL on the box (off most of the time): not usable for writes at any hour.

- No other alternatives are recorded for this project.

Consequences

- Lambda functions write only to always-on storage (engineering standard 16).
- The expiry is set by the nightly copy, so a failed copy never loses a row before it is copied.
- A bad write to the reports table cannot be undone; the copy in the data lake is the only other copy (ADR 0022).
- Definition of "area" for the counts: OQ-35.
- Where set-aside reports are stored is a design proposal (table `signal_report_set_aside`, OQ-21).
- The 30-day rule applies to the DynamoDB rows only. The lake and Snowflake copies (and the set-aside rows) follow the same 30-day rule, and the lake and Snowflake keep daily per-cell counts (OQ-43, ADR 0034).

`Will I have signal out there/adrs/0014-model-training-on-databricks-with-repo-copy.md`

0014. Train the confidence model in Databricks Free Edition and score with a copy kept in the repo

- Status: Accepted (John, Build Plan Decisions, Oct 9, 2026); export amended Oct 9, 2026 (Decided by John, Oct 9, 2026)
- Date: Oct 9, 2026
- Source: Build Plan Decisions "ML training"; build plan 1B.4 and "Decisions and open items" ("Databricks")

Context

Databricks Free Edition gives free managed MLflow and a model registry, but it is serverless only, has a daily fair-use quota, limited outbound internet, and is non-commercial.

Decision

- John uploads training data to Databricks through its Files API, and MLflow (a tool that tracks training runs) tracks every model version.
- After each training run the results and the chosen model are exported manually (downloaded from Databricks and committed to GitHub). Amended, Decided by John, Oct 9, 2026; this replaces the automatic copy step. An open question stays to test whether automation is possible (OQ-19).
- A small monthly job keeps the free account active.
- The pipeline always scores with the copy in the repo, so the site never depends on Databricks being up.

Reason recorded by John: it shows Databricks while the site never depends on it.

Alternatives considered

- Self-hosted MLflow: removed as a duplicate of the free managed MLflow.
- No other alternatives are recorded.

Consequences

- The upload is an untested assumption (OQ-3). Whether the export can be automated given limited outbound internet is open (OQ-19).
- The model artifact lives in git; its size is unknown until the model family is chosen (OQ-18).
- Databricks Free Edition is non-commercial.

0015. Run Project 1 marts on Snowflake over Glue, with SCD Type 2 coverage history and a refresh step

- Status: Accepted (John, Build Plan Decisions, Oct 9, 2026)
- Date: Oct 9, 2026
- Source: Build Plan Decisions "Warehouse" and "Snowflake"; build plan 1B.3; engineering standard 8 (keep business history in rows, not snapshots)

Context

Snowflake is the one data warehouse. It reads Iceberg tables in place through the AWS Glue catalog (a registry of tables), so no data is copied. FCC coverage changes between releases and business history must live in rows, not in Iceberg snapshots (snapshot expiry deletes old versions).

Decision

- Snowflake reads Iceberg through the Glue catalog; a refresh step runs at the end of each pipeline run so Snowflake sees new data.
- The dbt marts (finished tables built by dbt) run on Snowflake, with SCD Type 2 rows (history kept as rows with validity ranges) for carrier coverage changes between FCC releases.
- Metrics are defined in MetricFlow; one metric must give the same value from Snowflake and from DuckDB.
- A resource monitor suspends the warehouse at 3 credits a month (about $6 after the trial) and every query has a time limit. The 30-day trial with $400 of credits starts in Phase B step 1B.3.

Alternatives considered

- BigQuery as a warehouse: removed as a duplicate.
- No other alternatives are recorded.

Consequences

- If the Glue connection or refresh does not work as expected, the warehouse stays stale; the first-run check is OQ-4. Snowflake's docs list no edition requirement for the Glue connection; check in 1B.3.
- The site never reads Snowflake; it reads published Parquet through DuckDB.
- SCD Type 2 history depends on FCC release versions being identifiable (OQ-20).

0016. Publish orbit updates through Kafka and process them with an availableNow Structured Streaming job

- Status: Accepted (build plan 1B.2)
- Date: Oct 9, 2026
- Source: Build plan 1B.2; "When each tool is introduced"; engineering standards 3 (idempotent tasks) and 16 (round-the-clock Lambda writes only to always-on storage)

Context

Kafka (a message queue) runs only while the session box (the rented AWS server) is on. Project 1 introduces Kafka and Structured Streaming in Phase B at a low rate; Project 3 later uses them for a live stream.

Decision

The nightly run publishes each new element set to Kafka. A Structured Streaming job, run with Spark's availableNow trigger (process everything new since the last run, then stop), finds changed satellites and triggers re-propagation, with an explicit schema, checkpoints on S3, a watermark and dropDuplicatesWithinWatermark. The design note says plainly that this stream is low-rate.

Alternatives considered

- Flink: removed as a duplicate of Spark Structured Streaming.
- No other alternatives are recorded.

Consequences

- Replaying a day of updates must write the same rows (AC-1B.2).
- Checkpoints are temporary files removed after 7 days by the clean-up rules.
- Watermark value, topic and the definition of "changed" are decided in ADR 0028 (OQ-30).

`Will I have signal out there/adrs/0017-trip-assistant-metrics-tools-first.md`

0017. The trip-planning assistant answers through defined metric tools first, with logged raw SQL as a fallback

- Status: Accepted (John, Build Plan Decisions, Oct 9, 2026)
- Date: Oct 9, 2026
- Source: Build Plan Decisions "Agent queries", "Tracing", "Model", "AI spending", "AI placement"; build plan 1B.5 to 1B.7; engineering standard 13 (least privilege for agents)

Context

The assistant answers plain-language trip questions with a check-in plan. Wrong numbers are the main failure; agents that touch real systems are the main safety risk.

Decision

- An MCP server (Model Context Protocol, a standard tool interface for AI models) exposes the metrics as tools; raw SQL is a read-only, allowlisted fallback that is logged separately (reason recorded: fewer wrong numbers).
- Every model call is logged to S3 (model, prompt version, tokens, latency, cost) and every agent run is traced into AWS X-Ray, which is always on because agents run on Lambda at any hour.
- Prompts are versioned in git; every call uses structured output; only the cheapest model is used for now, behind a $10 monthly spend limit, a daily question cap and a token limit per answer (reason recorded: keeps cost low until the evals justify more). A stronger model is compared in Phase C.
- Amendment (Decided by John, Oct 9, 2026): the assistant answers only within the prediction span and says so for later days (ADR 0021).
- The assistant cites the trail points and times it used and says so when the data cannot answer.
- The harness fails an answer that states a number that is not in the query result.
- Source text is treated as data, never as instructions; the assistant has no write tool.

Alternatives considered

- Raw SQL first: not chosen; it risks more wrong numbers.
- A cross-project orchestrator: cut, because it was AI for show (platform-level cut; recorded for completeness).

Consequences

- Eval thresholds, the daily cap and the token limit are open (OQ-16, OQ-17).

- The assistant is limited to the prediction span (ADR 0021).
- Model and provider: the Anthropic API, cheapest Claude model (ADR 0034; OQ-39).

`Will I have signal out there/adrs/0018-contracts-and-weekly-source-check.md`

0018. Data contracts for every source and published table, plus a weekly test pull from every source

- Status: Accepted (John, Build Plan Decisions, Oct 9, 2026; contracts in Phase A, weekly pull from step 1B.9)
- Date: Oct 9, 2026
- Source: Build Plan Decisions "Source changes" and "Extras moved later"; build plan 1A.5, 1B.9; engineering standard 5 (a contract per source, schema diffed before loading); Documentation Plan "Interface specs"

Context

Upstream sources change format without notice. A change should fail loudly and early, before the nightly run fails.

Decision

- Each source has a contract with schema, rules, freshness target, owner and version (engineering standard 5), written in the Open Data Contract Standard (ODCS, a YAML format for data contracts). The schema is diffed before loading.
- Contracts exist from Phase A for CelesTrak, FCC, NPS trails and elevation.
- A weekly smoke test pulls one small sample from every source the project uses and alerts on any format change (step 1B.9).

Reason recorded by John: it catches format changes before the nightly run fails. The weekly test was moved to step 1B.9 so the first live page goes up sooner.

Alternatives considered

- None recorded.

Consequences

- Contract files live in `contracts/` and are checked in CI against sample files.
- Until the first-run checks run, source URLs and field lists are drafts (OQ-20).
- The ODCS version is pinned in ADR 0032 (OQ-32).

`Will I have signal out there/adrs/0024-opencellid-tower-source-and-remoteness-definition.md`

0024. Use OpenCelliD tower locations and define remoteness as distance to the nearest road or trailhead

- Status: Accepted (Decided by John, Oct 9, 2026); the remoteness data source is still to be chosen
- Date: Oct 9, 2026
- Source: Decided by John, Oct 9, 2026, to give the confidence model a tower-distance input and a definition of remoteness
- Amends: ADR 0008

Context

The confidence model uses distance to the nearest tower and remoteness, and the plan named no source for either.

Decision

- Tower locations come from OpenCelliD (an open database of cell tower locations), to compute the distance to the nearest tower.
- OpenCelliD is share-alike, so derived data carries the same license. This is recorded in the license register and on the page.
- Remoteness is the distance to the nearest road or trailhead, from published data; the source is set by ADR 0035 (TIGER/Line roads and NPS trail lines).

Alternatives considered

- No other tower source is recorded.

Consequences

- A source contract for OpenCelliD exists (contracts/opencellid_towers.yaml, draft).
- Share-alike also applies to published confidence values that use tower distance, in addition to the Ookla license (ADR 0013).
- The remoteness source is decided in ADR 0035 (OQ-18). Still open: the exact OpenCelliD attribution text and terms (OQ-22), and its fields (OQ-20).

`Will I have signal out there/adrs/0034-region-report-copies-and-model-provider.md`

0034. Region us-east-1; report copies deleted after 30 days, with daily counts kept; model provider Anthropic

- Status: Accepted (John, Oct 9, 2026, answering open questions OQ-42, OQ-43 and the provider part of OQ-39)
- Date: Oct 9, 2026
- Source: John's answers to the four major questions; the set-aside and call-log retention lines are defaults chosen by Claude, marked below

Context

Three decisions were needed before the platform and the assistant can be built: the AWS region, how long the lake and Snowflake keep individual hiker reports, and which model provider the trip assistant uses.

Decision

1. **AWS region (OQ-42).** Dev and live run in `us-east-1` (N. Virginia). Every resource name in design doc section 9.1 applies in this region.
2. **Report copies (OQ-43).** Raw report rows are deleted 30 days after the nightly copy in every store: DynamoDB, the lake table `signal_report`, and Snowflake. The lake and Snowflake keep one row per H3 resolution 9 cell and UTC day (table `report_cell_day`: the count of reports, and the count showing signal). That is enough for the label rule in ADR 0008, so model labels survive after the raw rows are gone.
3. **Set-aside rows (default, Claude).** The rows in `signal_report_set_aside` are raw reports, so they follow the same 30-day rule.
4. **Model provider (OQ-39).** The trip assistant and the LLM judge use the Anthropic API, on the cheapest Claude model. The exact model ID is pinned when the code is written. The $10 monthly spend limit is set in that account before the first call (FR-42).
5. **Call log retention (default, Claude).** The `llm_calls` log is kept 90 days, the same as pipeline logs. It holds no question text (ADR 0033).

Alternatives considered

- Region: `us-west-2` was the other option. John chose `us-east-1`.

- Report copies: 90 days or indefinitely. Rejected: they keep individual reports longer than the privacy promise needs.
- Model provider: another provider. Not chosen.

Consequences

- The `report_cell_day` table needs its own data contract before Phase B. It is not written yet.
- Snowflake and the lake lose individual reports after 30 days. Any analysis that needs them must be done inside that window.
- The Anthropic terms must be recorded in the licence register before the first model call (security and privacy, section 6).

`Will I have signal out there/adrs/0035-remoteness-from-tiger-roads-and-nps-trails.md`

0035. Remoteness is the distance to the nearest TIGER/Line road or NPS trail line

- Status: Accepted (John, Oct 9, 2026, choosing the first of three options laid out for OQ-18)
- Date: Oct 9, 2026
- Source: Open question OQ-18; ADR 0024 (definition); ADR 0031 (model family)

Context

The confidence model uses remoteness as an input (ADR 0008). ADR 0024 defined remoteness as the distance to the nearest road or trailhead, but left the data source open. The usual road data (OpenStreetMap) is under the ODbL share-alike licence. Its share-alike duty would reach the confidence values built on it, on top of the Ookla and OpenCelliD duties already there.

Decision

- **Roads.** U.S. Census Bureau TIGER/Line All Roads county files for the counties that contain the park and a buffer of 15 km around it (contract `tiger_roads`). The Census Bureau declares these files CC0 (public domain) on data.gov.
- **Trailheads.** The NPS Public Trails lines (contract `nps_trails`) stand in for trailheads. The distance is to the nearest point on any trail line, not only to a trailhead point. A separate trailhead dataset is not used, because the trail network is the data already loaded for Phase A.
- **Distance.** For each trail point, the distance in metres to the nearest road segment or trail line, computed after converting both to WGS84. Stored in the internal table `trail_point_remoteness` (proposed).

Conditions before the first load

- **Licence check (first-run check).** TIGER/Line files carry a rights flag of "otherRestrictions" on some county files. The Census Bureau page for the exact vintage must be read before the first download. If it restricts redistribution, only the computed distances are kept, never the road geometry. The distances are then derived data, and the credit to the Census Bureau stays on the page.
- **Attribution.** Credit the U.S. Census Bureau (TIGER/Line) and the National Park Service on the page and in the README. The NPS's own data-use wording must be checked for the exact dataset used. NPS GIS pages state that the data carry no warranty.

Alternatives considered

- OpenStreetMap roads. Rejected: share-alike (ODbL) would apply to the confidence values, on top of the existing share-alike terms.
- Dropping remoteness. Not chosen. It remains the fallback if the licence check fails.

Consequences

- The confidence model gains one input. Its value is tested in the evaluation (OQ-18 and the model card).
- The distance to a trail line is an approximation of the distance to a trailhead. It is recorded as such on the page's methodology notes.
- Two new raw sources (the Census files) need a manifest of URL, size and hash, under standard 1.

`Will I have signal out there/adrs/0036-horizon-computation-defaults.md`

0036. Terrain horizon computation: defaults for range, ray step, curvature, eye height and the AC-1A.2 margin

- Status: Accepted (John, Oct 9, 2026, approving the defaults proposed for OQ-24)
- Date: Oct 9, 2026
- Source: Open question OQ-24; ADR 0004 (method); AC-1A.2 (requirements, section 6.1)

Context

ADR 0004 fixed the method and the 36 sectors. OQ-24 listed the details the method still left open: the range, the ray step, curvature and refraction, the observer height, the vegetation question, whether 36 sectors is final, the interpolation, and the numeric margin behind "much higher" in AC-1A.2. Each detail changes which satellite windows are predicted, so each is set here.

Decision

Detail	Value	Why
Range	15 km from the trail point	Matches the range in ADR 0004 and the build plan
Ray step	30 m along each ray	The spacing of the Copernicus GLO-30 grid, so no grid cell is skipped
Observer height	1.7 m above the surface model	Typical eye height for a standing person
Earth curvature	Applied. At 15 km the surface falls about 17 m below the tangent, which lowers the apparent horizon	Leaving it out would overstate how far the ground can be seen
Atmospheric refraction	Not applied	Refraction raises distant objects slightly. Leaving it out makes the horizon a little higher, so the prediction is conservative: it predicts fewer windows, never more
Surface model	Kept as is, with vegetation and buildings counted as terrain	Conservative for the same reason. The effect is recorded as a limitation (ADR 0004)
Sectors	36 sectors of 10 degrees, sector 0 centred on north, numbered clockwise. This is final	Matches the Documentation Plan's example
Between sectors	`horizon_at` interpolates linearly between the two nearest sector centres	Avoids a jump when a satellite's azimuth crosses a sector edge
AC-1A.2 margin	The median horizon of the canyon-floor points is at least 5 degrees higher than the median horizon of the rim points in the test fixture	Gives "much higher" a number. The test fixture is the one AC-1A.2 already names

Alternatives considered

- Refraction applied. Rejected: its value depends on air conditions, which the pipeline does not model, and the prediction would be less conservative.
- Ray step of 10 m or 90 m. Rejected: 10 m oversamples the 30 m data, and 90 m skips grid cells.
- Nearest sector instead of interpolation. Rejected: it creates jumps in the satellite window edges.

Consequences

- The horizon is conservative: a satellite window can be missed where the surface model overstates a tree or a building, but a window is never predicted that the terrain blocks.
- The 5-degree margin is a test threshold. If the measured canyon and rim fixture shows a smaller difference, the threshold is revisited with a new ADR, not silently changed.
- Changing any value here changes the satellite windows, so each change needs a new ADR and a rerun of the affected days.

`Will I have signal out there/adrs/0037-credit-wording-and-licence-status.md`

0037. Credit wording for each data source, and how far each licence has been checked

- Status: Accepted. Two credit items are flagged as unverified; the page wording is decided here (OQ-22)
- Date: Oct 9, 2026
- Source: Open question OQ-22. Each licence was read from the provider's primary page where one could be reached. Secondary sources are marked as such.

Context

The page and the README must credit each source. The credit must match the licence, and some licences require exact wording. OQ-22 asked for the terms and the credit text for CelesTrak, FCC, NPS and OpenCelliD, and for the wording that says coverage is a map claim. Two of the primary pages could not be read from the build environment (HTTP 403), so those entries say so.

Decision

Verified from the primary licence or page

- **Copernicus GLO-30 (licence for COP-DEM-GLO-30-F, Articles 5 and 6).** The licence is free of charge. Public notice, for unmodified data: "© DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved." Our horizons are derived, so the notice for modified data applies: "produced using Copernicus WorldDEM-30 © DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved." When the data are distributed to the public, the licence also requires this statement: the organisations in charge of the Copernicus programme do not incur liability for any use of the data. Article 6(c) gives the wording.
- **NPS.** Material created by the NPS is generally in the public domain unless marked otherwise. Credit is appreciated and is not required. The NPS Arrowhead symbol must not be used. Some content carries third-party rights. Credit used: "Trail data: National Park Service."

Taken from secondary sources, to confirm before the page goes live

- **U.S. Census Bureau, TIGER/Line.** Declared CC0 (public domain) on data.gov. Some files carry a rights flag, "otherRestrictions", whose meaning is not stated. Confirm on the Census page for the exact vintage (first-run check, ADR 0035). Credit used: "Road data: U.S. Census Bureau, TIGER/Line."
- **OpenCelliD.** The licence is CC BY-SA 4.0 according to secondary sources. The OpenCelliD site returned HTTP 403, so the attribution page could not be read. Credit used: "Cell tower locations: OpenCelliD (opencellid.org), licensed CC BY-SA 4.0." Share-alike applies to the derived data (ADR 0024). **Unverified:** the exact credit text and terms, to read before publishing.
- **FCC mobile coverage.** Secondary sources report that the FCC asks for credit as a courtesy, not as a legal condition. The FCC broadband map page returned HTTP 403, so its terms could not be read. The carrier submissions behind the mobile map may carry their own restrictions. Credit used: "Coverage data: FCC National Broadband Map (broadbandmap.fcc.gov)." **Unverified:** the terms for carrier-level mobile data, to read before the Phase B coverage is published.

No terms found

- **CelesTrak.** No usage, terms or credit requirement was found on the CelesTrak webmaster page. Credit used, as a courtesy: "Orbital data: CelesTrak (celestrak.org), maintained by Dr. T.S. Kelso." The project publishes derived satellite windows, not the raw element sets. Before any raw element sets are republished, the maintainer is asked directly through the site.

Ookla

- Unchanged. The exact notice in ADR 0013 applies.

Page wording: coverage is a claim, and satellite texts are not guaranteed

- Shown on the trail page, next to the coverage strip: "Coverage shown is the FCC's modelled claim, not a measurement. Satellite texts are not guaranteed: they need T-Mobile service and a supported phone, and they can fail under cover that the terrain model does not show."
- The wording states the limits that the design already records. It is not a promise about any single message.

Alternatives considered

- Paraphrased credits. Rejected: the Copernicus licence gives exact wording, and it is used as written.
- Leaving the credits until the page is built. Rejected: the credit text is part of the acceptance check AC-1B.1 and the page's review.

Consequences

- Three entries (OpenCelliD, FCC mobile data, the Census rights flag) must be checked before publishing. They are listed as open first-run checks in the platform and security documents.
- The credit lines are the wording used on the page and in the README.

Part 8: Data contracts

Will I have signal out there/contracts/celestrak_gp.yaml

```
# Data contract (ODCS). Draft. ODCS version is a working placeholder (design-doc OQ-32).
# Upstream CelesTrak GP JSON field names are TO BE CONFIRMED (OQ-6, OQ-20) against a real file.
# The snake_case names below are the NORMALIZED lake column names. Mapping note: each lake column is filled from the
# upstream field named in its description (upstream name to be confirmed); the loader owns the rename.
apiVersion: v3.0.2
kind: DataContract
id: p1-source-celestrak-gp
name: celestrak_gp
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Orbital elements (GP data, meaning general perturbations elements, as JSON) for the Starlink satellites
listed by CelesTrak, used to select Direct-to-Cell satellites by the [DTC] name tag and to propagate them with SGP4.
  limitations: Pulled no more than once per 2-hour update. Any non-200 response stops the pull. Raw files are
immutable. The terms and credit text of CelesTrak are not recorded (OQ-22).
  usage: Input to the DTC selector (C2) and the satellite window engine (C5).
team:
  - username: John
    role: owner
servers:
  - server: celestrak-gp-json
    type: api
    description: CelesTrak Starlink list, GP data as JSON. Exact URL and query to be fixed after the first-run check
(OQ-20).
    location: "Public CelesTrak endpoint; URL fixed by the first-run check (OQ-20). Raw copies: s3://{env}-signal-
raw/celestrak/gp/. Lake table: {env}_signal.celestrak_gp (Glue catalog)."
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: One pull per nightly run at most; never more often than once per 2-hour update.
  - property: retention
    value: raw files kept
    description: Raw source files are kept (Decisions, storage clean-up). Live raw files are locked 30 days.
customProperties:
  - property: openQuestions
    value: OQ-6, OQ-20, OQ-22
  - property: adr
    value: adrs/0002, adrs/0018
schema:
  - name: celestrak_gp
    physicalName: celestrak_gp
    physicalType: table
    description: One row per satellite element set per pull. Partition per pull (run logical date). Carries source URL,
fetch time and content hash from the raw manifest.
    properties:
      - name: norad_cat_id
        logicalType: integer
        required: true
        customProperties:
          - property: upstreamField
            value: 'NORAD_CAT_ID (to be confirmed, OQ-6, OQ-20)'
        primaryKey: true
        primaryKeyPosition: 1
        description: Satellite catalog number.
      - name: epoch_utc
        logicalType: timestamp
        required: true
        customProperties:
          - property: upstreamField
            value: 'EPOCH (to be confirmed, OQ-6, OQ-20)'
        primaryKey: true
        primaryKeyPosition: 2
        description: Epoch of the element set, UTC.
      - name: object_name
        logicalType: string
        required: true
        customProperties:
          - property: upstreamField
```

```
          value: 'OBJECT_NAME (to be confirmed, OQ-6, OQ-20)'
      description: Satellite name; the [DTC] tag appears in this value for Direct-to-Cell satellites.
    - name: inclination_deg
      logicalType: number
      required: true
      customProperties:
        - property: upstreamField
          value: 'INCLINATION (to be confirmed, OQ-6, OQ-20)'
      description: Orbit inclination in degrees. Direct-to-Cell cross-check expects 43 or 53.
    - name: mean_motion_rev_per_day
      logicalType: number
      required: true
      customProperties:
        - property: upstreamField
          value: 'MEAN_MOTION (to be confirmed, OQ-6, OQ-20)'
      description: Mean motion; with eccentricity gives altitude for the cross-check (about 330 to 370 km).
    - name: eccentricity
      logicalType: number
      required: true
      customProperties:
        - property: upstreamField
          value: 'ECCENTRICITY (to be confirmed, OQ-6, OQ-20)'
    - name: ra_of_asc_node_deg
      logicalType: number
      required: true
      customProperties:
        - property: upstreamField
          value: 'RA_OF_ASC_NODE (to be confirmed, OQ-6, OQ-20)'
    - name: arg_of_pericenter_deg
      logicalType: number
      required: true
      customProperties:
        - property: upstreamField
          value: 'ARG_OF_PERICENTER (to be confirmed, OQ-6, OQ-20)'
    - name: mean_anomaly_deg
      logicalType: number
      required: true
      customProperties:
        - property: upstreamField
          value: 'MEAN_ANOMALY (to be confirmed, OQ-6, OQ-20)'
    - name: bstar
      logicalType: number
      required: true
      customProperties:
        - property: upstreamField
          value: 'BSTAR (to be confirmed, OQ-6, OQ-20)'
      description: Drag term used by SGP4.
    - name: source_url
      logicalType: string
      required: true
      description: URL the raw file came from.
    - name: fetched_at_utc
      logicalType: timestamp
      required: true
      description: Fetch time, UTC.
    - name: content_sha256
      logicalType: string
      required: true
      description: Hash of the raw file. Hash algorithm is a proposal (OQ-21).
  quality:
    - type: library
      metric: nullValues
      mustBe: 0
      description: Required fields are never null.
    - type: library
      metric: duplicateValues
      mustBe: 0
      arguments:
        properties:
          - norad_cat_id
          - epoch_utc
          - fetched_at_utc
      description: Unique per satellite, epoch and pull.
```

Will I have signal out there/contracts/copernicus_glo30.yaml

```
# Data contract (ODCS). Draft. ODCS version is a working placeholder (design-doc OQ-32).
apiVersion: v3.0.2
kind: DataContract
id: p1-source-copernicus-glo30
name: copernicus_glo30
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Copernicus GLO-30 elevation tiles on AWS, used to compute each trail point's terrain horizon.
  limitations: 30 m grid; heights on the EGM2008 datum; a surface model that includes vegetation and buildings; water
flattened. Free under the Copernicus licence with credit. Public archive that never changes, so only a manifest of URL,
size and hash is stored (engineering standard 1, which keeps a manifest for public archives).
  usage: Input to C4 (terrain horizon engine).
team:
  - username: John
    role: owner
servers:
  - server: copernicus-glo30-aws
    type: s3
    description: Copernicus GLO-30 on AWS open data. Bucket and key layout to be fixed in the first-run check (OQ-8,
OQ-20).
    location: "Public Copernicus GLO-30 tiles on AWS; tile list fixed by the first-run check (OQ-8, OQ-20). Manifest of
URL, size and hash: s3://{env}-signal-raw/copernicus/manifest/."
slaProperties:
  - property: frequency
    value: static archive
    description: The archive never changes; the manifest is created once and re-verified on demand.
  - property: retention
    value: manifest kept
customProperties:
  - property: openQuestions
    value: OQ-8, OQ-20, OQ-24
  - property: adr
    value: adrs/0004
schema:
  - name: glo30_tile_manifest
    physicalName: glo30_tile_manifest
    physicalType: table
    description: One row per tile that covers the park plus the horizon range. Tile raster data is read from the source
on demand; only the manifest is stored.
    properties:
      - name: tile_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Tile identifier (key in the source bucket).
      - name: source_url
        logicalType: string
        required: true
      - name: size_bytes
        logicalType: integer
        required: true
      - name: content_sha256
        logicalType: string
        required: true
      - name: fetched_at_utc
        logicalType: timestamp
        required: true
      - name: height_datum
        logicalType: string
        required: true
        description: Vertical datum of heights. Expected value EGM2008 (confirmed fact).
      - name: resolution_m
        logicalType: integer
        required: true
        description: Expected value 30 (confirmed fact).
    quality:
      - type: library
        metric: nullValues
```

```
        mustBe: 0
      - type: text
        description: Every tile needed for the park plus about 15 km margin is listed; first-run check confirms the
tiles open (OQ-8).
```

Will I have signal out there/contracts/dim_carrier_coverage_history.yaml

```
# Data contract (ODCS). Draft, Phase B. ODCS version is a working placeholder (design-doc OQ-32). Names are proposals
(OQ-21).
apiVersion: v3.0.2
kind: DataContract
id: p1-internal-dim-carrier-coverage-history
name: dim_carrier_coverage_history
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: SCD Type 2 history of carrier coverage per H3 cell across FCC releases, built by dbt on Snowflake.
  limitations: Release identifiers must be identifiable in the FCC files (OQ-20). Business history is kept in rows, not
Iceberg snapshots (engineering standard 8, which keeps business history in rows).
  usage: Feeds fct_trail_point_coverage.
team:
  - username: John
    role: owner
servers:
  - server: snowflake-signal-marts
    type: snowflake
    description: Snowflake mart; reads Iceberg tables through the Glue catalog.
    location: "Snowflake database SIGNAL_{ENV}, schema MARTS (names proposed, OQ-21), one per environment."
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Written or loaded in each nightly run unless stated otherwise.
customProperties:
  - property: openQuestions
    value: OQ-4, OQ-20
  - property: adr
    value: adrs/0015
schema:
  - name: dim_carrier_coverage_history
    physicalName: dim_carrier_coverage_history
    physicalType: table
    description: One row per cell, carrier and validity range.
    properties:
      - name: h3_cell_res9
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: carrier
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
      - name: covered
        logicalType: boolean
        required: true
      - name: valid_from_utc
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 3
      - name: valid_to_utc
        logicalType: timestamp
        required: false
        description: Null for the current row.
      - name: fcc_release
        logicalType: string
        required: true
    quality:
```

```
      - type: library
        metric: nullValues
        mustBe: 0
        description: Applies to required fields.
```

Will I have signal out there/contracts/dim_time.yaml

```
# Data contract (ODCS). Draft. Published gold table (Parquet). The grain is not decided (OQ-33).
apiVersion: v3.0.2
kind: DataContract
id: p1-published-dim-time
name: dim_time
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Time dimension of the small star schema (trail point, time, coverage and satellite windows, build plan
1A.5).
  limitations: The grain is one row per UTC minute (OQ-33, ADR 0028). Only the UTC value is defined here. Local time is
computed on the page, never stored (engineering standard 15, which stores timestamps in UTC).
  usage: Joined from fct_satellite_window.
team:
  - username: John
    role: owner
servers:
  - server: published-parquet
    type: s3
    description: Published gold Parquet. Separate for dev and live.
    location: "s3://{env}-signal-lake/published/dim_time/"
slaProperties:
  - property: frequency
    value: 1
    unit: d
customProperties:
  - property: openQuestions
    value: OQ-33
schema:
  - name: dim_time
    physicalName: dim_time
    physicalType: table
    description: One row per time value at the grain decided in OQ-33.
    properties:
      - name: time_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: time_utc
        logicalType: timestamp
        required: true
        description: UTC value of the time bucket.
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
      - type: library
        metric: duplicateValues
        mustBe: 0
        arguments:
          properties:
            - time_key
```

Will I have signal out there/contracts/dim_trail_point.yaml

```
# Data contract (ODCS). Draft. Published gold table (Parquet). Column names are proposals (OQ-21).
apiVersion: v3.0.2
kind: DataContract
id: p1-published-dim-trail-point
name: dim_trail_point
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
```

```
  purpose: One row per sampled trail point (every 50 meters) with its trail identity and location. The dimension for
the trail page strip map.
  limitations: Phase A covers Grand Canyon's main trails only (ADR 0001). The list of main trails is open (OQ-36).
  usage: Read by the serving Lambda (DuckDB) behind the trail page.
team:
  - username: John
    role: owner
servers:
  - server: published-parquet
    type: s3
    description: Published gold Parquet in the site or lake bucket. Path set by the publisher (C10). Separate for dev
and live.
    location: "Published Parquet: s3://{env}-signal-lake/published/dim_trail_point/. Dev and live buckets are
separate."
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Rebuilt in each nightly run (10:00 UTC).
  - property: retention
    value: published data kept
customProperties:
  - property: openQuestions
    value: OQ-21, OQ-36
  - property: adr
    value: adrs/0001, adrs/0004
schema:
  - name: dim_trail_point
    physicalName: dim_trail_point
    physicalType: table
    description: One row per trail point.
    properties:
      - name: trail_point_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: trail_id
        logicalType: string
        required: true
      - name: trail_name
        logicalType: string
        required: false
        description: From TRLNAME.
      - name: unit_code
        logicalType: string
        required: true
        description: From UNITCODE.
      - name: seq
        logicalType: integer
        required: true
        description: Order along the trail, starting at 0 (convention proposed).
      - name: lon
        logicalType: number
        required: true
      - name: lat
        logicalType: number
        required: true
      - name: elevation_m
        logicalType: number
        required: true
        description: From the surface model (EGM2008 datum).
      - name: h3_cell_res9
        logicalType: string
        required: true
        description: Resolution 9 cell containing the point (join key to FCC coverage).
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
      - type: library
        metric: duplicateValues
        mustBe: 0
        arguments:
          properties:
            - trail_point_id
```

```
      - type: library
        metric: rowCount
        mustBeGreaterThan: 0
        description: Empty results are rejected by the publish gate.
```

Will I have signal out there/contracts/fcc_mobile_coverage.yaml

```
# Data contract (ODCS). Draft. ODCS version is a working placeholder (design-doc OQ-32).
# FCC file layout beyond what the Documentation Plan confirms is open (OQ-20, OQ-7).
apiVersion: v3.0.2
kind: DataContract
id: p1-source-fcc-mobile-coverage
name: fcc_mobile_coverage
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Mobile coverage published by the FCC (U.S. Federal Communications Commission), downloaded per provider,
state and technology as a Shapefile or GeoPackage; most datasets include H3 resolution 9 hexagons alongside the raw
modeled coverage. Used to flag each trail point covered or not for a carrier.
  limitations: Coverage is a modeled claim, not a measurement; the confidence model (Phase B) judges how far to trust
it. Phase A loads one carrier in Arizona; Phase B loads every carrier for the park. The release cadence and terms are
not recorded (OQ-20, OQ-22).
  usage: Input to C6 (coverage flag) and C13 (all carriers); source of SCD Type 2 carrier coverage history in Snowflake
(Phase B).
team:
  - username: John
    role: owner
servers:
  - server: fcc-coverage-download
    type: api
    description: FCC coverage file download per provider, state and technology. Exact location to be fixed in the
first-run check (OQ-20).
    location: "Public FCC download; file names fixed by OQ-20. Raw copies: s3://{env}-signal-raw/fcc/. Lake table:
{env}_signal.fcc_coverage_cell."
slaProperties:
  - property: frequency
    value: release cadence not recorded
    description: Freshness target depends on the FCC release schedule (OQ-20). Reference data refresh cadence is OQ-25.
  - property: retention
    value: raw files kept
customProperties:
  - property: openQuestions
    value: OQ-7, OQ-11, OQ-20, OQ-22, OQ-25
  - property: adr
    value: adrs/0007, adrs/0015
schema:
  - name: fcc_coverage_cell
    physicalName: fcc_coverage_cell
    physicalType: table
    description: One row per H3 resolution 9 cell, carrier, technology and release. Partition per pull.
    properties:
      - name: h3_cell_res9
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: H3 resolution 9 cell index from the file's hexagons.
      - name: carrier
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: Provider as named in the FCC file. The Phase A carrier is T-Mobile (ADR 0020); the provider name
as written in the file is OQ-20.
      - name: technology
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 3
        description: Technology as named in the FCC file.
      - name: release
        logicalType: string
        required: true
        primaryKey: true
```

```
        primaryKeyPosition: 4
        description: FCC release identifier or date; basis for SCD Type 2 rows in Phase B. Format open (OQ-20).
      - name: source_url
        logicalType: string
        required: true
      - name: fetched_at_utc
        logicalType: timestamp
        required: true
      - name: content_sha256
        logicalType: string
        required: true
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
      - type: library
        metric: duplicateValues
        mustBe: 0
        arguments:
          properties:
            - h3_cell_res9
            - carrier
            - technology
            - release
```

Will I have signal out there/contracts/fct_coverage_confidence.yaml

```
# Data contract (ODCS). Draft, Phase B. Published gold table (Parquet). Column names are proposals (OQ-21).
apiVersion: v3.0.2
kind: DataContract
id: p1-published-fct-coverage-confidence
name: fct_coverage_confidence
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: How likely the FCC coverage claim is to be right at a trail point and carrier, or the statement that there
are not enough measurements.
  limitations: A confidence level exists only where enough Ookla tests exist nearby (threshold OQ-14). Elsewhere the
status is not_enough_measurements and no number is shown. Derived from Ookla data (CC BY-NC-SA 4.0, ADR 0013) and
OpenCelliD tower locations (share-alike, ADR 0024), so it carries both licenses. Inputs are terrain, distance to the
nearest tower, remoteness (distance to the nearest TIGER/Line road or NPS trail line; ADR 0035).
  usage: Read by the serving Lambda; shown on the trail page; used by the assistant through the metrics tools.
team:
  - username: John
    role: owner
servers:
  - server: published-parquet
    type: s3
    description: Published gold Parquet. Separate for dev and live.
    location: "s3://{env}-signal-lake/published/fct_coverage_confidence/"
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Scored in each nightly run with the model copy in the repo.
customProperties:
  - property: license
    value: "Includes data derived from Speedtest by Ookla® open data, licensed under CC BY-NC-SA 4.0. Derived data
published here is shared under the same license."
  - property: openQuestions
    value: OQ-12, OQ-13, OQ-14, OQ-18, OQ-19
  - property: adr
    value: adrs/0008, adrs/0013, adrs/0014, adrs/0024
schema:
  - name: fct_coverage_confidence
    physicalName: fct_coverage_confidence
    physicalType: table
    description: One row per trail point and carrier.
    properties:
      - name: trail_point_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
```

```
      - name: carrier
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
      - name: status
        logicalType: string
        required: true
        description: One of confidence_available or not_enough_measurements.
      - name: confidence
        logicalType: number
        required: false
        description: Probability that the FCC claim is right. Null when status is not_enough_measurements. Units and
level labels are open (OQ-21).
      - name: nearby_test_count
        logicalType: integer
        required: true
        description: Number of tests that counted as nearby.
      - name: model_version
        logicalType: string
        required: true
        description: MLflow-tracked version of the model copy in the repo.
      - name: scored_at_utc
        logicalType: timestamp
        required: true
    quality:
      - type: sql
        description: Confidence is null exactly when status is not_enough_measurements.
        query: SELECT COUNT(*) FROM fct_coverage_confidence WHERE (status = 'not_enough_measurements') <> (confidence
IS NULL)
        mustBe: 0
      - type: library
        metric: duplicateValues
        mustBe: 0
        arguments:
          properties:
            - trail_point_id
            - carrier
```

Will I have signal out there/contracts/fct_report_counts_area.yaml

```
# Data contract (ODCS). Draft, Phase B. Published gold table (Parquet). Column names are proposals (OQ-21).
apiVersion: v3.0.2
kind: DataContract
id: p1-published-fct-report-counts-area
name: fct_report_counts_area
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Counts of hiker signal reports per area, shown on the trail page.
  limitations: An area is the H3 resolution 6 parent of the report cell (OQ-35, ADR 0030). Counts are shown only where
at least 5 reports exist in 7 days. Reports set aside as impossible are not counted. No visitor identifiers exist in
this table.
  usage: Read by the serving Lambda and shown on the page.
team:
  - username: John
    role: owner
servers:
  - server: published-parquet
    type: s3
    description: Published gold Parquet. Separate for dev and live.
    location: "s3://{env}-signal-lake/published/fct_report_counts_area/"
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Rebuilt after the nightly report copy.
customProperties:
  - property: openQuestions
    value: OQ-35
  - property: adr
    value: adrs/0009, adrs/0012
schema:
  - name: fct_report_counts_area
```

```
    physicalName: fct_report_counts_area
    physicalType: table
    description: One row per area, carrier and signal answer.
    properties:
      - name: area_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Area key, an H3 resolution 6 cell (ADR 0030).
      - name: carrier
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
      - name: had_signal
        logicalType: boolean
        required: true
        primaryKey: true
        primaryKeyPosition: 3
      - name: report_count
        logicalType: integer
        required: true
      - name: counted_through_utc
        logicalType: timestamp
        required: true
        description: Latest report hour included.
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
      - type: library
        metric: duplicateValues
        mustBe: 0
        arguments:
          properties:
            - area_id
            - carrier
            - had_signal
```

Will I have signal out there/contracts/fct_satellite_window.yaml

```
# Data contract (ODCS). Draft. Published gold table (Parquet). Column names are proposals (OQ-21).
apiVersion: v3.0.2
kind: DataContract
id: p1-published-fct-satellite-window
name: fct_satellite_window
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: When a Direct-to-Cell satellite clears both the minimum elevation and the terrain horizon for each trail
point, stored as start and end times (never one row per minute).
  limitations: The run predicts 26 hours (24 plus the 2-hour nightly maximum, ADR 0019); the page says the next 24
hours. Predictions are valid until valid_until_utc (the end of the prediction span). When the data is late the page
keeps showing it with a late notice and the time of the last good run (ADR 0006, amended); it does not hide it. A
satellite window's edges are accurate to the 1-minute propagation step. The row limit is set from the measured size
(OQ-1, OQ-15). The definition of "late" for the notice is OQ-23.
  usage: Read by the serving Lambda; shown on the page with a late notice when late.
team:
  - username: John
    role: owner
servers:
  - server: published-parquet
    type: s3
    description: Published gold Parquet. Separate for dev and live.
    location: "Published Parquet: s3://{env}-signal-lake/published/fct_satellite_window/. Dev and live buckets are
separate."
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Rebuilt in each nightly run at 10:00 UTC, hard maximum 2 hours.
  - property: latency
    value: 2
```

```
    unit: h
    description: Nightly maximum run time; usually about 45 minutes. Publish must complete inside it.
  - property: retention
    value: published data kept
customProperties:
  - property: openQuestions
    value: OQ-1, OQ-9, OQ-15, OQ-21, OQ-23
  - property: adr
    value: adrs/0003, adrs/0005, adrs/0006, adrs/0019
schema:
  - name: fct_satellite_window
    physicalName: fct_satellite_window
    physicalType: table
    description: One row per trail point, satellite and satellite window. Partition per run logical date (UTC);
rerunning a day overwrites only that partition.
    properties:
      - name: trail_point_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: satellite_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: Satellite catalog number as text.
      - name: window_start_utc
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 3
        description: First minute the satellite is reachable.
      - name: window_end_utc
        logicalType: timestamp
        required: true
        description: Last minute the satellite is reachable. Must be on or after window_start_utc.
      - name: min_elevation_deg
        logicalType: number
        required: true
        description: Minimum elevation used for this run (10 in the starting configuration, ADR 0003).
      - name: run_logical_date
        logicalType: date
        required: true
        description: Partition key.
      - name: valid_until_utc
        logicalType: timestamp
        required: true
        description: Time the run's predictions stop being valid (end of the 26-hour prediction span).
      - name: run_started_utc
        logicalType: timestamp
        required: true
        description: Start time of the last good run that produced these rows; shown in the late notice.
      - name: loaded_at_utc
        logicalType: timestamp
        required: true
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
      - type: library
        metric: duplicateValues
        mustBe: 0
        arguments:
          properties:
            - trail_point_id
            - satellite_id
            - window_start_utc
            - run_logical_date
      - type: sql
        description: Window end is on or after satellite window start.
        query: SELECT COUNT(*) FROM fct_satellite_window WHERE window_end_utc < window_start_utc
        mustBe: 0
```

```
      - type: sql
        description: valid_until_utc is later than every satellite window start in the run.
        query: SELECT COUNT(*) FROM fct_satellite_window WHERE valid_until_utc < window_start_utc
        mustBe: 0
```

Will I have signal out there/contracts/fct_trail_point_coverage.yaml

```
# Data contract (ODCS). Draft. Published gold table (Parquet). Column names are proposals (OQ-21).
apiVersion: v3.0.2
kind: DataContract
id: p1-published-fct-trail-point-coverage
name: fct_trail_point_coverage
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Whether each trail point is covered for a carrier, according to the FCC modeled coverage (a claim, not a
measurement).
  limitations: Phase A has one carrier; Phase B adds every carrier for the park. The release identifier supports SCD
Type 2 history in Snowflake (ADR 0015). The Phase A carrier is T-Mobile (ADR 0020); the Phase B carrier list is open
(OQ-11).
  usage: Read by the serving Lambda for the strip map.
team:
  - username: John
    role: owner
servers:
  - server: published-parquet
    type: s3
    description: Published gold Parquet. Separate for dev and live.
    location: "Published Parquet: s3://{env}-signal-lake/published/fct_trail_point_coverage/. Dev and live buckets are
separate."
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Rebuilt in each nightly run; the FCC input changes only with a new release (OQ-25).
  - property: retention
    value: published data kept
customProperties:
  - property: openQuestions
    value: OQ-11, OQ-21, OQ-25
  - property: adr
    value: adrs/0007, adrs/0015
schema:
  - name: fct_trail_point_coverage
    physicalName: fct_trail_point_coverage
    physicalType: table
    description: One row per trail point and carrier.
    properties:
      - name: trail_point_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: carrier
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
      - name: covered
        logicalType: boolean
        required: true
        description: True when the point's H3 resolution 9 cell is in the carrier's FCC coverage.
      - name: fcc_release
        logicalType: string
        required: true
        description: FCC release used (format open, OQ-20).
      - name: loaded_at_utc
        logicalType: timestamp
        required: true
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
      - type: library
```

```
        metric: duplicateValues
        mustBe: 0
        arguments:
          properties:
            - trail_point_id
            - carrier
      - type: text
        description: Every trail point has a coverage flag for the Phase A carrier (AC-1A.4).
```

Will I have signal out there/contracts/feedback_votes.yaml

```
# Data contract (ODCS). Draft, Phase B. ODCS version is a working placeholder (design-doc OQ-32). Names are proposals
(OQ-21).
apiVersion: v3.0.2
kind: DataContract
id: p1-source-feedback-votes
name: feedback_votes
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Thumbs-up and thumbs-down votes on assistant answers, stored in DynamoDB and used for the A/B test of two
prompt versions.
  limitations: Whether a vote carries any visitor identity, and its retention, are not recorded.
  usage: Eval and A/B analysis (C20).
team:
  - username: John
    role: owner
servers:
  - server: dynamodb-feedback-votes
    type: custom
    description: DynamoDB, always on; provisioned within the free tier.
    location: "DynamoDB table {env}-signal-feedback (name proposed, OQ-21)."
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Written or loaded in each nightly run unless stated otherwise.
customProperties:
  - property: openQuestions
    value: OQ-39, OQ-38
  - property: adr
    value: adrs/0017
schema:
  - name: feedback_votes
    physicalName: feedback_votes
    physicalType: table
    description: One item per vote. No visitor ID stored (ADR 0033).
    properties:
      - name: vote_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: answer_id
        logicalType: string
        required: true
      - name: vote
        logicalType: string
        required: true
        description: up or down.
      - name: prompt_version
        logicalType: string
        required: true
      - name: vote_hour_utc
        logicalType: timestamp
        required: true
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
        description: Applies to required fields.
```

Will I have signal out there/contracts/hiker_signal_reports.yaml

```
# Data contract (ODCS). Draft, Phase B. ODCS version is a working placeholder (design-doc OQ-32).
# Table and attribute names are proposals (OQ-21).
apiVersion: v3.0.2
kind: DataContract
id: p1-source-hiker-signal-reports
name: hiker_signal_reports
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Hiker reports "I had signal here" or "I had no signal here", written by Lambda to DynamoDB, copied nightly
to the lake and Snowflake, used as labels for the confidence model and for per-area counts.
  limitations: A report holds only the H3 resolution 9 cell, carrier, signal yes or no and the time rounded to the
hour. It never carries a visitor ID or exact coordinates. Raw rows delete themselves 30 days after the nightly copy.
Point-in-time recovery is OFF for this table, so deletion is final at 30 days (ADR 0022, Decided by John, Oct 9, 2026).
  usage: Nightly copy (C23); model labels (C17); counts per area on the page.
team:
  - username: John
    role: owner
servers:
  - server: dynamodb-signal-reports
    type: custom
    description: DynamoDB, point-in-time recovery off for the reports table (ADR 0022), provisioned within the free
tier. Table names and capacity are set in Terraform (proposed names).
    location: "DynamoDB tables {env}-signal-reports (signal_reports) and {env}-signal-report-limits
(report_limit_counters); names proposed (OQ-21). Lake copy: {env}_signal.signal_report."
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Copied to the lake in each nightly run.
  - property: retention
    value: 30
    unit: d
    description: Raw rows expire 30 days after the nightly copy (ADR 0012).
customProperties:
  - property: openQuestions
    value: OQ-12, OQ-26, OQ-29, OQ-35, OQ-38, OQ-41
  - property: adr
    value: adrs/0009, adrs/0010, adrs/0011, adrs/0012, adrs/0022, adrs/0023
schema:
  - name: signal_reports
    physicalName: signal_reports
    physicalType: table
    description: One item per accepted report. No visitor ID is stored on this table.
    properties:
      - name: report_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Server-generated unique key. Not derived from the visitor ID (proposal).
      - name: h3_cell_res9
        logicalType: string
        required: true
        description: H3 resolution 9 cell computed on the phone.
      - name: carrier
        logicalType: string
        required: true
        description: Carrier the hiker reports on. Allowed values follow the carrier list (OQ-11).
      - name: had_signal
        logicalType: boolean
        required: true
        description: True for "I had signal here", false for "I had no signal here".
      - name: report_hour_utc
        logicalType: timestamp
        required: true
        description: Time of the report rounded to the hour, UTC.
      - name: copied_at_utc
        logicalType: timestamp
        required: false
        description: Set by the nightly copy.
      - name: expires_at_epoch
```

```
        logicalType: integer
        required: false
        description: Expiry attribute set by the nightly copy to 30 days after the copy.
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
        description: Applies to required fields.
      - type: text
        description: The h3_cell_res9 value is a valid resolution 9 H3 index; reports outside any mapped area are set
aside by the nightly copy, not rejected at write time.
  - name: report_limit_counters
    physicalName: report_limit_counters
    physicalType: table
    description: Limit counters kept apart from reports. They hold the random visitor ID and expire 72 hours after
their UTC day (ADR 0026). No report attribute is stored here besides what is needed for the one-per-cell-per-24-hours
rule. One system-wide daily counter with no visitor ID enforces the 50 a day cap. Whether point-in-time recovery is off
for this table is OQ-41.
    properties:
      - name: counter_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Visitor ID plus day (and cell), or the system-wide day key. Days are counted by the report's own
hour, not arrival time (ADR 0011). Exact composition and day boundary are proposals (OQ-21, OQ-26, OQ-38).
      - name: count
        logicalType: integer
        required: true
      - name: expires_at_epoch
        logicalType: integer
        required: true
        description: Expiry time, 72 hours after the end of the UTC day (ADR 0026).
    quality:
      - type: text
        description: Limits enforced (Decided by John, Oct 9, 2026) - 10 reports a day per visitor ID, one per cell per
visitor every 24 hours, and at most 50 reports a day for the whole system; caps are raised as traffic grows.
```

Will I have signal out there/contracts/nps_trails.yaml

```
# Data contract (ODCS). Draft. ODCS version is a working placeholder (design-doc OQ-32).
apiVersion: v3.0.2
kind: DataContract
id: p1-source-nps-trails
name: nps_trails
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: National Park Service Public Trails ArcGIS service for Grand Canyon, giving trail polylines used to sample
points every 50 meters. Phase A uses Bright Angel, South Kaibab and North Kaibab (ADR 0020).
  limitations: GeoJSON output is supported, 2,000 records per request, so the pull pages. Other fields beyond TRLNAME
and UNITCODE are not recorded (OQ-20). Public data from data.gov; credit on the page (OQ-22).
  usage: Input to C3 (trail loader and sampler).
team:
  - username: John
    role: owner
servers:
  - server: nps-public-trails
    type: api
    description: NPS Public Trails ArcGIS service (polylines, GeoJSON, 2,000 records per request). Service URL to be
fixed in the first-run check (OQ-8, OQ-20).
    location: "Public NPS Public Trails ArcGIS service; query and UNITCODE fixed by OQ-20 and OQ-36. Raw pages:
s3://{env}-signal-raw/nps/."
slaProperties:
  - property: frequency
    value: refresh cadence open
    description: See OQ-25.
  - property: retention
    value: raw files kept
customProperties:
  - property: openQuestions
    value: OQ-8, OQ-20, OQ-22, OQ-25, OQ-36
  - property: adr
    value: adrs/0004, adrs/0018
```

```
schema:
  - name: nps_trail_line
    physicalName: nps_trail_line
    physicalType: table
    description: One row per trail line. Partition per pull.
    properties:
      - name: trail_line_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Source key for the line (the service's object identifier, name to be confirmed, OQ-20).
      - name: trlname
        logicalType: string
        required: false
        description: Trail name (field TRLNAME).
      - name: unitcode
        logicalType: string
        required: true
        description: Park code (field UNITCODE). The value for Grand Canyon is confirmed in the first-run check (OQ-
20).
      - name: geometry_geojson
        logicalType: object
        required: true
        description: Polyline geometry as GeoJSON (WGS 84 longitude and latitude is the expected reference; confirm,
OQ-20).
      - name: source_url
        logicalType: string
        required: true
      - name: fetched_at_utc
        logicalType: timestamp
        required: true
      - name: content_sha256
        logicalType: string
        required: true
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
        description: Applies to required fields.
      - type: library
        metric: duplicateValues
        mustBe: 0
        arguments:
          properties:
            - trail_line_id
            - fetched_at_utc
```

Will I have signal out there/contracts/ookla_open_data.yaml

```
# Data contract (ODCS). Draft, Phase B. ODCS version is a working placeholder (design-doc OQ-32).
apiVersion: v3.0.2
kind: DataContract
id: p1-source-ookla-open-data
name: ookla_open_data
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Ookla open mobile speed-test data for Arizona (quarterly, about 600-meter tiles, GPS-accurate tests only),
used as measured evidence for the coverage confidence model.
  limitations: License CC BY-NC-SA 4.0 (non-commercial, credit, share-alike). Derived data published here carries the
same license. Speed tests exist only where phones had signal, so dead zones are missing. It is a public archive, so a
manifest of URL, size and hash is stored instead of a full copy (engineering standard 1, which keeps a manifest for
public archives). File size unknown (OQ-7).
  usage: Input to C13 and C17.
team:
  - username: John
    role: owner
servers:
  - server: ookla-open-data-aws
    type: s3
    description: Ookla open data on AWS. Location to be fixed in the first-run check (OQ-10, OQ-20).
    location: "Ookla open data on AWS; quarterly file list fixed by OQ-10 and OQ-20. Manifest: s3://{env}-signal-
raw/ookla/manifest/. Lake table: {env}_signal.ookla_tile."
```

```
slaProperties:
  - property: frequency
    value: 1
    unit: quarter
    description: Quarterly release.
  - property: retention
    value: manifest kept
customProperties:
  - property: license
    value: "Includes data derived from Speedtest by Ookla® open data, licensed under CC BY-NC-SA 4.0. Derived data
published here is shared under the same license."
  - property: openQuestions
    value: OQ-2, OQ-7, OQ-10, OQ-20
  - property: adr
    value: adrs/0007, adrs/0013
schema:
  - name: ookla_tile
    physicalName: ookla_tile
    physicalType: table
    description: One row per tile and quarter, with the H3 resolution 8 cell assigned at load (engineering standard 9,
which picks the H3 resolution from the source). Field names from the Ookla file are confirmed in the first-run check
(OQ-10).
    properties:
      - name: quarter
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Quarter of the release.
      - name: tile_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: Tile identifier from the source (name to be confirmed, OQ-20).
      - name: h3_cell_res8
        logicalType: string
        required: true
        description: H3 resolution 8 cell for the tile (assignment rule in the loader, ADR 0007).
      - name: test_count
        logicalType: integer
        required: true
        description: Number of tests in the tile; basis for "enough tests nearby" (threshold OQ-14).
      - name: speed_metrics
        logicalType: object
        required: true
        description: Measured speed fields as provided by the source (names to be confirmed, OQ-20).
      - name: source_url
        logicalType: string
        required: true
      - name: fetched_at_utc
        logicalType: timestamp
        required: true
      - name: content_sha256
        logicalType: string
        required: true
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
      - type: library
        metric: duplicateValues
        mustBe: 0
        arguments:
          properties:
            - quarter
            - tile_key
```

Will I have signal out there/contracts/opencellid_towers.yaml

```
# Data contract (ODCS). Draft, Phase B. ODCS version is a working placeholder (design-doc OQ-32).
# Field names and source location are to be confirmed in the first-run check (OQ-20).
apiVersion: v3.0.2
kind: DataContract
id: p1-source-opencellid-towers
name: opencellid_towers
```

```
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: OpenCelliD cell tower locations, used to compute the distance to the nearest tower, one input of the
coverage confidence model (Decided by John, Oct 9, 2026, ADR 0024).
  limitations: OpenCelliD is share-alike, so derived data carries the same license. This is recorded in the license
register and on the page. Attribution text and terms are not recorded (OQ-22). File size, format and update cadence are
not recorded (OQ-7, OQ-20).
  usage: Input to C13 (loader) and C17 (confidence model features).
team:
  - username: John
    role: owner
servers:
  - server: opencellid-download
    type: api
    description: OpenCelliD download. Location, access method and any key (which would live in Parameter Store) to be
fixed in the first-run check (OQ-20).
    location: "Public OpenCelliD download; URL fixed by OQ-20. Raw copies: s3://{env}-signal-raw/opencellid/. Lake
table: {env}_signal.opencellid_tower."
slaProperties:
  - property: frequency
    value: refresh cadence open
    description: See OQ-25.
  - property: retention
    value: raw files kept
customProperties:
  - property: license
    value: Share-alike. Derived data carries the same license (credit wording in ADR 0037; OpenCelliD terms to verify).
  - property: openQuestions
    value: OQ-7, OQ-20, OQ-22, OQ-25
  - property: adr
    value: adrs/0024, adrs/0018
schema:
  - name: opencellid_tower
    physicalName: opencellid_tower
    physicalType: table
    description: One row per tower record for the area around the park. Field names to be confirmed (OQ-20).
    properties:
      - name: tower_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Source key for the tower record (name to be confirmed).
      - name: lon
        logicalType: number
        required: true
      - name: lat
        logicalType: number
        required: true
      - name: h3_cell_res9
        logicalType: string
        required: true
        description: Resolution 9 cell of the tower (join key, ADR 0007).
      - name: source_url
        logicalType: string
        required: true
      - name: fetched_at_utc
        logicalType: timestamp
        required: true
      - name: content_sha256
        logicalType: string
        required: true
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
      - type: library
        metric: duplicateValues
        mustBe: 0
```

```
        arguments:
          properties:
            - tower_id
            - fetched_at_utc
```

Will I have signal out there/contracts/orbit_update_stream.yaml

```
# Data contract (ODCS). Draft, Phase B. ODCS version is a working placeholder (design-doc OQ-32). Names are proposals
(OQ-21).
apiVersion: v3.0.2
kind: DataContract
id: p1-internal-orbit-update-stream
name: orbit_update_stream
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Kafka topic of new CelesTrak element sets published by the nightly run, read by the availableNow Structured
Streaming job to find changed satellites.
  limitations: Low-rate stream with an explicit schema, S3 checkpoints, a watermark and dropDuplicatesWithinWatermark.
Topic name, watermark and the meaning of changed are open.
  usage: Input to C14 and the trigger for re-propagation in C5.
team:
  - username: John
    role: owner
servers:
  - server: kafka-orbit-element-sets
    type: kafka
    description: Kafka topic on the session box of each environment; low-rate (ADR 0016).
    location: "Kafka topic orbit-element-sets on the session box (topic name proposed, OQ-30)."
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Written or loaded in each nightly run unless stated otherwise.
customProperties:
  - property: openQuestions
    value: OQ-30
  - property: adr
    value: adrs/0016
schema:
  - name: orbit_update_stream
    physicalName: orbit_update_stream
    physicalType: table
    description: One message per element set. Fields mirror celestrak_gp plus the pull time.
    properties:
      - name: norad_cat_id
        logicalType: integer
        required: true
      - name: epoch_utc
        logicalType: timestamp
        required: true
      - name: fetched_at_utc
        logicalType: timestamp
        required: true
      - name: elements
        logicalType: object
        required: true
        description: Orbital elements as in celestrak_gp (column list there).
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
        description: Applies to required fields.
```

Will I have signal out there/contracts/report_cell_day.yaml

```
# Data contract (ODCS v3.0.2). Draft, Phase B. Lake table that replaces raw report rows after 30 days (ADR 0034).
apiVersion: v3.0.2
kind: DataContract
id: p1-internal-report-cell-day
name: report_cell_day
version: 0.1.0
status: draft
```

```
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Daily counts of hiker reports per H3 resolution 9 cell. These counts are kept after the raw report rows are
deleted 30 days after their nightly copy, and they are the input to the label rule (N reports in one cell on one day,
ADR 0008).
  limitations: Holds counts only. No visitor ID, no coordinates and no hour, so a count cannot identify one hiker. A
cell-day with fewer reports than the label threshold N is still stored as a count, and only cell-days that pass the
threshold become labels. Cells set aside as impossible are not counted.
  usage: Label input for the confidence model (C17) and the source for area counts (C23).
team:
  - username: John
    role: owner
servers:
  - server: lake-report-cell-day
    type: s3
    description: Iceberg table in the lake, written by the nightly copy (C23).
    location: "Lake table {env}_signal.report_cell_day in s3://{env}-signal-lake/. Table name proposed (OQ-21)."
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Updated in each nightly run, after the report copy.
  - property: retention
    value: kept
    description: Counts are kept. They contain no personal data (ADR 0034).
customProperties:
  - property: openQuestions
    value: OQ-12, OQ-13, OQ-43
  - property: adr
    value: adrs/0034, adrs/0008, adrs/0012
schema:
  - name: report_cell_day
    physicalName: report_cell_day
    physicalType: table
    description: One row per H3 resolution 9 cell and UTC day.
    properties:
      - name: h3_cell_res9
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: H3 resolution 9 cell computed on the phone.
      - name: report_date_utc
        logicalType: date
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: UTC day of the report's hour (ADR 0026).
      - name: report_count
        logicalType: integer
        required: true
        description: Number of accepted reports in the cell on that day.
      - name: signal_yes_count
        logicalType: integer
        required: true
        description: How many of those reports said the hiker had signal.
      - name: signal_no_count
        logicalType: integer
        required: true
        description: How many of those reports said the hiker had no signal.
      - name: built_at_utc
        logicalType: timestamp
        required: true
        description: When the nightly copy wrote this row.
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
      - type: library
        metric: duplicateValues
        mustBe: 0
        arguments:
          properties:
            - h3_cell_res9
            - report_date_utc
```

```
      - type: sql
        description: The yes and no counts add up to the report count.
        query: SELECT COUNT(*) FROM report_cell_day WHERE signal_yes_count + signal_no_count <> report_count
        mustBe: 0
```

Will I have signal out there/contracts/signal_report_set_aside.yaml

```
# Data contract (ODCS). Draft, Phase B. ODCS version is a working placeholder (design-doc OQ-32). Names are proposals
(OQ-21).
apiVersion: v3.0.2
kind: DataContract
id: p1-internal-signal-report-set-aside
name: signal_report_set_aside
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Reports set aside by the nightly copy as impossible patterns (outside any mapped area, bursts that
contradict nearby reports) for review.
  limitations: Excluded from counts and labels. Carries no visitor ID. Retention not recorded; follows the 30-day rule
of raw reports unless John decides otherwise (recovery is off for the reports table, ADR 0022).
  usage: Review by John; not published.
team:
  - username: John
    role: owner
servers:
  - server: iceberg-signal-report-set-aside
    type: s3
    description: Iceberg table in the lake, for review.
    location: "Lake table {env}_signal.signal_report_set_aside in s3://{env}-signal-lake/ (table name proposed, OQ-
21)."
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Written or loaded in each nightly run unless stated otherwise.
customProperties:
  - property: openQuestions
    value: OQ-35
  - property: adr
    value: adrs/0012
schema:
  - name: signal_report_set_aside
    physicalName: signal_report_set_aside
    physicalType: table
    description: One row per set-aside report.
    properties:
      - name: report_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: h3_cell_res9
        logicalType: string
        required: true
      - name: carrier
        logicalType: string
        required: true
      - name: had_signal
        logicalType: boolean
        required: true
      - name: report_hour_utc
        logicalType: timestamp
        required: true
      - name: reason
        logicalType: string
        required: true
        description: Which rule set it aside.
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
        description: Applies to required fields.
```

```
# Data contract (ODCS v3.0.2). Draft, Phase B. Source for the remoteness input (ADR 0035).
# Field names are the TIGER/Line names as published; they are confirmed against a real file in the first-run check (OQ-20).
apiVersion: v3.0.2
kind: DataContract
id: p1-source-tiger-roads
name: tiger_roads
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: Road centre lines from the U.S. Census Bureau TIGER/Line All Roads county files, used to compute how far each trail point is from the nearest road (remoteness, ADR 0035).
  limitations: Covers roads in the counties that contain the park and a buffer of 15 km around it. Roads are used only as distance targets; nothing from this file is published on its own. The file's rights field carries "otherRestrictions" on some county files. The licence of the exact vintage used is checked on the Census Bureau page before the first load (first-run check, ADR 0035).
  usage: Input to the remoteness calculation (C17) only.
team:
  - username: John
    role: owner
servers:
  - server: census-tiger-all-roads
    type: api
    description: U.S. Census Bureau TIGER/Line All Roads county-based shapefiles, one file per county. Download URLs fixed by the first-run check (OQ-20).
    location: "Public Census Bureau download. Raw copies with URL, fetch time and hash: s3://{env}-signal-raw/tiger/roads/. Lake table: {env}_signal.tiger_road (Glue catalog)."
slaProperties:
  - property: frequency
    value: reference data
    description: Downloaded when a new TIGER/Line vintage is chosen. Not nightly (ADR 0028).
  - property: retention
    value: raw files kept
    description: Raw source files are kept. Live raw files are locked for 30 days.
customProperties:
  - property: openQuestions
    value: OQ-20
  - property: licence
    value: "Declared CC0 1.0 public domain on data.gov, with a rights flag of otherRestrictions on some files. Checked on the Census Bureau page before the first load."
  - property: adr
    value: adrs/0035, adrs/0031
schema:
  - name: tiger_roads
    physicalName: tiger_road
    physicalType: table
    description: One row per road segment per vintage. Partition per vintage year.
    properties:
      - name: tlid
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        customProperties:
          - property: upstreamField
            value: 'TLID (to be confirmed, OQ-20)'
        description: Census permanent line identifier of the segment.
      - name: vintage_year
        logicalType: integer
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: TIGER/Line vintage the segment comes from.
      - name: full_name
        logicalType: string
        required: false
        customProperties:
          - property: upstreamField
            value: 'FULLNAME (to be confirmed, OQ-20)'
        description: Road name as published. May be empty.
      - name: feature_class_code
        logicalType: string
```

```
        required: true
        customProperties:
          - property: upstreamField
            value: 'MTFCC (to be confirmed, OQ-20)'
        description: Census feature class code, for example the code for a primary road or a local street.
      - name: county_fips
        logicalType: string
        required: true
        description: Five-digit county code that the file was downloaded for.
      - name: geometry_wkt
        logicalType: string
        required: true
        description: Segment geometry as well-known text, in the coordinate system the file publishes (NAD83).
Converted to WGS84 before any distance is computed.
      - name: source_url
        logicalType: string
        required: true
        description: URL the raw file came from.
      - name: fetched_at_utc
        logicalType: timestamp
        required: true
        description: Fetch time, UTC.
      - name: content_sha256
        logicalType: string
        required: true
        description: Hash of the raw file.
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
        description: Required fields are never null.
      - type: library
        metric: duplicateValues
        mustBe: 0
        arguments:
          properties:
            - tlid
            - vintage_year
        description: Unique per segment and vintage.
```

Will I have signal out there/contracts/trail_point_horizon.yaml

```
# Data contract (ODCS). Draft. Internal Iceberg table, written by Spark (C4). Not a source or a published table;
# included because the horizon is one of the four trickiest parts of the design.
apiVersion: v3.0.2
kind: DataContract
id: p1-internal-trail-point-horizon
name: trail_point_horizon
version: 0.1.0
status: draft
domain: project1-signal-planner
dataProduct: signal-planner
description:
  purpose: For every trail point, how high terrain rises in each compass direction out to about 15 km, stored as 36
horizon angles.
  limitations: Built from a surface model that includes vegetation and buildings. Horizon details (range cutoff,
curvature, observer height, direction convention) are open (OQ-24).
  usage: Input to the satellite window engine (C5) through horizon_at.
team:
  - username: John
    role: owner
servers:
  - server: iceberg-trail-point-horizon
    type: s3
    description: Iceberg table in the lake, written by Spark.
    location: "Lake table {env}_signal.trail_point_horizon in s3://{env}-signal-lake/."
slaProperties:
  - property: frequency
    value: reference data
    description: Recomputed when trail or elevation inputs change, or nightly if OQ-25 decides so. Partition overwrite
makes either safe.
customProperties:
  - property: openQuestions
    value: OQ-24, OQ-25
  - property: adr
    value: adrs/0004
```

```
schema:
  - name: trail_point_horizon
    physicalName: trail_point_horizon
    physicalType: table
    description: One row per trail point.
    properties:
      - name: trail_point_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: horizon_angles_deg
        logicalType: array
        required: true
        description: Exactly 36 angles in degrees, one per 10-degree compass sector (sector 0 centered on north,
clockwise; convention is a proposal, OQ-24).
        quality:
          - type: text
            description: Array length is 36. Each value is between 0 and 90 degrees (expected range; confirm margin,
OQ-24).
      - name: computed_at_utc
        logicalType: timestamp
        required: true
      - name: elevation_source_manifest_sha256
        logicalType: string
        required: true
        description: Hash of the tile manifest used, so a result can be replayed.
    quality:
      - type: library
        metric: nullValues
        mustBe: 0
      - type: library
        metric: duplicateValues
        mustBe: 0
        arguments:
          properties:
            - trail_point_id
```

Part 9: API specification

Will I have signal out there/openapi.yaml

```
# OpenAPI 3 specification for Project 1. Draft.
# - The OpenAPI version (3.0.3 here) is a working placeholder; pinned in ADR 0032.
# - Paths, field names and operation ids are proposals (OQ-21).
# - Phase A operations: listTrails, getTrailSignal (without confidence fields).
# - Phase B operations: submitReport, getReportCounts, askTripQuestion, submitFeedback; confidence fields on
getTrailSignal.
# - API type: API Gateway HTTP API for public pages. The REST type is needed only where API keys and per-key usage
plans are used; this project issues no API keys.
# - Lambda memory, timeout, concurrency and throttling values are not recorded (OQ-28).
openapi: 3.0.3
info:
  title: Will I have signal out there? API
  version: 0.1.0
  description: |
    Serving API for the trail page, the hiker report endpoint and the trip-planning assistant.
    All timestamps are UTC in ISO 8601 form and are converted to local time only on the page.
    The first request after a period of inactivity may take up to 10 seconds while the service initializes.
servers:
  - url: https://TBD.example/
    description: Placeholder host. The real base URLs for dev and live are Terraform outputs, in us-east-1 (OQ-42, ADR
0034).
tags:
  - name: trails
    description: Phase A
  - name: reports
    description: Phase B
  - name: assistant
    description: Phase B
paths:
  /trails:
    get:
      tags: [trails]
      operationId: listTrails
      summary: List the main trails the page can show
      responses:
        '200':
          description: Trails
          content:
            application/json:
              schema:
                type: object
                required: [trails]
                properties:
                  trails:
                    type: array
                    items:
                      $ref: '#/components/schemas/TrailSummary'
        '503':
          $ref: '#/components/responses/Unavailable'
  /trails/{trail_id}/signal:
    get:
      tags: [trails]
      operationId: getTrailSignal
      summary: Coverage, satellite windows and (Phase B) confidence along one trail
      description: |
        Returns every sampled point of the trail with its coverage flag per carrier and its satellite windows.
        The response carries the run's valid until time and the time of the last good run. When the data is late, the
page still shows it with a visible late notice and does not hide it (ADR 0006, amended).
      parameters:
        - name: trail_id
          in: path
          required: true
          schema:
            type: string
      responses:
        '200':
          description: Trail signal
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/TrailSignal'
```

```
        '404':
          $ref: '#/components/responses/NotFound'
        '503':
          $ref: '#/components/responses/Unavailable'
  /reports:
    post:
      tags: [reports]
      operationId: submitReport
      summary: Submit one hiker signal report (Phase B)
      description: |
        The body holds only the H3 resolution 9 cell, the carrier, whether the hiker had signal and the time rounded to
the hour.
        Exact coordinates never reach the server. The random visitor ID is used only for rate limits and is not stored
with the report.
        Limits (Decided by John, Oct 9, 2026, ADR 0011): 10 reports a day per visitor ID, one per cell per visitor
every 24 hours, and at most 50 reports a day for the whole system. They are applied by the hour the report was made
(report_hour_utc), not by arrival time, so a batch sent after being offline is judged by the report hours.
        The day boundary is midnight UTC; reports older than 48 hours are refused (ADR 0026).
      parameters:
        - $ref: '#/components/parameters/VisitorId'
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ReportIn'
      responses:
        '202':
          description: Accepted
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ReportAccepted'
        '400':
          $ref: '#/components/responses/BadRequest'
        '429':
          $ref: '#/components/responses/LimitExceeded'
        '503':
          $ref: '#/components/responses/Unavailable'
  /report-counts:
    get:
      tags: [reports]
      operationId: getReportCounts
      summary: Report counts per area (Phase B)
      description: An area is the H3 resolution 6 parent of the report cell; counts from fewer than 5 reports are not
shown (ADR 0030).
      parameters:
        - name: trail_id
          in: query
          required: false
          schema:
            type: string
      responses:
        '200':
          description: Counts
          content:
            application/json:
              schema:
                type: object
                required: [counts]
                properties:
                  counts:
                    type: array
                    items:
                      $ref: '#/components/schemas/ReportCount'
        '503':
          $ref: '#/components/responses/Unavailable'
  /assistant/questions:
    post:
      tags: [assistant]
      operationId: askTripQuestion
      summary: Ask a plain-language trip question and get a cited check-in plan (Phase B)
      description: |
        Subject to a daily question cap and a token limit per answer (values open, OQ-17).
        The answer cites the trail points and times it used and says so when the data cannot answer.
        The assistant answers only within the prediction span and says so for later days (ADR 0021, Decided by John,
```

```
Oct 9, 2026).
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/QuestionIn'
      responses:
        '200':
          description: A plan or a statement that the data cannot answer
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/AssistantAnswer'
        '400':
          $ref: '#/components/responses/BadRequest'
        '429':
          $ref: '#/components/responses/LimitExceeded'
        '503':
          $ref: '#/components/responses/Unavailable'
  /feedback:
    post:
      tags: [assistant]
      operationId: submitFeedback
      summary: Thumbs-up or thumbs-down vote on an answer (Phase B)
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/FeedbackIn'
      responses:
        '202':
          description: Accepted
        '400':
          $ref: '#/components/responses/BadRequest'
        '429':
          $ref: '#/components/responses/LimitExceeded'
        '503':
          $ref: '#/components/responses/Unavailable'
components:
  parameters:
    VisitorId:
      name: X-Visitor-Id
      in: header
      required: true
      description: A 128-bit random value created on the phone, written as 32 hexadecimal characters. Used only for
rate limits, and sent only over HTTPS (ADR 0025).
      schema:
        type: string
  responses:
    BadRequest:
      description: The request is invalid
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
    NotFound:
      description: Unknown trail
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
    LimitExceeded:
      description: A limit was exceeded (report limits, or the assistant daily cap)
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
    Unavailable:
      description: Published data or a dependency is unavailable
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
  schemas:
```

```
    Error:
      description: Error body. Codes are listed in design-doc.md, section 10 (proposed, OQ-21).
      type: object
      required: [error]
      properties:
        error:
          type: object
          required: [code, message]
          properties:
            code:
              type: string
            message:
              type: string
    TrailSummary:
      type: object
      required: [trail_id, trail_name, unit_code]
      properties:
        trail_id:
          type: string
        trail_name:
          type: string
          nullable: true
          description: From TRLNAME.
        unit_code:
          type: string
          description: From UNITCODE.
    TrailSignal:
      type: object
      required: [trail_id, run_logical_date, valid_until_utc, last_good_run_utc, min_elevation_deg, points]
      properties:
        trail_id:
          type: string
        run_logical_date:
          type: string
          format: date
        valid_until_utc:
          type: string
          format: date-time
          description: Predictions are not valid after this time (end of the 26-hour prediction span).
        last_good_run_utc:
          type: string
          format: date-time
          description: Start time of the last good run, shown in the late notice.
        min_elevation_deg:
          type: number
          description: Minimum elevation used by the run.
        carriers:
          type: array
          items:
            type: string
        points:
          type: array
          items:
            $ref: '#/components/schemas/PointSignal'
    PointSignal:
      type: object
      required: [trail_point_id, seq, lon, lat, coverage, windows]
      properties:
        trail_point_id:
          type: string
        seq:
          type: integer
        lon:
          type: number
        lat:
          type: number
        coverage:
          type: array
          items:
            $ref: '#/components/schemas/CarrierCoverage'
        windows:
          type: array
          items:
            $ref: '#/components/schemas/WindowOut'
    CarrierCoverage:
      type: object
```

```
      required: [carrier, covered]
      properties:
        carrier:
          type: string
        covered:
          type: boolean
          description: FCC modeled claim, not a measurement.
        confidence:
          type: object
          nullable: true
          description: Phase B only.
          required: [status]
          properties:
            status:
              type: string
              enum: [confidence_available, not_enough_measurements]
            value:
              type: number
              nullable: true
              description: Null when status is not_enough_measurements. The page then shows "Not enough measurements
here; the coverage map's claim is unverified."
    WindowOut:
      type: object
      required: [satellite_id, start_utc, end_utc]
      properties:
        satellite_id:
          type: string
        start_utc:
          type: string
          format: date-time
        end_utc:
          type: string
          format: date-time
    ReportIn:
      type: object
      required: [cell, carrier, had_signal, report_hour_utc]
      additionalProperties: false
      properties:
        cell:
          type: string
          description: H3 resolution 9 cell index computed on the phone.
        carrier:
          type: string
          description: Allowed values follow the carrier list (T-Mobile in Phase A, ADR 0020; Phase B list OQ-11).
        had_signal:
          type: boolean
        report_hour_utc:
          type: string
          format: date-time
          description: Time rounded to the hour, UTC.
    ReportAccepted:
      type: object
      required: [status]
      properties:
        status:
          type: string
          enum: [accepted]
    ReportCount:
      type: object
      required: [area_id, carrier, had_signal, report_count]
      properties:
        area_id:
          type: string
        carrier:
          type: string
        had_signal:
          type: boolean
        report_count:
          type: integer
    QuestionIn:
      type: object
      required: [question]
      additionalProperties: false
      properties:
        question:
          type: string
```

```
        description: Plain-language question. Maximum length is open (OQ-17).
    AssistantAnswer:
      type: object
      required: [answerable, prompt_version]
      properties:
        answerable:
          type: boolean
          description: False when the data cannot answer.
        message:
          type: string
          description: Explanation when answerable is false.
        plan:
          $ref: '#/components/schemas/CheckInPlan'
        prompt_version:
          type: string
        answer_id:
          type: string
          description: Used for a later vote.
    CheckInPlan:
      type: object
      required: [summary, citations]
      properties:
        summary:
          type: string
        citations:
          type: array
          items:
            type: object
            required: [trail_point_id]
            properties:
              trail_point_id:
                type: string
              time_utc:
                type: string
                format: date-time
              satellite_id:
                type: string
              carrier:
                type: string
    FeedbackIn:
      type: object
      required: [answer_id, vote]
      additionalProperties: false
      properties:
        answer_id:
          type: string
        vote:
          type: string
          enum: [up, down]
```

Part 10: Changelog

`Will I have signal out there/changelog.md`

Project 1 changelog

This file records two things: changes to the documents, and, once code exists, changes to each phase of the project. The newest entry is first.

Oct 9, 2026: defaults confirmed

- John confirmed ADRs 0025 to 0033 (visitor ID, day boundary, late-data rule, refresh and time grain, limits, area counts, model family, versions, votes). Their status is now Accepted.

Documents

Oct 9, 2026: horizon defaults, credit wording, assistant spend limit

- OQ-24 (horizon details) is decided in ADR 0036: 15 km range, 30 m ray step, 1.7 m eye height, Earth curvature applied, refraction not applied, the surface model kept, 36 sectors final, linear interpolation, and a 5-degree margin for AC-1A.2.
- OQ-22 (credit wording) is decided in ADR 0037. Copernicus and NPS were checked against their own licence and policy text. CelesTrak has no terms, so a courtesy credit is used. OpenCelliD and FCC mobile data are unverified, because their primary pages returned HTTP 403 from the build environment. The page wording that coverage is a claim, and that satellite texts are not guaranteed, is also set there.
- OQ-17 is decided in ADR 0038 by John: a monthly spend limit of $10, replacing $15 everywhere in the set. Derived limits: 20 questions a day, 2,500 input and 800 output tokens per answer, with a price check before the first call.
- New contract: none in this step. The licence register and the credit wording were updated.
- Open questions: 19 open and 1 partly answered (OQ-39). ADRs total 39.

Oct 9, 2026: remoteness source, report-cell contract, and the state of open questions

- Remoteness (OQ-18) is decided by John: U.S. Census TIGER/Line roads plus NPS trail lines (ADR 0035). Before the first load, the licence of the exact Census vintage must be checked, because some files carry a rights flag, "otherRestrictions". The check is a first-run check.
- New contracts: `tiger_roads` (the road source for remoteness) and `report_cell_day` (the daily per-cell counts that replace raw reports after 30 days, ADR 0034). Contracts now total 21.
- New internal table in the design doc: `trail_point_remoteness`. New lake table: `tiger_road`.
- Glossary and the licence register gained the Census source and the remoteness term.
- Open questions: 16 answered, 1 partly answered (OQ-39, the provider's terms), and 22 open. The 22 open ones need measurements, real files or accounts, or a decision that only John can make (thresholds, caps, N, attribution wording). Each has a GitHub issue still to be created.
- ADRs total 36 (0001 to 0035, with 0006a superseded).

Oct 9, 2026: minor open questions decided (security first)

John asked for minor questions with an obvious best-practice answer to be decided, security first, and for major ones to be left for him. Thirteen questions were decided this way. Each has an ADR (0025 to 0033) marked as a default that John can change. Each answered row in design doc section 16 keeps its question and gains its answer, so existing references still resolve.

- Decided: OQ-11, OQ-21 (names accepted as working names), OQ-23, OQ-25, OQ-26, OQ-28, OQ-29, OQ-30, OQ-32, OQ-33, OQ-35, OQ-38, OQ-41. Partly decided: OQ-18 (model family) and OQ-39 (votes and question text).

- Decided by John, Oct 9, 2026 (ADR 0034): AWS region us-east-1 (OQ-42); report copies deleted 30 days after the copy in every store, with daily per-cell counts kept (OQ-43); model provider the Anthropic API on the cheapest Claude model (OQ-39, partly).
- New ADRs: 0025 visitor ID; 0026 day boundary and counter lifetime; 0027 late-data rule; 0028 reference refresh, stream and time grain; 0029 limits, retention and log window; 0030 area counts; 0031 confidence model family and carriers; 0032 pinned versions; 0033 votes and assistant data.
- Affected text was updated in the requirements, the design doc, ADRs 0001, 0008, 0011, 0016, 0018, 0020, 0022 and 0024, the contracts, the API file, the security document, the operations document and the model card.

Oct 9, 2026: review pass before coding

Every item below was checked against the build plan, the Build Plan Decisions, the answered decisions and the open-question log.

Added - README.md: what the project is, the document map, the reading order, the build order by dependency and the accounts needed. - ../Platform/platform-requirements.md: the platform parts Project 1 depends on, so the project can be built from the documents alone. - operations.md: schedules, freshness targets, monitoring, cost, 17 runbooks, incidents and backups. - security-and-privacy.md: threat model, personal-data inventory with retention, access matrix, licence and attribution register. - model-card.md: the confidence model and the trip assistant, with the evaluation plan and known failure cases. No results are claimed. - This changelog.

Changed - ADR 0006 existed twice. The superseded version (0006a) was removed on Oct 9, 2026 as a duplicate old version. The amended version keeps the number 0006. - ADR 0005 said the window run covered 24 hours. It now says the 26-hour prediction span (ADR 0019). - ADR 0011 and ADR 0022 said the limit counters expire after 24 hours. The build plan gives that value. Whether it covers the offline delay is open (OQ-26). - ADR 0012 now says the 30-day rule covers DynamoDB only. The lake and Snowflake copies are open (OQ-43). - ADR 0023 had a stray reference to another project's step. Removed. - All 19 data contracts now have a server with a concrete storage location. Eleven contracts had `TBD`, and eight had no server block. The naming convention is in design-doc.md, section 9.1. - The design doc, requirements and API specification now link to ../Platform/platform-requirements.md instead of placeholder paths to a platform folder that does not exist. - The design doc gained: section 9.1 (storage names); section 10 (error codes, and a note that the vote endpoint's 429 has no recorded limit); section 14 (a table of where decisions changed the build plan); section 12.2 (a proposed rule that access logs exclude the visitor ID header); and open questions OQ-42 (AWS region) and OQ-43 (retention of report copies). OQ-39 was extended to cover vote limits. - Glossary: duplicate entries removed; ODCS, residual risk, runbook and threat model added. - Requirements: typos in the feature name "satellite window" corrected; FR-55 notes the open retention question.

Not changed - Requirements, the 17-section design doc structure, the 24 ADRs, the 19 contracts' schemas, and the API paths and operations. These were already complete. - The root-level files `ANSWERED_DECISIONS.md` and `DECISIONS_NEEDED.md`. Each Project 1 item in `DECISIONS_NEEDED.md` is either answered in the design doc or narrowed to an open sub-question (design doc, section 16). Answers come from `ANSWERED_DECISIONS.md`.

Still open at this revision - 24 open questions remain, counting the two partly decided ones (OQ-18 and OQ-39). Each has a GitHub issue still to be created. - The platform's own document set (not written). - The reference generated from code (written when the code exists). - John's approval of each document.

Oct 9, 2026: first complete set

- Requirements (76 functional requirements, 22 quality requirements, acceptance criteria for every feature in Phases A and B).
- Design doc, 17 sections, following the design doc template in the Documentation Plan.
- ADRs 0001 to 0024.
- Data contracts (ODCS v3.0.2 working version) for the 19 published, source and internal tables.
- API specification (OpenAPI 3.0.3 working version) with six operations.
- Status: draft for review.

Phases

Phase	Status	Released
Platform skeleton (before Project 1)	Not started	Not released
Phase A: live (Grand Canyon, T-Mobile, next 24 hours)	Not started	Not released
Phase B: depth (every carrier, warehouse, confidence model, assistant, reports)	Not started	Not released
Phase C: scale (second round)	Not started. Acceptance checks not yet written	Not released

Entries for each phase are added here when the phase ships. Each entry lists the features that went live, the checks that passed, and any known gaps.